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P *indicates a recommendation or evidence relevant to pediatric care.*

MEDICAL ABBREVIATIONS & ACRONYMS

AANA – American Association of Nurse Anesthetists
AHNA – American Holistic Nurses Association
ASA – American Society of Anesthesiologists
BIS – Bispectral index
CAM – Complementary and alternative medicine
CPM – Continuous passive motion
ERAS – Enhanced Recovery After Surgery
FDA – US Food and Drug Administration
GI – Gastrointestinal
ICC – Induction Compliance Checklist
ICU – Intensive care unit
IKDC – International Knee Documentation Committee Subjective Knee Form
INVR – Index of Nausea, Vomiting, and Retching
IU – International units
LOHS – Length of hospital stay
LOS – Length of stay
MRI – Magnetic resonance imaging
NIN – Noninvasive interactive neurostimulation
NSAID – Nonsteriodal anti-inflammatory drug
PACU – Postanesthesia care unit
PEMF – Postoperative pulsed electromagnetic field
PENS – Percutaneous electrical nerve stimulation
PDNV – Post-discharge nausea and vomiting
POD – Postoperative day
PONV – Postoperative nausea and vomiting
PT – Physical therapy
RCT – Randomized controlled trial
RN – Registered nurse
ROM – Range of motion
TCM – Traditional Chinese medicine
tDCS – Transcranial direct current stimulation
TENS – Transcutaneous electrical nerve stimulation
TKA – Total knee arthroplasty
VR – Virtual reality

GUIDELINE FOR COMPLEMENTARY CARE

The Guideline for Complementary Care was approved by the AORN Guidelines Advisory Board and became effective as of June 10, 2021. Information about the systematic review supporting this guideline, including a description of the evidence review, the PRISMA flow diagram, the evidence rating model, and the evidence summary table is available at https://www.aorn.org/evidencetables/.

Purpose

This document provides guidance to perioperative registered nurses (RNs) for determining what **complementary** care interventions will be used in the organization and how to implement them in the perioperative setting. Optimal perioperative nursing practice promotes patient well-being, and implementing patient-centered complementary care interventions can improve the perioperative experience for patients, their families, and health care workers, as well as reduce health care costs. The perioperative experience begins when an operative or invasive procedure is scheduled and ends when the patient is released from follow-up surgical care. Patients can experience preoperative stress, intraoperative discomfort, and problems with postoperative healing, some of which can be addressed with complementary care interventions.

Assessing patients' individual values, health beliefs, and health experiences concerning physical, mental, emotional, spiritual, and environmental factors that can affect well-being and healing is essential when planning and implementing perioperative complementary care interventions. The use of complementary care has increased in many clinical areas, and an increasing number of researchers are studying perioperative outcomes that can be improved with complementary care interventions. As such, this guideline revision expands on the previous version, which was focused on reducing perioperative pain and anxiety, by including **holistic** methods for optimizing the overall health and well-being of perioperative patients through the use of a variety of complementary care interventions that can be implemented throughout the perioperative care plan.

The ability to provide complementary care interventions depends on several patient and organizational factors. Patient factors include the patient's acceptance and engagement, individual values, health beliefs, and experiences with health care. Organizational factors include the clinical experience, the education and competency of the perioperative team to provide complementary care interventions, and procedural and facility constraints. Considerations for cost and feasibility are also discussed in this guideline.

The variable nature of complementary care intervention studies (eg, implementation techniques, assessment, evaluation) limits the reliability and validity of the literature. Researchers have concluded that additional research studies are needed to confirm initial findings and conclusions because intervention techniques, outcomes, and settings have been inconsistent and limit generalizability.[1-10] Complementary care interventions may not be feasible or suitable for every patient; therefore, the perioperative RN plays an integral role in facilitating an optimal perioperative experience for the patient by individualizing complementary care based on the evidence-based practice recommendations in this guideline.

Researchers have examined current use of **complementary and alternative medicine** (CAM) and age-specific differences in use. In national surveys[11-14] conducted from 2012 to 2016, adults, **older adults**, and Hispanic respondents reported increased use of complementary care interventions. Clarke et al[13] found that use of dietary and herbal interventions was the most popular response. Ho et al[12] found that respondents born between 1946 and 1964 were more likely to use complementary care than other groups. Black et al[11] found that older pediatric patients were more likely to use complementary care interventions (eg, yoga). De Moura et al[14] found that younger pediatric patients were more likely to report postoperative pain and 42% of pediatric patients reported having preoperative anxiety. P

The recommendations and interventions described in this guideline are intended to complement **conventional treatment** and not intended to be alternatives to conventional surgical care[10] or **integrative treatments** for conditions, diseases, or illnesses. The National Institutes of Health defines complementary as an adjunct to conventional medicine and alternative as a replacement for conventional medicine.[10] Complementary and alternative medicine includes the use of natural products (eg, herbs, vitamins and minerals, probiotics, dietary supplements) and mind and body practices (eg, deep breathing, yoga, meditation, massage, homeopathy, relaxation, guided imagery). The definition of complementary care is expanding as new interventions continue to be the focus of study.

Topics outside the scope of this document are

- conventional medical treatment,
- nontraditional forms of conventional medicine prescribed by a licensed practitioner (eg, alternative medicine, functional medicine, integrative medicine),
- interventions that have limited use in the perioperative setting (eg, animal therapy, yoga),
- interventions that are implemented outside of the perioperative setting (eg, rehabilitation, physical therapy),

- interventions intended for the purpose of treating a condition or disease (eg, natural hormones, alternative medication, ERAS), and
- interventions to improve health care workers' well-being.

A discussion of **enhanced recovery after surgery (ERAS)**[15] and other conventional medical care guidelines is outside the scope of this guideline, but many elements of ERAS can be found in these recommendations.

1. Selecting Interventions

1.1 Convene an interdisciplinary team to review, select, and evaluate complementary care interventions that will be used in the health care organization.[16] *[Recommendation]*

Using an interdisciplinary team approach can facilitate buy-in from perioperative team members and other health care personnel.[16]

1.1.1 The interdisciplinary team may include

- **surgeons,**
- **anesthesia professionals,**
- **perioperative leaders,**
- **perioperative nurses (eg, preoperative RNs, RN circulators, postoperative RNs),**
- **perioperative team members (eg, scrub personnel, transporters/aides),**
- **quality improvement professionals,**
- **risk management and legal professionals,**
- **resource personnel (eg, registered dietitian, social workers, case managers), and**
- **other stakeholders identified by the organization.**

[Conditional Recommendation]

Including representatives of all perioperative team members as well as other disciplines and departments facilitates safe and efficacious care when complementary care interventions involve multimodal techniques and settings for implementation. Many times, complementary care interventions can be performed preoperatively to positively influence postoperative outcomes, and this necessitates effective team communication, especially when adverse events are a concern. For instance, preoperative interventions are implemented through collaboration of multiple disciplines (eg, schedulers, preadmission personnel, preoperative RNs, intraoperative RNs, anesthesia professionals, surgeons, postoperative RNs); intraoperative interventions will necessitate collaboration from the OR team (eg, parental presence during induction); postoperative interventions will necessitate collaboration with other departments (eg, case managers scheduling follow ups). Some health care organizations have resources such as complementary care and alternative medicine practitioners and integrative medicine professionals who can be excellent resources during selection and planning of complementary care interventions in the perioperative setting.

1.2 Follow applicable local, state (eg, scope of practice[16]), and federal laws when defining and implementing complementary care interventions that will be used in the organization.[16] *[Regulatory Requirement]*

Some complementary care interventions require formal education, training, and licensure (eg, hypnosis, **biofield interventions**, massage, **reflexology**). An American Nurses Association position statement reflects the expanding role of nurses' practice based on education, evidence, and experience.[16] As informed decision makers, nurses can collaborate with patients and their families to implement complementary care interventions that they are suitably educated, trained, and competent to perform.[16]

1.3 When selecting complementary care interventions that may be used in the organization, review the

- **patient population,**
- **types of procedures performed, and**
- **types of anesthesia administered.**

[Conditional Recommendation]

Complementary care interventions are used in diverse populations and settings. Researchers study interventions in a specific population under controlled conditions. Interventions that have been studied and are believed to be beneficial for a specific population under specific conditions may be beneficial for other populations and conditions. The interdisciplinary team can weigh the benefits and harms of the intervention to determine whether it is generalizable to other patient populations, different procedures, or perioperative settings that were not studied.

1.4 When selecting individualized complementary care interventions, take into account information obtained from a comprehensive patient assessment (See Recommendation 2.1).[1,15,17-21] *[Recommendation]*

High-quality evidence[17-20] and guidelines[1,15,21] support using individualized (eg, age-specific[20]) complementary care that may increase patients' satisfaction with their care[22] and that may reduce negative perioperative outcomes.[20] Low-quality evidence supports regular assessment of the patient's preference and comfort with complementary care interventions.[23] Guidelines[1,15,21] emphasize that patient care should honor the patient's values, beliefs, and experiences, including a focus on healing the whole person's well-being and interrelationship with the environment[23] by integrating self-care, self-responsibility, spirituality, and reflection of life.[24]

An American Holistic Nurses Association (AHNA) position statement for the role of nurses in complementary and integrative health approaches includes several recommendations for collaborating on the plan of care with the patient (eg, preferences, family, culture, health beliefs, values) and integration steps (ie, identifying needs, finding providers, facilitating through education, coordinating, and evaluating effectiveness).[23]

1.5 Standardize protocols for implementing complementary care interventions.[1-9,15,25-27] *[Recommendation]*

Standardizing implementation protocols including education, assessments, interventions, and evaluations, can increase health care organizations' and nursing researchers' ability to implement evidence-based interventions[15] for patients who will receive the most benefit. Standardization of education and evaluations,[1,3,25] assessments,[1-3,9,26] and interventions[1,4-7,26,27] increases the feasibility for perioperative nurses to implement practice changes and for health care organizations to accurately assess efficacy and cost-effectiveness of each intervention.

1.5.1 Elements to standardize for complementary care interventions may include

- **Intervention techniques,**[1,4-7,26,27]
- **outcome-specific assessments,**[1-3,9,26]
- **documentation,**[28] **and**
- **perioperative team education and competency evaluation.**[1,3,25]

[Conditional Recommendation]

High-quality evidence supports standardization of complementary care in the perioperative setting.[4-9] Researchers have concluded that lack of standardized skills,[3] assessments,[3,9] and interventions,[4-8] have limited the study of complementary care interventions. The researchers agree that standardization could facilitate study of interventions and increase the reliability of their benefits in the perioperative setting.[1-9] Low-quality evidence[3] and guidelines[1,2] support standardization for optimal implementation of complementary care interventions in the perioperative setting.

1.5.2 Select and implement standardized and validated patient assessment[2] **and evaluation**[1,9,26] **tools for complementary care interventions.**[3] *[Recommendation]*

High-quality evidence supports using standardized assessment tools for baseline measurements[26] and standardized evaluation[9] tools for complementary care interventions. Several authors have concluded that using standardized assessment tools would increase the reliability of their conclusions and improve patient care.[3,9] The authors of one meta-analysis could not compare the efficacy of interventions because baseline assessments were not always included in the studies.[26] Low-quality evidence[3] and guidelines[1,2] support conducting a standardized assessment of the patient to facilitate gathering sufficient information for optimal perioperative care.

2. Patient Assessment

2.1 Before implementing complementary care interventions, assess the patient's

- **capability and willingness to participate in care**[15]**;**
- **preference, tolerance, and willingness**[29-31] **to use complementary care interventions**[13,32-34]**;**
- **understanding of and current use of complementary care interventions (eg, benefits, potential harms);**
- **developmental age, cognitive status (eg, delay, impairment) and understanding**[20]**;**
- **mental state (eg, highly anxious, spiritual)**[24]**;**
- **functional status;**
- **laboratory test results (eg, albumin as an indicator of nutritional status, cortisol as in indicator of stress response)**[17-19,35]**;**
- **nutritional status (eg, malnourished)**[17-19,35]**;**
- **current use of dietary and herbal supplements;**
- **understanding of potential drug-supplement interactions;**
- **history of postoperative nausea and vomiting (PONV); and**
- **risk for**
  - **falls when considering early ambulation,**
  - **pressure injury related to wearable devices,**[36] **and**
  - **surgical burns related to wearable devices when electrosurgery is used.**[37]

[Recommendation]

Assessing the patient's current use of complementary care practices and willingness to use complementary care interventions in the perioperative setting can facilitate creating the individualized plan of care.[29-31] Patients may already use complementary care interventions; however, patient surveys[29,30] and guidelines[31] report that few patients discuss their complementary or non-Western practices with their health care providers. Omitting this information in the assessment can be detrimental if the perioperative plan of care creates contraindications with the patient's complementary care practices. For example, many herbs can interfere with some anesthetics, increase adverse effects of the herbs or the conventional medicine, and increase the potential for perioperative adverse events.[31] One national survey[12] found that the patient's biological sex may influence their choice about whether to use complementary care interventions and which they prefer to use.

Patients may be more willing to try complementary care interventions if they believe the interventions are beneficial. Patients who are interested in complementary care are likely to be more open to learning about and consenting to specific interventions. Patients who have used complementary care will have additional insight into what worked for them and can collaborate to reduce the time, money, and effort of implementing interventions that they have no interest in pursuing. Patients who are already using complementary care interventions may not be aware of suitable use, additional interventions, and potential synergistic or maladaptive interactions between their current interventions and the complementary care interventions proposed in the perioperative plan of care. Additional patient education may be required to promote safe perioperative care.[13,32-34]

Moderate-quality evidence supports identifying needs and risk factors related to perioperative care.[38] High-quality evidence[39] and guidelines[1,15,21] recommend assessing risk factors for PONV to determine the plan of care, including for whom and when each complementary care intervention may be used.

Low-quality evidence[29,30,40] and guidelines[31,41] support that many perioperative patients are unaware that dietary and herbal interventions can interact with their conventional medical plan of care, especially the consumption of herbs. Patients may not disclose this information if they perceive that their herbal supplementation is unrelated to their medical condition; however, herbal use may increase the patient's risk for adverse events and negative outcomes in the perioperative setting, regardless of the herbs indicated or desired intent. Sankar-Maharaj et al[40] found that patients who used cannabis preoperatively required active warming postoperatively.

In a survey of 526 patients at one facility, Levy et al[29] found that of 44% of patients who reported using dietary and herbal supplements, only 16.5% understood that these supplements could interact negatively with anesthesia or increase their risk for injury. The researchers concluded that perioperative management and health care workers' education about supplements should be emphasized to promote safe use.

The American Association of Nurse Anesthetists (AANA) recommends that anesthesia providers interview patients preoperatively and be aware of potential adverse effects, drug interactions, and complications associated with dietary and herbal interventions.[31] Though there is no official guidance, a patient education brochure from the AANA recommends that the patient discontinue taking herbal supplements at least 1 to 2 weeks before undergoing an operative or invasive procedure.[31] The AANA recommends requesting full disclosure of natural remedies and discussing the conventional surgical plan of care with patients and their families during the anesthesia preoperative visit.[31]

The American Society of Anesthesiologists (ASA) brochure on dietary and herbal interventions identifies that half of Americans take some herbal or dietary supplement.[41] These are not regulated by the US Food and Drug Administration (FDA), and some dietary and herbal interventions have been associated with prolonged anesthesia metabolism, increased bleeding, hypertension, interference with medications, and heart problems in some individuals. The brochure advises that patients should bring all supplement bottles to preoperative appointments because anesthesia professionals may recommend discontinuing use at least 2 weeks before an operative or invasive procedure.[41]

2.2 **Assess the patient's response to complementary care interventions and adjust as needed based on the patient's input and an evaluation of intended outcomes.**[28,42-46] ***[Recommendation]***

Changes in the patient's condition may change the patient's response to complementary care interventions, resulting in a need for interventions to be modified or discontinued. The Institute of Medicine recommends evaluating CAM intervention implementation and efficacy.[28] Intervention methods may change in phases of care or depending on the patient's condition preferences, and potential interactions of the patient's current and desired complementary care interventions with the patient's individualized conventional surgical plan.

High-quality evidence supports that interventions may have synergistic effects in combination, though some effects may not be enhanced but diminished.[42-46] Perioperative RNs can help guide the patient to prioritize interventions that provide the most benefit and reduce conditions (eg, cost, time, mental energy) that may occur with additional interventions if one intervention has no effect or less effect than another.

Low-quality evidence supports regular assessment of the patient's desire for and the effectiveness and efficacy of complementary care interventions.[23]

In a qualitative study of perioperative nurses' perceptions of the preoperative assessment, Malley et al[38] organized the findings into four themes: understanding the patient's vulnerabilities, multidimensional communication, managing patients' expectations, and the nurses' role in compensating for gaps. The researchers concluded that the perioperative nurses' use of the preoperative assessment to identify the patients' needs and risk factors was associated with a positive perioperative care trajectory.

A position statement from the AHNA about the role of nurses in complementary and integrative health approaches recommends collaborating on the plan of care with the patient (eg, preferences, family, culture, health beliefs, values) and integration steps (ie, identifying needs, finding providers, facilitating through education, coordinating, and evaluating effectiveness).[23]

3. Planning Interventions

3.1 **Use individualized, multimodal complementary care interventions based on an assessment of the patient.**[1,12,15,17,19-21,28,46-62] *[Recommendation]* P

Clinical guidelines support individualized, multimodal interventions to improve desired outcomes.[1,15,21,28]

The Institute of Medicine CAM guidelines recommend an evaluation of the patient, history, physical examination, and conventional and other methods of diagnosis; documentation of medical options discussed, offered, tried, and refused; referral of treatment; risks and benefits; and extent of potential interference with other recommended or ongoing treatment.[28]

The ERAS guidelines are developed through a review of literature for multimodal conventional, CAM, and surgical techniques for specific patient populations to improve clinical and nonclinical outcomes.[15]

High-quality evidence also supports implementing individualized multimodal complementary care interventions together with conventional surgical care (eg, **essential oils** plus medications for PONV) to improve the patient's perioperative experience.[12,17,19,20,46-62] Specific patient populations may respond to complementary care interventions differently than others, and high-quality evidence supports additional research to determine the influence of age on complementary care intervention efficacy.[63]

Though there is limited research, evidence supports individualizing interventions based on patient factors, including

- age (eg, pediatric,[20,47,53-55,62] older adult[17,19,28,48,57-60]),
- nutrition status (eg, malnutrition),[49-51] and
- baseline anxiety scores.[46,52,56]

3.1.1 **Complementary care interventions specific to reducing perioperative anxiety in pediatric patients may be used.**[15,20,28,53,56,62] *[Conditional Recommendation]* P

High-quality evidence supports using complementary care interventions for pediatric patients.[20,53,56,62] Based on the limitations of their results, researchers found different levels of effectiveness and concluded that additional research can determine whether pediatric patients' ages are associated with the effectiveness of different interventions.[20,54,55] Guidelines also support age-suitable interventions, as pediatric populations may experience perioperative complementary care interventions differently than adult-only populations.[15,28]

In a systematic review of 33 trials and quasi-experimental studies, Bice and Wyatt[62] found four holistic comfort interventions (eg, music [2003-2007], n = 287; amusement and entertainment [1994-2015], n = 1,210; caregiver facilitation [2001-2013] n = 349; and a combination of interventions [2000-2015], n = 569) that increased comfort for pediatric patients undergoing a variety of procedures (eg, venipuncture, port access, injections, burn wound care). A study criterion was that the patient sample must include children 4 to 7 years of age. The researchers concluded that various distraction techniques were beneficial for decreasing anxiety, distress, fear, and pain. The researchers suggested that future studies are needed to explore the children's experience of painful procedures, including those that are performed in the perioperative setting, to determine the benefits of each complementary care intervention.

In a randomized controlled trial (RCT), Liguori et al[53] provided 20 pediatric patients ages 6 to 11 years with a 6-minute video on a mobile app the night before their surgical procedures. The video depicted two surgeons dressed as clowns who gave a comical and informative tour of the operating room (OR) with the intention of reducing preoperative anxiety among the children who viewed it. Children who viewed the video had significantly lower anxiety scores before induction of general anesthesia compared to a control group that received standard care (n = 20). The researchers concluded that the mobile application–delivered video could substitute for education and tours delivered by health care workers to decrease costs. The researchers recommended that future studies control for age, sex, and history of previous surgery.

In an RCT, Kassai et al[56] gave pediatric patients ages 6 to 17 years a comic information leaflet before surgery (n = 54). The researchers found that compared to a group that received standard preoperative education (n = 57), anxiety was significantly lower in the intervention group. Patients with higher baseline anxiety scores had more significantly reduced anxiety, and the researchers concluded that these patients benefited more from the intervention.

In a quasi-experimental study that included 83 pediatric patients ages 9 to 18 years, Aytekin et al[20] found that tailoring the distraction technique (eg, games, music, cartoons, books) to the age group significantly decreased anxiety and separation anxiety in participants who received individualized interventions.

3.1.2 **Complementary care interventions specific to older adult patients may be used.**[15,17,19,28,48,57-60] *[Conditional Recommendation]*

High-quality evidence[17,19,48,58-60] supports specific interventions for the older adult patient including nutritional support,[57,59] acupoint for gastrointestinal (GI) functional recovery after GI procedures,[58] and cognitive health support.[48,60] Older adult patients are more at-risk for nutritional and cognitive deterioration, and perioperative complications. Nutrition status deficits among older adults can be improved with preoperative dietary supplementation for malnutrition[57,59] and electrical acupoint stimulation for GI functional recovery.[58] Music[48] and cognitive education and skills training[60] for older adult patients can significantly improve postoperative cognitive function.

Guidelines[15,28] also support age-specific interventions, as older adult patients may experience perioperative complementary care interventions differently than younger patients, including having an increased risk for adverse events (eg, medication interactions,[19] fractures[17,18]).

In a systematic review that included retrospective case series totaling 15 trials, the researchers found that preoperative nutrition status (eg, low serum albumin, more than 10% weight loss in the previous 6 months) was a significant indicator of postoperative outcome in older adult patients. The researchers concluded that older adult patients are subject to nutrition deterioration and complications and would benefit from dietary interventions.[57]

In an RCT, of 126 older adult patients with hip fractures, Myint et al[59] found oral liquid nutritional support significantly shortened length of stay (LOS) in the rehabilitation ward and decreased the infection rate. The researchers found that both the intervention and control groups had a decrease in body mass index, but the weight loss was less in the group that received oral liquid nutritional support (n = 65) compared with a group that received the standard hospital diet (n = 61). There were no changes in albumin level, functional status, or mobility. The researchers concluded there were nutritional benefits but no rehabilitation benefits to protein supplementation for older adult perioperative patients.

In an RCT conducted by Lili et al,[58] 40 older adult patients undergoing GI procedures were given electrical acupoint stimulation for 2 minutes every postoperative day (POD). The researchers found significant improvement in gastrin levels on POD 3 and 5, improved motilin levels on POD 3, shorter time to flatus, and a lower incidence of complications (eg, abdominal pain, distention, and diarrhea), as well as improved the electrocardiogram frequency and amplitude on POD 5. The researchers concluded that **acupuncture** stimulation promoted GI function recovery and could decrease complications efficiently and painlessly in older adult patients.

In an RCT conducted by Çetinkaya[48] in Turkey of 60 older adult patients undergoing total joint procedures, Turkish music was played for 20 minutes three times a day on the first, second, and third postoperative days. Patients' cognitive scores were significantly improved on POD 3 compared with POD 1. The researcher concluded that music prevented cognitive dysfunction, though the study sample was limited to patients in one facility and only one type of music was used.

In an RCT pilot, Kulason et al[60] found that providing 12 older adult postoperative patients with cognitive education and skills training (ie, **simple calculation and reading aloud** [SCRA]) drills 3 to 5 times a week for 30 minutes significantly improved the patients' motor function and quality of life. The researchers concluded that preoperative cognitive education and skills improved the cognitive and emotional state (eg, level of distress) of postoperative older adult patients.

In an RCT, Seidi et al[19] assigned patients to receive either one dose of 1 g of oral ginger (n = 40), two doses of 500 mg of oral ginger (n = 42), or a placebo (n = 40) before undergoing cataract procedures under general anesthesia. The researchers found significantly lower scores for nausea and vomiting for the participants who received two smaller doses. They concluded that oral ginger is safe and effective to reduce PONV, and two smaller doses may be more beneficial than one large dose. However, the researchers noted potential adverse effects in older adults and potential for adverse events with certain medications. They recommended that additional research include a longer preoperative consumption time to confirm the benefits of oral ginger.

In an RCT conducted by Bischoff-Ferrari et al,[17] older adult patients (65 years or older) with hip fractures were given extended postoperative physical therapy (PT) and either 2,000 international units (IU) or 800 IU of cholecalciferol per day (2,000 IU + standard PT [n = 42], 2,000 IU + extended PT [n = 44], 800 IU + standard PT [n = 44], 800 IU + extended PT [n = 43]). The researchers found a 25% reduction in falls and 39% reduction in hospital readmissions in both groups that received 2,000 IU. The patients who received 2,000 IU of cholecalciferol had a lower risk of hip fractures, lower vitamin D deficiency, and lower facility readmission rates that the patients who received 800 IU.

In an RCT, Chapuy et al[18] provided 1.2 g elemental calcium and 800 IU vitamin D3 (n = 1,634) compared to placebos (n = 1,636) to women ages 69 to 106 years who were living in nursing homes. The researchers found that calcium and vitamin D3 supplementation significantly lowered the incidence of hip and other nonvertebral fractures.

3.1.3 **Plan complementary care interventions for patients who are identified as being highly anxious at the baseline measurement.**[46,56] *[Recommendation]* P

High-quality evidence suggests that patients who are assessed as highly anxious receive greater benefit from complementary care interventions than those who are not identified as highly anxious in the preoperative assessment.[46,56]

In an RCT, Kassai et al[56] provided pediatric patients ages 6 to 17 years with a comic information leaflet before surgery (n = 54). The researchers found that compared to a group that received standard preoperative education (n = 57), anxiety was significantly lower in the intervention group. The researchers concluded that those with higher baseline anxiety scores had more significantly reduced anxiety and benefited more from the intervention.

In a quasi-nonequivalent controlled trial, Kim et al[46] evaluated the effects of intraprocedural handholding with or without sharing of verbal step-by-step information on the anxiety of patients undergoing percutaneous vertebroplasty under local anesthesia. Ninety-four patients received either standard care (n = 30), handholding (n = 30), or handholding and procedural information (n = 34) during their procedures. The researchers found a significant decrease in systolic blood pressure in the control group and significant decrease in anxiety among participants that received both handholding and verbal step-by-step information shared throughout the procedure. The researchers concluded that this study demonstrates the value of nursing care, specifically handholding, for anxious patients during a surgical intervention.

3.2 **In collaboration with the patient, plan complementary care interventions as early as possible after a surgical intervention is scheduled.**[1,21,23,38] *[Recommendation]*

Moderate-quality evidence supports that early assessment and collaboration with patients facilitates creating a complementary care plan that fits with the patient's needs, lifestyle, and perioperative plan of care by identifying the patient's physical, spiritual, or social needs.[23,38] Well-established complementary care interventions are associated with a positive effect on perioperative adverse outcomes (eg, PONV,[1,21] anxiety related to isolation, uncertainty about the procedure, pain or discomfort).

If there is limited preoperative time, some interventions may not be feasible, whereas delaying the assessment may prevent implementation of time-dependent interventions (eg, **prehabilitation**, massage, nutritional supplementation) or resource-heavy interventions (eg, stress diversion, music, aromatherapy, acupoint, massage).

In a qualitative study of perioperative nurses' perceptions of the preoperative assessment, Malley et al[38] organized the findings into four themes: understanding the patient's vulnerabilities, multidimensional communication, managing patients' expectations, and the nurses' role in compensating for gaps. The researchers concluded that perioperative nurses use the preoperative assessment to identify the patients' desires and risk factors associated with the perioperative care trajectory.

An AHNA position statement for the role of nurses in complementary and integrative health approaches includes recommendations for collaborating on the plan of care with the patient (eg, preferences, family, culture, health beliefs, values) and integration steps (ie, identifying needs, finding providers, facilitating through education, coordinating, and evaluating effectiveness).[23]

3.2.1 **Additional time may be scheduled for consultation[28] with practitioners in other disciplines,[16] to obtain any needed orders, and to implement the intervention in the indicated phase of care.** *[Conditional Recommendation]*

The time frame may extend months in advance of the planned procedure (ie, prehabilitation).

3.2.2 **Include the patient's priorities for surgical outcomes and expectations of complementary care interventions in the perioperative plan of care.**[3,28,64] *[Recommendation]*

Discussing priorities and expectations of complementary care interventions with the patient is essential when creating the perioperative plan of care. Moderate-quality evidence[3,64] and guidelines[28] support assessing expectations and priorities and educating patients on postoperative outcomes to promote realistic expectations and care plan goal setting.

Silva et al[64] conducted a survey in which patients reported high expectations for pain, thirst, and hunger discomfort. However, after their procedures, the patients reported that their discomfort regarding thirst, hunger, physical weakness, and coldness far surpassed their expectations.

In a survey of patients evaluating perception of nursing care, Fatma and Serife[3] established that an expectation for pain management and evaluating specific interventions for efficacy can improve the patient's experience and future experiences. The researchers suggested that perceived and actual patient needs, expectations, and responses to interventions may change for different events as well as throughout perioperative experiences and their lives.

3.3 **Obtain informed consent before implementing complementary care interventions.**[11-14] *[Recommendation]*

Complementary care interventions can have both benefits and potential harms to the patient, and these can be different for each patient. For this reason, a discussion of the benefits and potential harms is important as part of obtaining informed consent.

Moderate-quality evidence supports using a patient-centered approach when communicating the risks and benefits of complementary care interventions and when incorporating these interventions into the conventional surgical plan of care.[11-14] Based on national trends[11-14] and patient education[65] and preference,[12] nurses can suggest common interventions with which patients are more likely to have experience, that they may be willing to try, and that are likely to provide a benefit. Patients who are educated in complementary care intervention risks and benefits may be more willing to accept the interventions or decide which interventions would not improve their perioperative experience.

3.4 **Limit the intervention exposure (eg, music,[66] aromatherapy[67-69]) to the patient for whom it is intended.** *[Recommendation]* P

Some interventions have the risk of affecting more than just the intended patient (eg, unwanted transfer of aromatherapy scent to other patients,[68] patient-selected music being heard by other patients, music contributing to noise in the care environment[66]). Restricting the exposure of interventions to only the intended patient preserves patients' autonomy for their care.

In a systematic review of 73 RCTs conducted between 1988 to 2013, Hole et al[66] evaluated research on perioperative music intervention benefits for postoperative pain, anxiety, use of analgesia, and patient satisfaction. The researchers found improvements in all of these outcomes and concluded that perioperative music interventions were beneficial. The researchers found that music was effective even when patients were under a general anesthetic. However, the benefits of allowing patients to choose the music and a suitable volume for music were unclear. Also unclear were conclusions about whether the use of headphones hindered communication between patients and perioperative health care workers and whether use of headphones alleviated imposing the music on the health care workers.

Aromatherapy delivery devices are meant to limit exposure of the aromatherapy intervention to a single user. High-quality evidence supports using aromatherapy delivery devices in perioperative settings to reduce PONV[68] and post-discharge nausea vomiting (PDNV).[67,69] Essential oil room-diffusers may pose challenges, and researchers in one study reported the difficulty of restricting aromatherapy interventions through a handheld device.[68] Creative methods may be required to restrict aromatherapy to its intended area.

Kiberd et al[68] found that a limitation of a pilot RCT of pediatric patients (N = 162) ages 4 to 16 years who used a handheld aromatherapy delivery device was that the researchers were unable to contain the aromatherapy intervention in a way that the control group was not also exposed to the aromatherapy. Even though research was limited by not having a true control group, the researchers did find that there was a significant desire to continue use of the handheld aromatherapy delivery device among the intervention group.

In an RCT conducted by Stallings-Welden et al,[67] patients received a handheld aromatherapy delivery device after surgery (n = 108) versus standard care (n = 113). The aromatherapy used was a blend of pure essential oils including spearmint, peppermint, ginger, and lavender. There was no significant difference in PONV incidence, antiemetic dose, or patient satisfaction in the intervention group prior to patient discharge from the perioperative setting with the product for at-home use. However, the aromatherapy intervention was 100% effective in treating PDNV among the 11 patients who reported PDNV. The researchers concluded that identifying patients at risk for PONV and planning a holistic comfort approach to address PONV that included aromatherapy was beneficial.

A quasi-experimental study also supported use of aromatherapy delivery devices to reduce PONV. Lee and Shin[69] gave 60 patients either ginger essential oil or normal saline in a necklace delivery device in the postoperative setting. The researchers found significantly lower PONV measured by the Index of Nausea, Vomiting, and Retching (INVR) scale in the first 6 hours. The researchers concluded that even though there is no documented optimal delivery technique for aromatherapy, use of the necklaces decreased PONV, and the intervention was inexpensive and easy to administer

3.5 **When devices are used to deliver complementary care interventions (eg, headphones for music, a wristband for acupoint), follow the manufacturer's instructions for use.** *[Conditional Recommendation]*

3.6 **Establish a process for the care and cleaning of reusable devices.** *[Recommendation]*

3.7 **Encourage the patient's autonomy by having the patient select specific preferences related to each intervention (eg, music type,[43,70] stress diversion[71]) in each perioperative setting, when possible.** *[Recommendation]*

High-quality evidence supports encouraging patients' autonomy by allowing them to express a preference regarding complementary care interventions, when possible.[43,70,71]

In an RCT conducted by Nagata et al,[43] patients (N = 224) either listened to instrumental music, received aromatherapy, or both compared to a control group while undergoing colonography procedures (n = 56 in each group). The results for pain scores and vital signs were nonsignificant between the groups, though a significant number of patients asked to use the same intervention for the next screening. The researchers concluded that additional research including music chosen according to patient preference may be beneficial.

In an RCT that included 124 male patients undergoing flexible cystoscopy awake procedures, Zhang et al[70] compared an intervention with patient-selected music (n = 62) to standard care (n = 62). The researchers found heart rates, pain scores, and anxiety scores were significantly reduced in the intervention group.

3.8 **In collaboration with surgeons and anesthesia professionals, determine which dietary and herbal supplements should be discontinued before an operative or other invasive procedure, when they should be discontinued, and when the patient can resume taking dietary and herbal supplements after the procedure.**[23,31,41,72] *[Recommendation]*

Discontinuing use of dietary and herbal supplements preoperatively can mitigate the risk for adverse events. Low-quality evidence[72] and instructions from the AANA[31] include discontinuing supplements before procedures to reduce potential complications, though the recommended time frames vary.

Patients may not be aware of the interactions of their current practices with medications they may receive in the perioperative setting, and may not consider them relevant to disclose during the preoperative assessment. Perioperative nurses can facilitate open communication by delving into the patient's interests and background[23] and reviewing common complementary care practices and potential adverse events associated with them.

The AANA recommends that surgical patients discontinue herbal remedies 1 to 2 weeks before an operative or invasive procedure to reduce the risk for complications (eg, prolonged anesthesia metabolism, increased bleeding, hypertension, interference with medications, heart issues), inform their provider of historical or current use, and bring supplement containers to their appointments. The AANA also recommends that the patient tell their family members about their herbal supplement and medication use in case of an emergency when the patient cannot communicate this information.[31]

The ASA brochure on dietary and herbal interventions identifies that 50% of Americans take some herbal or dietary supplement. These are not regulated by the FDA, and some dietary and herbal interventions have been associated with prolonged anesthesia metabolism, increased bleeding, hypertension, interference with medications, and heart problems in some individuals. The brochure advises that patients should bring all supplement bottles to preoperative appointments because anesthesia professionals may recommend discontinuing use at least 2 weeks before an operative or invasive procedure.[41]

In a case report that was included in a literature review, Pedroso et al[72] described one patient's chronic use of ginkgo biloba for a variety of conditions (ie, cognitive impairment, tinnitus, vertigo, asthma, allergies). The researchers found that these were an ill-founded use of the herb and that the patient required hospital admission for spontaneous cerebral bleeding attributed to chronic use of the herb, which improved after discontinuation.

4. Supportive Education Interventions

4.1 **Schedule time to provide patients with unrushed supportive education before the procedure[25,73] (eg, the night before surgery,[74,75] the morning of surgery[75,76]) and the opportunity to ask questions about the perioperative experience.** *[Recommendation]*

4.1.1 **Select and implement techniques for perioperative supportive education delivery that are tailored to the patient's age[77-80] and cognitive ability.[57]** *[Recommendation]* P

Moderate-quality evidence supports tailoring counseling based on the patient's age (eg, pediatric,[77-80] older adult[57]). Factors other than age can also affect understanding (eg, learning disabilities, cognitive disorders, distractors).

In a systematic review that included qualitative surveys, Perry et al[77] found that reducing preoperative anxiety in pediatric populations required age-specific teaching interventions, including the information that is shared, when and how it is shared, and who provides the information. They found that role-play or play therapy with dolls and toys was more efficacious for younger children, and OR tours, coloring books, or video games were more efficacious for school-aged children. They also found that patients between 3 and 6 years were less anxious when given interventions less than 7 days before the procedure, while patients older than 6 years needed interventions at least 5 days in advance and patients younger than 3 may not benefit from preoperative teaching interventions.

In an RCT, Rhodes et al[78] evaluated anxiety and patient satisfaction in 65 pediatric and young adult patients (ages 11 to 21 years) who received either standard care (n = 39) or standard care with the inclusion of 30 minutes of education (n = 26) before undergoing a posterior spinal fusion procedure to correct scoliosis. The researchers found that the education intervention did not lower anxiety as hypothesized, and the intervention group had significantly higher perioperative anxiety; however, both the patients' and caregivers' overall satisfaction was significantly higher. The researchers speculated that having more information increased the patients' perioperative anxiety but improved their postoperative expectations and experience, which could be associated with higher satisfaction as a result of the increased trust and stronger patient-family-caregiver relationship created by increased preoperative interaction.

In an RCT, Fernandes et al[79] assigned 125 pediatric patients ages 8 to 12 years to one of seven different preoperative education groups: education and entertainment via an education board (n = 15), video (n = 15), or booklet (n = 15); entertainment only via an education board (n = 15), video (n = 15), or booklet (n = 15); or a control group that received no materials (n = 35). The researchers found that regardless of the technique, receiving education significantly reduced the patients' anxiety compared to receiving entertainment alone. However, the was no difference in parental anxiety scores regardless of the child's intervention group. The researchers concluded that lower resource–intensive interventions could be beneficial, though they also recommended that additional studies include longer preoperative time frames and patients undergoing major inpatient procedures.

In an RCT of 42 pediatric patients ages 5 to 12 years, Batuman et al[80] found that preoperative information provided by video (n = 21) instead of verbally (n = 21) significantly lowered anxiety and postoperative maladaptive behaviors. The researchers concluded that future research should focus on comparing when the video is viewed and composition of video interventions.

4.1.2 **Deliver supportive education interventions in the patient's preferred language.[81,82]** *[Recommendation]*

High-quality evidence supports providing supportive education interventions in the patient's preferred language.[81,82]

In an RCT, West et al[81] found that when an instructional video in Spanish was used for patients whose preferred language was Spanish, they had significantly decreased anxiety scores and increased satisfaction. The researchers concluded that though there were no objective measurements, videos provided in the patient's preferred language could be a beneficial substitution (eg, less cost, less time) when language interpreters are not available. The researchers of another RCT recommended that future studies regarding the efficacy of video education involve multiple languages.[82]

4.2 **Select and implement preoperative supportive education interventions for anxiety.** *[Recommendation]*

High-quality evidence supports providing patients with preoperative supportive education to decrease anxiety. Researchers found that compared to receiving no materials, receiving verbal,[83,84] written,[85] and audio-visual[81,82,86] materials designed to increase knowledge of the perioperative setting decreased patients' preoperative anxiety and could improve other clinical outcomes. Researchers also found improved cortisol levels with the use of structured verbal materials[84] and improved patient satisfaction with the use of structured verbal[83] and audio visual[86] materials.

In an RCT, Zhang et al[83] found that patients undergoing elective coronary artery bypass grafting who received 3 days of nurse-initiated preoperative structured education (n = 20) had significantly lower rates of complications and less anxiety than patients who were admitted only 1 day before surgery (n = 20). Patients in the intervention group had lower rates of postoperative lower-extremity edema, urinary retention, and constipation than those in the control group.

In an RCT, Amini et al[85] assigned patients undergoing elective hernia or cholecystectomy procedures to receive either preoperative verbal education (n = 20), education via a booklet (n = 20), or standard care (n = 20). There was a significant decrease in anxiety after the intervention in both interventional groups, and the researchers concluded that a well-designed booklet could decrease preoperative anxiety and reduce the resources required for in-person education. However, limitations include the pseudo-randomization in which patient groups were predictable to the researchers and that the results can be generalized only to patients undergoing hernia and cholecystectomy procedures.

In a RCT conducted by Bahrami et al[84] in Iran, patients undergoing elective gynecological procedures who participated in anxiety-reduction techniques (eg, deep breathing, distraction, repeating religious words) the night before their procedure (n = 30) had significantly lower serum cortisol levels on the morning of the procedure than those who received standard care (n = 30). However, no significant difference in vital signs was found between the intervention and control groups. The researchers concluded that individualized interventions to reduce patient anxiety may be included in the nursing plan of care.

In a pilot RCT, Tou et al[82] studied adult patients scheduled to undergo operative colorectal procedures (n = 16) who watched a 13-minute cartoon video step-by-step guide to colorectal surgery compared to a control group (n = 15). Fourteen (88%) of the patients in the intervention group reported that the video was helpful, and the researchers found a significant decrease in immediate and discharge anxiety in the group that watched the video compared to the control group. The researchers concluded that videos are beneficial and recommended that future studies involve videos for other procedures and in other languages.

In an RCT of 100 adults undergoing a variety of operative procedures (eg, upper abdominal surgery, lower abdominal surgery, other surgery [eg, thyroidectomy mastectomy, lower limb]) with either general or spinal anesthesia, Lin et al[86] had patients view an 8-minute video on perioperative anesthesia education (n = 50) or receive a verbal briefing by an anesthesia professional (n = 50) during their visit to the preoperative clinic for preanesthetic assessment. The video included information about the preoperative assessment, differences between general and spinal anesthesia, medication administration, and recovery management, whereas the verbal information included the process and protocols for anesthesia procedures. The researchers found no significant differences in opioid use in the immediate postoperative period and no difference in PONV incidence between the intervention and control groups. However, the patients in the video group had significantly lowered anxiety scores and increased patient satisfaction compared to those who received an 8-minute verbal briefing by an anesthesia professional.

In an RCT, West et al[81] found that an instructional video in Spanish for Spanish-speaking adults significantly decreased anxiety scores and increased patient satisfaction of the knowledge provided compared to patients whose immediate preoperative anesthesia evaluation was conducted through an interpreter. The researchers concluded that this interim study demonstrated that though the outcomes were subjective (ie, patient reported), and the patients' baseline literacy could influence the results, videos could be beneficial to supplement interpreter services in a cost-effective manner.

4.2.1 **Additional supportive education time (eg, the night before surgery,[74,75] the morning of surgery[75,76]) may be scheduled for patients with a high level of anxiety.** *[Conditional Recommendation]*

Moderate-quality evidence[74-76] supports decreasing preoperative patient anxiety and improving the perioperative experience with additional preoperative visits from an OR health care worker (eg, perioperative nurse, surgical technician, nurse practitioner) based on when patients are admitted to the facility. Researchers have found

- decreased anxiety with additional visits on the night before[74,75] and morning of the procedure,[75,76]

- decreased complications with additional visits on the night before the procedure,[74] and
- improved quality of life scores with additional visits on the morning of the procedure.[76]

In an RCT, Sadati et al[74] assigned women undergoing laparoscopic cholecystectomy to receive either routine nursing care (n = 50) or two preoperative nursing visits that included patient education about the OR, anesthesia, the surgical procedure, and pre- and postoperative care (n = 50). The first visit occurred on the day before the procedure and the second occurred after transfer to the preoperative area. The researchers found no difference in anxiety scores between the groups on admission to the preoperative area; however, there was a significant decrease in anxiety just before patients entered the OR after the second nursing visit. Postoperative outcomes were also significantly better in the intervention group, including mean time to consciousness (ie, Aldrete score of 9), incidence of PONV, level of postoperative pain, duration of vital sign stabilization, and time to ambulation. The researchers concluded that additional nursing visits should be integrated into standard care for laparoscopic cholecystectomy patients.

In a quasi-experimental study conducted by Bagheri et al,[75] 70 adults undergoing elective hernia procedures in Iran received two preoperative visits from the same surgical technologist, first on the night before the procedure and again after transfer to the perioperative suite through anesthesia induction. The surgical technologist provided information about the surgeon, surgery, time of surgery, duration of surgery, OR setting, and duration of the immediate postoperative unit stay using simple and understandable language. Patients were given the opportunity to ask questions, and the surgical technologist reassured the patient that he would be present in the OR for the surgery. Although the study was underpowered, the researchers found a significant decrease in the patients' anxiety scores and concluded that preoperative visitation could decrease anxiety for patients undergoing elective procedures.

In a quasi-experimental study conducted by Donker et al,[76] 214 older adults undergoing elective vascular procedures at one facility in the Netherlands received either standard care (ie, consultation by the surgeon alone [n = 81]) or additional education from a nurse practitioner involved with the operative procedures (n = 133). Both groups had a significant increase in quality of life scores and a decrease in anxiety scores, but no difference in depression scores. Researchers concluded that nurse practitioners could provide education as efficaciously as the surgeon. They also concluded that future studies should measure the effect of involving a nurse practitioner on patient satisfaction.

4.3 **When applicable, provide preoperative patient education for alcohol and smoking cessation.**[15,87]
[Recommendation]

ERAS guidelines[15] and an RCT[87] support cessation programs to reduce the potential for negative postoperative outcomes related to suboptimal preoperative patient conditions (eg, alcohol consumption, smoking.) Optimization for perioperative care guidelines, including ERAS,[15] may include minimizing or abstaining from alcohol and participating in smoking cessation programs, as these habits are known deterrents to incision healing.[15]

In an RCT, Lindstrom et al[87] assigned patients to receive either standard care (n = 54) or an 8-week smoking cessation program with 4 weeks before and 4 weeks after the procedure (n = 48). The researchers found a significant decrease in the complication rate (eg, requiring additional medical or surgical treatment) in the intervention group (21%) compared with the standard care group (41%). The researchers found a significant decrease in the overall complication rates 30 days post surgery in the intervention group, and they concluded that a 4-week preoperative program may be as effective as the 8-week program.[87]

4.4 **When applicable, provide information to patients about accessing postoperative support, including**
- **postoperative support groups[88] and**
- **postoperative cognitive skills programs for older adult patients.[60]**

[Conditional Recommendation]

High-quality evidence[60] supports postoperative cognitive skills training to improve older adult patients' mental and physical function and quality of life, and moderate-quality evidence[88] supports that patients sharing their experiences can increase coping postoperatively.

Increased coping skills may benefit all perioperative patients, especially those who undergo major procedures. Grouping patients in support groups according to procedure, age, and sex can increase participation and comradery among those who participate.[60,88] The potential barriers to implementing support groups include the challenge of scheduling multiple small groups, barriers to meeting in person (eg, geographic location) and limitations of online platforms, and challenges with buy-in resulting in the need for the support group to be participant-led and health care organization facilitated.

In an RCT pilot conducted by Kulason et al,[60] 12 older adult postoperative patients who received SCRA cognitive skills training had significantly improved motor function and quality of life scores. The researchers concluded that even with the study limitations (ie, performance bias and subjective motivation with a small sample and no placebo group), preoperative cognitive skills training can improve the cognitive and emotional state (ie, distress) of postoperative older adult patients.

In a nonrandomized case controlled trial, 34 of 67 women undergoing radiotherapy were included in a postoperative support group.[88] Emilsson et al[88] found a significant improvement in measures of coping; however, anxiety and depression measures were not significantly different between the groups. The researchers concluded that sharing the mutual experiences increased levels of coping resources, though future research of the long-term effects is needed for better understanding of the benefits.

4.5 **Information may be provided to the patient about how to prepare for operative and other invasive procedures, including**

- **strength exercises to improve quality of life,**[73]
- **walking to improve cognitive recovery (eg, social function, emotional role**[89]**) and functional recovery (eg, increase walking capacity**[89,90]**), and**
- **a combination of activities (eg, walking, diet, relaxation skills).**[91]

[Conditional Recommendation]

High-quality evidence[89-91] and guidelines[15] support the use of prehabilitation interventions to improve postoperative outcomes. Prehabilitation is mental or physical conditioning that the patient commits to completing before a scheduled procedure that is intended to improve outcomes during the postoperative recovery phase.

In an RCT, Gillis et al[91] assigned patients to receive either a combination of prehabilitation interventions (eg, exercise, diet, relaxation techniques) (n = 38) or standard postoperative rehabilitation (n = 39). The researchers found that at postoperative week 4, 50% of both groups were still below their preoperative functional exercise capacity baseline. After 8 weeks, the prehabilitation participants had significantly higher function on a 6-minute walking test compared to the rehabilitation group, who remained below their baseline. Though there were no differences in complications or LOS, the researchers found that there were clinically meaningful benefits to prehabilitation interventions.

In an RCT, 35 patients undergoing liver resection procedures participated in 4 weeks of high intensity cycle interval exercise before the scheduled procedure.[73] Dunne et al[73] found that the patients experienced preoperative improvements in oxygen and peak exercise levels and significant improvement of quality of life and overall mental health in the postoperative setting. The researchers concluded that prehabilitation interventions that include high-intensity exercise on the cycle improved the postoperative patient experience and that the additional time spent with patients and education provided to patients during prehabilitation interventions was also beneficial.

In an RCT conducted by Carli et al,[90] 112 patients received a structured exercise prehabilitation intervention (eg, strengthening [n = 58] compared with walking exercises [n = 54]) before undergoing a colorectal procedure. There was no difference in mean functional walking capacity between the groups before the prehabilitation intervention or at the first postoperative follow-up, however the walking capacity of the walking group was significantly improved compared to the strengthening group. The researchers concluded there was an unexpected benefit for patients in the walking intervention group though there was no baseline or control for at-home adherence. The researchers recommended assessing prehabilitation responders' characteristics to determine which patients would benefit most from the intervention.

In a quasi-experimental pilot study, Li et al[89] compared the outcomes of 42 patients who were assigned prehabilitation walking exercises with outcomes from a 45-patient cohort before prehabilitation was implemented. The researchers found significant improvement in walking in postoperative weeks 4 and 8 and recovery at week 8 for the prehabilitation group compared with the standard care cohort (81% versus 40%). The intervention group had significantly higher scores for social function, emotional role, and mental wellness. The researchers concluded that prehabilitation improved postoperative recovery.

4.6 **Provide the patient with information on the use of postoperative cold temperature interventions (eg, cryotherapy,**[92-94] **ice packs,**[95] **cold application plus aromatherapy**[45]**).** *[Conditional Recommendation]*

Some examples of postoperative cold temperature interventions include

- cryotherapy to decrease pain,[93,94,96] improve function,[93,96] improve temperature,[92,94] and decrease blood loss[93];
- ice packs to decrease pain and decrease opioid use[95]; or
- a combination (eg, cold application and lavender oil) to decrease pain and anxiety.[45]

High-quality evidence supports using cold therapy in the postoperative period,[45,92-96] although the authors of one systematic review found conflicting evidence of the benefits of cryotherapy interventions compared with the cost.[93] High-quality evidence also supports the use of cold therapy combined with other interventions (eg, lavender oil).[45]

In an RCT, Quinlan et al[95] compared targeted cold therapy of 1 L of ice to the lower back for 20 minutes (n = 74) with side-lying alone after spinal fusion procedures (n = 74). The researchers found both interventions decreased patients' pain, though there was a significant decrease in morphine-equivalent consumption and cumulative patient-controlled analgesia consumption from time point 7 to 12 hours in the group that received cold therapy. The researchers concluded that cold therapy is a beneficial adjunct to optimize pain management and reduce opioid consumption.

In a quasi-experimental study, Murata et al[92] compared 36 patients who received a cryotherapy icing system after undergoing spinal procedures (n = 16) to a control group (n = 20). The researchers found a significantly lower temperature of the wound and a positive correlation between the depth and temperature (ie, the deeper the wound the colder the temperature); however, cryotherapy had no effect on pain scores or bleeding outcomes. The researchers concluded that the benefits of cryotherapy are limited to decreasing the wound temperature.

Adie et al[93] performed a Cochrane review that included 12 trials (N = 809) in which cryotherapy was provided for 48 hours after total knee arthroplasty (TKA) procedures compared with standard care, the researchers found that cryotherapy had a significant but small benefit to reducing blood loss, decreasing pain scores at postoperative hour 48, and improved range of motion (ROM) at discharge. The researchers concluded that the results were not clinically significant, and the potential benefits could not justify the inconvenience and expense of the intervention.

In an RCT of 66 patients who underwent TKA procedures, Demoulin et al[94] investigated the effects of gaseous cryotherapy (n = 22) compared with standard cryotherapy (n = 22) or cold packs (n = 22). Gaseous cryotherapy significantly decreased wound temperature immediately in the postoperative setting and on POD 1. All groups reported an increase in pain immediately after the procedure, but pain scores in the standard cryotherapy group did not significantly increase. The researchers concluded that gaseous cryotherapy technology is not more beneficial than standard cryotherapy after TKA procedures.

In an RCT of 80 patients undergoing chest tube removal days after a coronary artery bypass grafting procedure, Hasanzadeh et al[45] compared three interventions, including cold application and lavender oil combined, cold application only, lavender oil only, and a control group (n = 20 in each group). The researchers found a significantly lower pain intensity and anxiety report for 15 minutes after chest tube removal in both groups in which cold interventions were applied. There was a significant decrease in anxiety in the combined intervention group compared to the group that received cold application alone, though the difference in pain scores was not significant between these groups.

5. Dietary and Herbal Interventions

5.1 Provide information to the patient about dietary interventions that may affect perioperative outcomes, including

- **the value of consulting with specialized personnel (eg, a registered dietitian, an herbologist) in the planning process[15];**
- **the benefits of following recommendations provided by specialized personnel[15] before and after an operative or other invasive procedure; and**
- **the benefits and risks (eg, potential for adverse interactions, bleeding risk) associated with dietary and herbal supplementation.[15,17,18,29,35,49,97-101]**

[Conditional Recommendation]

Guidelines support optimizing physical health during the perioperative period, including assessing nutritional status and using dietary and herbal interventions and glycemic control.[15] Several dietary and herbal interventions have been associated with improved perioperative outcomes that may outweigh the associated risks. Patients who are knowledgeable about risks and benefits can better discuss their current practices or potential implementation with their providers.

Fish Oil Supplementation

High-quality evidence supports preoperative and postoperative fish oil supplementation to improve postoperative outcomes[49,98,99] (eg, infection rate,[99] length of hospital stay [LOHS] and intensive care unit [ICU] stay[99]) without an associated increased risk of adverse events. However, the benefits were only studied in specific procedures (eg, abdominal,[99] esophagogastric[49]) and generalizability may be limited.

In a systematic review of 52 studies, Begtrup et al[98] found that fish oil supplementation reduced platelet aggregation but did not increase the risk for intraoperative bleeding or the postoperative blood transfusion rate. The researchers concluded that fish oil supplementation does not need to be discontinued before surgery.

In a meta-analysis that included 13 RCTs (N = 8,992), Chen et al[99] evaluated the effects of fish oil supplementation on the outcomes of adult patients who underwent abdominal procedures. The researchers found fish oil supplementation had a positive effect on LOHS, ICU LOS, and postoperative infection rates and no effect on mortality rates. The researchers concluded fish oil supplementation was a safe and effective way to improve clinical outcomes.

In an RCT, 195 patients received Omega-3 fatty acid enteral supplementation to improve their immunity compared with standard enteral nutrition before and after undergoing esophagogastric procedures.[49] Sultan et al[49] found a significantly higher Omega-3 concentration (ie, ratio of Omega 6 to Omega 3) in the intervention group. There was a significant increase in protein and energy consumption in the intervention group, as the volume needed to achieve target caloric and protein levels was lower and easier for patients to consume. Though there were no clinical or significant improvements in infection complications, ICU LOS, LOHS, morbidity, or mortality rate, the researchers concluded that patients who are malnourished or unable to consume enteral nutrition should supplement their dietary intake to optimize their postoperative healing.

Symbiotics

High-quality evidence supports that use of symbiotics[100] and probiotics[100,101] may decrease the patient's risk for developing sepsis. In a meta-analysis of 15 RCTs that included 1,201 patients undergoing GI procedures, the researchers found that probiotics and symbiotics significantly decreased the risk of sepsis.[101] In a meta-analysis conducted of 13 RCTs (including nine trials that were part of the previous meta-analysis[91]) (N = 962), symbiotics were found to be more beneficial than probiotics in reducing sepsis incidence.[100]

Vitamin D and Calcium

High-quality evidence[17,18] used to support clinical guidelines for orthopedic surgeons[35] indicates that postoperative vitamin D and calcium supplementation may decrease postoperative fracture complications in older adult nonoperative patients.

In an RCT, adult patients older than 65 years with hip fractures received extended physical therapy (n = 87) or standard physical therapy (n = 86) combined with either 2,000 IU (n = 86) or 800 IU of cholecalciferol per day (n = 87).[17] Bischoff-Ferrari et al[17] found a 25% reduction in falls and 39% reduction in readmissions in the groups that consumed 2,000 IU. The risk of facility readmission was lower for patients who received 2,000 IU of cholecalciferol.

In an RCT, Chapuy et al[18] provided either 1.2 g elemental calcium and 800 IU of vitamin D3 (n = 1,634) or placebos (n = 1,636) to female patients ages 69 to 106 years who were living in nursing homes. The researchers found that calcium and vitamin D3 supplementation lowered the incidence of nonvertebral and hip fractures.

Bleeding Risk

Low-quality evidence supports evaluating the patient's use of complementary practices and over-the-counter supplements for potential interactions with the surgical and anesthetic plans of care.[29,97] In a systematic review, Wang et al[30] evaluated the effects of dietary and herbal interventions on coagulation and bleeding among patients undergoing operative and other invasive procedures. The researchers found 11 common herbal medications (ie, echinacea, ephedra, garlic, ginger, ginkgo, ginseng, green tea, kava, saw palmetto, St John's wort, and valerian) and four supplements (ie, coenzyme Q10, glucosamine and chondroitin sulfate, fish oil, and vitamins) that may increase the risk for bleeding and cardiovascular instability.

In a cross-sectional cohort study, Levy et al[29] evaluated the effects of dietary supplements on general anesthesia and bleeding in surgery. The researchers found that patients who used sage as a dietary supplement had increased sedative toxicity and decreased blood level of anesthesia. The researchers also found that use of other dietary supplements had additional effects related to general anesthesia: chamomile prolonged sedation, green tea reduced anesthesia hypnotic effects, and lemon balm increased sedation. Several dietary supplements were identified as having effects on patient coagulation potential, including additive antiplatelets from Omega-3 fish oil, green tea, magnesium, rosemary, and flaxseed. Other platelet inhibitors were chamomile, ginger, and sage.

In a systematic review of nine trials including two observational studies, Marx et al[97] found that oral ginger reduced platelet aggregation in four of the studies. In these studies, varied delivery methods (eg, capsules, raw, powder, dried) and doses of ginger (eg, 10 g, 5 g, 625 mg four times a day, 1 g) or combinations with other medications (eg, nifedipine) decreased the ability of researchers to conclude whether one form of oral ginger had higher adverse associations with platelet aggregation. Evidence further conflicts as the remaining studies found no association with adverse events for varied routes (eg, capsules, raw, capsules with warfarin) and doses (eg, 4 g, 15 g, 3.6 g, 500 mg four times).

5.2 Dietary and herbal interventions may be implemented for prevention of postoperative ileus complications. *[Conditional Recommendation]*

Moderate-quality evidence,[102-104] and guidelines[15] support using dietary and herbal interventions to prevent postoperative ileus complications. Dietary and herbal supplement interventions that may prevent or reduce the incidence and severity of postoperative ileus complications include gum chewing (eg, LOHS, time to flatus, time to bowel movement)[102] and Dai-kenchu-to, a Japanese herbal medicine (eg, time to flatus,[103] health care organization costs, LOHS[104]).

Short et al[102] performed a Cochrane review of 81 RCTs (N = 9,072) in which patients received chewing gum to promote postoperative GI function versus a control group. Though the results were not statistically significant, the researchers found that gum chewing decreased time to flatus, slightly decreased the LOHS and time to bowel sounds, and improved postoperative healing after colorectal procedures.

In a quasi-experimental study patients received either 7.5 g of Dai-kenchu-to for 7 postoperative days (n = 15) or standard care (n = 15).[103] Yoshikawa et al[103] found that patients in the intervention group had a significantly shorter time to flatus and lower C-reactive protein on POD 3.

In a retrospective matched cohort study, Yasunaga et al[104] administered Dai-kenchu-to to 144 patients with adhesive small bowel obstruction. The researchers found that compared to a control group (n = 144), the intervention group had fewer requirements for long-tube decompression insertion as well as time to remove the tube and discharge the patient, which significantly lowered facility charges. Though the results were not adjusted for patient characteristics, the researchers concluded that Dai-kenchu-to was an effective intervention to decrease long-tube decompression requirements and associated costs.

5.3 Postoperative interventions that can reduce thirst discomfort may be used in the postoperative phase of care.[105-107] *[Conditional Recommendation]*

Moderate-quality evidence supports using interventions that may reduce a patient's thirst discomfort during postoperative recovery,[105-107] including menthol chewing gum[107] and oral care using menthol in normal saline swabs. Water interventions may include aromatherapy water, cold water, and menthol water.

In an RCT, Garcia et al[107] investigated the effects of menthol chewing gum (n = 51) versus standard care (n = 51) on thirst discomfort in the postoperative recovery area. The researchers found a significant decrease in the intensity and discomfort of postoperative thirst in the group that received menthol gum and concluded that the intervention was effective and safe.

In a quasi-experimental study, 60 patients received a menthol in normal saline oral care swab compared to standard oral care. Al Sebaee and Elhadary[105] found a significant decrease in thirst intensity and improvement in saliva and tongue health. The researchers concluded that adding menthol to the water used for oral care significantly improved postoperative recovery as part of an oral care bundle for patients recovering from abdominal procedures.

In a systematic literature review of 10 studies including one observational study, Garcia et al[106] found that frozen gauze, ice chips, cold water, menthol, chewing gum, acupressure, sipping liquids through a thin straw, and early oral fluids decreased thirst discomfort. Though they concluded that combining strategies can increase patient satisfaction, the researchers found that decreasing the temperature of the water was the most predominant and effective intervention to decrease thirst discomfort.

5.4 Oral ginger may be used for postoperative nausea.[1,19,21,108,109] *[Conditional Recommendation]*

High-quality evidence[19,108,109] and guidelines[1,21] support that ginger decreases postoperative nausea,[19,108,109] but conflict on differences in vomiting incidence[19,109] and administration (ie, dose, timing, and route).[19,108]

Tóth et al[108] performed a meta-analysis of 10 RCTs (N = 918) in which patients were given 100 mg to 1,500 mg of oral ginger. They found a significant decrease in nausea and vomiting severity, though PONV incidence and use of rescue medications were not statistically improved. The researchers concluded that underdosing may have contributed to nonsignificant results and that oral ginger is a safe and well-tolerated method for reducing PONV severity.

In an RCT of 100 patients undergoing cholelithiasis who received oral ginger, Soltani et al[109] found nausea severity was significantly lower at postoperative hours zero, 2, and 4, but the incidence of vomiting was only significantly lower at hour 16. The researchers concluded that oral ginger given 1 hour before surgery is an inexpensive, effective, and safe method to decrease PONV.

In another RCT, 122 patients received either one dose of 1 g or two doses of 500 mg of oral ginger or a placebo before undergoing cataract procedures under general anesthesia. Seidi et al[19] found significantly lower scores for nausea and vomiting with two smaller doses compared to one large dose or the placebo. The researchers concluded that oral ginger is a safe and effective intervention to reduce PONV, and two separate, smaller doses may be more beneficial than one larger dose. They recommended this intervention for older adult patients, for whom the risk of adverse events with antiemetic drug consumption is higher.

5.4.1 **Use oral ginger with caution in patients who are at risk for bleeding.**[97] *[Recommendation]*

Moderate-quality evidence found conflicting results on the potential adverse events (eg, bleeding risk associated with increased platelet aggregation) related to oral ginger use, depending on the dose.[97] Additional research is required to determine the optimal dose of oral ginger to avoid increasing platelet aggregation.

In a systematic review of nine trials including two observational studies, Marx et al[97] found oral ginger reduced platelet aggregation in four of the studies. In these studies, varied delivery methods (eg, capsules, raw, powder, dried) and doses of ginger (eg, 10 g, 5 g, 625 mg four times a day, 1 g) or combinations with other medications (eg, nifedipine) decreased the ability of researchers to conclude whether one form of oral ginger had higher adverse associations with platelet aggregation. Evidence further conflicts as the remaining studies found no association with adverse events for varied routes (eg, capsules, raw, capsules with warfarin) and doses (eg, 4 g, 15 g, 3.6 g, 500 mg four times).

5.5 **Do not give patients essential oils for oral ingestion.**[110] *[Regulatory Requirement]*

The FDA classifies essential oils as Generally Recognized as Safe as a regulated indirect additive for flavor, and they are not recognized as safe for any other indication.[110]

6. Music Interventions

6.1 **Music interventions may be used in the perioperative environment.**[66] *[Conditional Recommendation]*

In a high-quality systematic review of 73 RCTs conducted from 1988 to 2013, Hole et al[66] evaluated research on the benefits of perioperative music interventions for postoperative pain, anxiety, use of analgesia, and patient satisfaction. The researchers found improvements in all of these outcomes and concluded that perioperative music interventions were beneficial.

6.1.1 **When music interventions will be used, determine**

- **whether the music will be patient selected**[66,111,112] **or assigned,**[48,113]
- **what types of music will be used,**[27,48,55,66,111-117] **and**
- **what delivery mode will be used.**[113,114]

[Conditional Recommendation] P

High-quality evidence conflicts regarding using patient-selected music in the perioperative setting.[27,48,55,66,111-117] Researchers have found that patients desired to choose music based on their personal preferences[111,112] and, conversely, that a high percentage of patients requested to use again the same music genre that was assigned to them previously.[113] Other researchers found inconclusive evidence on whether music or music volume jeopardizes patient safety in the OR.[66]

Potential harms of music selection or volume can include negatively affecting the patient's response to anesthesia induction and emergence and volume at a level that introduces noise into the intraoperative environment and interrupts perioperative team communication.[66] Limitations to selecting a music intervention include the cost-effectiveness of using specific types of music (ie, **interactive music** interventions that require additional education and training)[27]; however, none of the researchers included a cost analysis in their studies.

In a systematic review of 73 RCTs, Hole et al[66] found a reduction in postoperative pain, anxiety, and use of analgesia and an increase in patient satisfaction with the use of music interventions. However, the researchers concluded that allowing patients to choose the music and suitable volume were unclear limitations. It was also unclear whether the use of headphones created a risk of distracting patients from communicating with perioperative health care workers or a benefit of alleviating imposition of the music on the health care workers.

In an RCT, Palmer et al[114] assigned 207 female patients undergoing breast procedures to receive either patient-selected live music (eg, in-person guitar or keyboard) (n = 69), recorded music (n = 70), or usual preoperative area noises as a control in the preoperative setting (n = 68). In the OR, both music groups received therapist-selected intraoperative music and the control group received earmuffs. The researchers found a significant decrease in anxiety scores in the music groups, though the preoperative live music intervention resulted in a shorter recovery time, as indicated by discharge readiness. The researchers concluded that live music was more beneficial than recorded music; and preoperative music was safe and time efficacious to reduce anxiety.

In an RCT of 180 pediatric patients ages 3 to 15 years undergoing dental procedures, Ozkalayci et al[55] assigned participants to one of three groups: sound-isolating headphones with music, isolation headphones without music, or standard care (n = 60 in each group). The researchers found that there was a significantly prolonged recovery (ie, time to Aldrete discharge parameters) in the groups that did not receive a music intervention and concluded that use of a music intervention could decrease recovery time.

In an RCT conducted by Çetinkaya[48] in Turkey of 60 older adult patients undergoing total joint procedures, Turkish music was played for 20 minutes three times a day on the first, second, and third postoperative days. Cognitive scores significantly improved on POD 3 compared with POD 1. The researcher concluded that music prevented cognitive dysfunction, though the study sample was limited to patients in one facility and only one type of music was used.

In an RCT, Ko et al[115] found that patients undergoing colonoscopy without sedation (n = 138) had a significant decrease in anxiety when **Kern light music** was played intraoperatively. The researchers concluded it was a safe, convenient, and effective intervention.

In an RCT, Forooghy et al[112] compared patients who listened to 20 to 40 minutes of light instrumental music intraoperatively after local anesthesia was injected during percutaneous transluminal coronary angioplasty (n = 32) to a control group (n = 32). The researchers found a significant decrease in anxiety scores and a significant decrease in blood pressure immediately after the intervention and up to 30 minutes after the procedure and intervention were completed. Limitations included environmental noise and potential confounders related to patient's psychological or personality characteristics, attitudes, and beliefs. The researchers concluded that additional research is needed to determine the benefit of intraoperative light instrumental music.

In a pilot study of 34 patients undergoing colonoscopy procedures who received either muted headphones (n = 17) or listened to a loop of Bach music on headphones (n = 17) before and during their procedures, Martindale et al[113] found that anxiety scores decreased in both groups. However, all of the patients in the music group reported a desire to receive music for repeat procedures compared to 50% in the control group who said they would wear muted headphones again. Of interest, 64.7% of patients in the music group reported that they did not want to choose their own music, and the researchers concluded that any music genre may be beneficial to reduce anxiety.

In an RCT conducted with 98 patients, Weeks and Nilsson[116] found a significant postintervention reduction in anxiety and increase in well-being with use of a music intervention, regardless of whether music was provided by an audio pillow or loudspeaker during coronary angiographic procedures.

In an RCT conducted by Padmanabhan et al,[117] 100 patients in three groups preoperatively received either **binaural beat** music (n = 35), music without binaural tones (n = 34), or no intervention as a control group (n = 35). The researchers found a significant decrease in anxiety for both music groups and a significantly lower increase in anxiety in the binaural tones group compared to the control group. The researchers concluded that the potential to decrease acute preoperative anxiety warranted use of music interventions.

In a quasi-experimental study conducted by Hsu et al,[111] 49 older adults who had undergone TKA procedures completed 25 minutes of postoperative **continuous passive motion** (CPM) rehabilitation in the facility ward. Each patient performed CPM both with and without music during the course of two postoperative days. Patients reported significantly decreased pain and increased knee flexion angle when they listened to music during CPM rehabilitation. The researchers concluded that playing patient-selected music and decreasing environmental noise could improve the effectiveness of CPM rehabilitation.

6.2 **Establish a process for the care and cleaning of reusable music delivery devices.** *[Recommendation]*

6.3 **Music intervention techniques may be used in all perioperative phases of care for anxiety and pain reduction for pediatric patients.**[27,118] *[Conditional Recommendation]* P

High-quality evidence supports using music during all perioperative phases of care to decrease anxiety and pain in pediatric patients.[27,118]

In a systematic review and meta-analysis of 19 RCTs that included 1,513 pediatric patients ages 1 month to 18 years, Klassen et al[118] found a significant decrease in pain and anxiety when music was implemented in the perioperative setting. The researchers concluded that music was an effective adjunct to reduce pharmacologic requirements, pain, and anxiety.

In an RCT conducted to investigate an interactive music therapy intervention in pediatric patients ages 3 to 7 years, Kain et al[27] found that patients who received midazolam (n = 34) had significantly lower anxiety than patients in the music intervention group (n = 51) or control group (n = 38) on induction. The midazolam group had significantly lower anxiety even after the researchers controlled for the results for one music therapist whose group had significantly lower anxiety during preoperative parental separation and entrance to the OR. The researchers concluded that interactive music therapy may be helpful depending on the therapist, but not during anesthesia induction, and therefore, the intervention was not cost-effective.

6.4 Preoperative music interventions may be used for adult patients.[27,113,114,117,119] ***[Conditional Recommendation]***

High-quality evidence supports the use of preoperative music interventions to improve vital signs[119] and decrease anxiety[113,114,117,119] and the length of time in the postanesthesia care unit (PACU).[114] Limitations of music interventions include the cost for a music therapist[114] and logistics for interactive music interventions.[27]

In a systematic review of 26 trials (N = 2,051 patients) that included quasi-experimental studies, Bradt et al[119] found that pre-recorded music in the preoperative setting decreased anxiety scores and had a small effect on patients' heart rate and diastolic blood pressure but not on systolic blood pressure, respiratory rate, or skin temperature. The researchers concluded that these results were consistent with the results of other Cochrane reviews that found preoperative music may decrease patient anxiety.

In a pilot study, Martindale et al[113] assigned 34 patients undergoing colonoscopy procedures to receive either muted headphones (n = 17) or to listen to a loop of Bach music (n = 17) before and during their procedures. During the study, anxiety scores decreased, and 100% of the patients in the music group reported a desire to receive music for repeat procedures compared to 50% in the control group that would wear muted headphones again. Limitations include the potential confounding factor of patient-controlled volume.

In an RCT conducted by Padmanabhan et al,[117] 104 patients received either binaural beat audios (n = 35), music without binaural tones (n = 34), or no intervention as a control group (n = 35). The researchers found a significant decrease in anxiety for both music groups and a significantly lower increase in anxiety in the group with the binaural beat audios. The researchers concluded that the potential to decrease acute preoperative anxiety warrants use of binaural tones.

In an RCT, Palmer et al[114] assigned 207 female patients undergoing breast procedures to listen to either patient-selected live music (ie, in-person guitar or keyboard) (n = 69), recorded music (n = 70), or usual preoperative area noises as a control in the preoperative setting (n = 68). The researchers found a significant decrease in anxiety with live music, resulting in a shorter recovery as indicated by discharge readiness. The researchers concluded that live music is safe and time efficacious, though determining the ideal timing of the intervention requires additional research.

6.5 Intraoperative music interventions may be used.[66,112,113,115,120-122] ***[Conditional Recommendation]***

High-quality evidence supports that intraoperative music interventions may

- decrease analgesia use,[66,120]
- decrease anxiety,[66,112,113,115,120,121]
- decrease adrenaline and noradrenaline levels,[112,120,121]
- improve subjective coping,[122]
- improve vital signs,[112,120,121]
- increase patient satisfaction,[66,113,122] and
- decrease subjective stress.[121]

In a systematic review of 73 RCTs conducted between 1988 to 2013, Hole et al[66] found that use of a music intervention reduced postoperative pain, anxiety, and analgesia use and increased patient satisfaction. The researchers concluded that music interventions were effective in improving the studied outcomes and recommended the use of music in the perioperative setting.

In an RCT, Bashiri et al[120] evaluated music and anesthesia type for effects on patient anxiety scores. Group 1 received conscious sedation without music (n = 25), group 2 received conscious sedation with music (n = 33), group 3 received deep sedation without music (n = 55), and group 4 received deep sedation with music (n = 41). In each group that had a music intervention, the researchers found a decrease in anxiety and propofol use, increased patient satisfaction, and a desire to listen to music during future procedures. Heart rate and mean arterial pressure were also significantly lower during and after the procedure in the music groups. The researchers concluded that music interventions improved the comfort of patients undergoing procedures without general anesthesia.

In an RCT, Jiménez-Jiménez et al[121] provided either classical music (n = 20) or standard care (n = 20) to patients undergoing great saphenous vein stripping procedures. The researchers found significantly lower anxiety scores, lower serum adrenaline and noradrenaline levels, and improved vital signs in the group that listened to classical music. The researchers concluded that listening to the selected music reduced intraoperative anxiety.

In an RCT that included 26 women who underwent elective surgical abortions with either self-selected music and local anesthesia (n = 11) or local anesthesia alone (n = 13), Wu et al[122] found significantly better postoperative coping and less anxiety immediately after the procedure in the group that listened to music. The researchers concluded that music interventions as an adjunct to local anesthesia are associated with less postprocedure anxiety.

In a pilot study, Martindale et al[113] assigned 34 patients undergoing colonoscopy to either receive muted headphones (n = 17) or listen to a loop of Bach music (n = 17) before and during their procedures. Anxiety scores were reduced in both groups, and

100% of the patients in the music group reported a desire to receive music for repeat procedures compared to 50% in the control group who said they would wear muted headphones again. The researchers concluded that even though the music did not significantly reduce anxiety, pain scores, or medication doses (ie, midazolam, fentanyl), the patients expressed a clear preference to listen to music during future procedures.

In an RCT, Ko et al[115] found that 138 patients undergoing colonoscopy procedures without sedation had significantly decreased anxiety scores after listening to Kern light music. The researchers concluded it is a safe, convenient, and effective intervention.

In an RCT, Forooghy et al[112] compared patients who listened to 20 to 40 minutes of light instrumental music intraoperatively after receiving local anesthesia at the start of their percutaneous transluminal coronary angioplasty procedure (n = 32) to a control group (n = 32). The researchers found a significant decrease in postprocedure anxiety scores in the music group compared with the standard care group and concluded that music therapy is a safe, simple, inexpensive, and noninvasive nursing intervention that can significantly alleviate patient anxiety during coronary angioplasty.

6.6 **Postoperative music interventions may be used.**[48,66,111,123] *[Conditional Recommendation]*

High-quality evidence supports that using music therapy postoperatively for older adult patients may decrease their pain and cognitive decline.[48,66,123] Moderate-quality evidence supports postoperative use of music to decrease objective and subjective pain and anxiety outcomes and increase patient satisfaction.[111] Evidence conflicts on the ability of music to reduce analgesic consumption.[66,123]

Postoperative music interventions may

- decrease anxiety,[66]
- improve cognitive function,[48]
- improve physical function,[111]
- decrease pain,[66,111] and
- improve coping and stress.[123]

In a systematic review of 73 RCTs, Hole et al[66] found a reduction in postoperative pain, anxiety, and use of analgesia and an increase in patient satisfaction with the use of music interventions. The researchers concluded that allowing patients to choose the music and suitable volume were unclear limitations. It was also unclear whether the use of headphones created a risk of distracting patients from communicating with perioperative health care workers or a benefit of alleviating imposition of the music on the health care workers.

In an RCT of 163 patients, Easter et al[123] studied the effects of a music intervention on respiratory rate, oxygen saturation, and opioid consumption in the PACU after multiple outpatient procedures (eg, gastroenterology, ophthalmology, general, gynecology, neurology, oral, orthopedic, urology). The researchers found significantly lower respiratory rates with higher satisfaction in the music group (n = 111) compared to the control group (n = 102). There were no significant differences between the groups in oxygen saturation or opioid consumption in the PACU setting. The researchers concluded that music could be an alternative method to cope with postoperative pain.

In an RCT conducted by Çetinkaya[48] of 60 older adult patients undergoing total joint replacement procedures, Turkish music was played for 20 minutes three times a day on the first, second, and third postoperative days. Cognitive scores significantly improved on POD 3 compared with POD 1. The researcher concluded that music prevented cognitive dysfunction, though the study sample was limited to patients in one facility and only one type of music was used.

In a quasi-experimental study conducted by Hsu et al,[111] 49 older adults who had undergone TKA procedures completed 25 minutes of postoperative CPM rehabilitation in the facility ward. Each patient performed CPM both with and without music during the course of two postoperative days. Patients reported significantly decreased pain and increased knee flexion angle when they listened to music during CPM rehabilitation. The researchers concluded that playing patient-selected music and decreasing environmental noise could improve the effectiveness of CPM rehabilitation.

7. Stress Diversion Interventions

7.1 **Select and implement perioperative stress diversion interventions based on the patient's age.**[20,53,54,56,71,124-140] *[Conditional Recommendation]* P

Types of stress diversion include

- preoperative humor,[53,56,132] play,[124] video,[71,125] and virtual reality (VR)[126-130] interventions;
- intraoperative distraction and a low stimulus environment[133,134];
- parental presence during emergence from anesthesia[135-137]; and
- postoperative patient visitation for both adult[139,140] and pediatric patients.[54,138]

High-quality evidence supports that preoperative stress diversion interventions can reduce anxiety,[53,56,71,124-131] including reducing the anxiety of the parents of pediatric patients.[20]

In a quasi-experimental study that included 83 pediatric patients ages 9 to 18 years, Aytekin et al[20] found that participating in distractions tailored to their age group (eg, games, music, cartoons, books) significantly decreased patients' anxiety and separation anxiety compared to patients in a control group. The researchers concluded that patients may benefit from different distraction methods geared toward developmental age.

7.2 Preoperative humorous interventions may be used to decrease preoperative anxiety in pediatric patients.[53,56,132] *[Conditional Recommendation]* P

High-quality evidence supports using humor (eg, costumes, toys, clowns, comical informational materials) to decrease preoperative anxiety in pediatric patients and their mothers.[53,56,132]

In an RCT, Kassai et al[56] gave a comic information leaflet to pediatric patients ages 6 to 17 years before surgery (n = 55). Anxiety scores were significantly decreased in the intervention group compared to a group that received standard care (n = 55). The researchers also found that those with a higher baseline anxiety scores benefited more from the intervention.

In an RCT, Liguori et al[53] provided pediatric patients ages 6 to 11 years with either a 6-minute video (ie, clown physicians giving a comical and informative tour of the OR) the night before the procedure (n = 20) or standard care (n = 20). Children who viewed the video, which was provided via an app, had significantly lower preoperative anxiety scores and significantly lower changes in anxiety before induction than those who did not view the video. The researchers concluded that the app could be used as a substitute when there is not enough time for health care workers to provide education and to decrease costs associated with staffing.

In a quasi-experimental study conducted by Berger et al,[132] 42 pediatric patients ages 4 to 17 years were distracted through humor, costumes, and toys while playing with their mothers. The researchers found significantly lower anxiety scores for the children on admission to the perioperative suite and significantly lower anxiety just before the procedure in both the children and their parents. The researchers concluded that costume and humor distraction was effective and improved the family perioperative experience.

7.2.1 Do not use clown humor interventions if the patient has an aversion to clowns.[132] *[Recommendation]*

If the patient has an aversion to clowns, using this type of humor as a stress diversion intervention may negatively affect the patient's perioperative experience and may increase anxiety instead.[132]

7.2.2 Preoperative play interventions may be used to decrease preoperative anxiety in pediatric patients.[124] *[Conditional Recommendation]* P

Moderate-quality evidence supports distracting pediatric patients preoperatively by providing a separate area dedicated to play.[124]

In an RCT, Hosseinpour and Memarzadeh[124] found that providing access to a preoperative playroom 30 minutes before a surgical procedure significantly decreased the anxiety scores for pediatric patients ages 2 to 6 years, compared with standard care.

7.3 Audio-video interventions may be used for preoperative anxiety reduction.[71,125] *[Conditional Recommendation]* P

High-quality evidence[71,125] supports using videos to distract pediatric patients' and to reduce preoperative anxiety during high-anxiety preoperative periods, such as during transport to the OR[125] and during induction.[71,125] The benefits for pediatric patients could apply to other age groups.

In an RCT, Kerimoglu et al[125] assigned 96 pediatric patients ages 4 to 9 years to one of three groups: video glasses for distraction for 20 minutes, midazolam only, or video glasses for distraction combined with midazolam (n = 32 in each group). The interventions were implemented before transport and during anesthetic gas induction of general anesthesia. There was a significant decrease in anxiety during transport in the midazolam with video glasses group and significant increase in anxiety in the midazolam-only group, though the researchers did not find this to be clinically significant. The researchers concluded that the use of video glasses alone was not inferior to use of midazolam alone for decreasing preoperative anxiety. Limitations of the study include potential confounders that were not measured, such as **emergence delirium** and behavioral dysfunction.

In an RCT conducted by Mifflin et al[71] of 89 pediatric patients ages 2 to 10 years, patients streamed a video during induction (n = 42) or participated in traditional distraction methods (n = 47) including nonprocedural talk and game playing. The researchers found significantly lower anxiety scores and a significantly smaller increase in anxiety from the holding area to anesthesia induction time for those patients who watched the video. The researchers concluded that video distraction is useful, though future research should focus on confounding factors such as premedication with sedatives and parental presence during induction, as well as longer term effects of stress diversions used in recovery areas.

7.4 **Virtual reality (eg, OR experience, gamification) audio-visual interventions may be used for preoperative anxiety reduction.**[126-130] *[Conditional Recommendation]* P

High-quality evidence supports that a short VR tour of the OR can decrease preoperative anxiety in both adult[126] and pediatric populations (ie, between 4 and 12 years of age).[126-130] Virtual reality gamification also significantly reduced patients' anxiety and improved induction compliance in one study; however, the cost of gamification (ie, adding gaming elements to non-game contexts) within VR interventions may not outweigh the benefits compared with general VR interventions.[129]

Evidence supports that VR interventions may benefit perioperative outcomes including decreased anxiety,[126-130] better induction compliance,[129,130] and increased patient satisfaction.[126] Evidence conflicts about effects on patient distress[126,127] and parental satisfaction.[127,130]

In an RCT, Bekelis et al[126] provided patients undergoing cranial and spinal procedures with a VR experience (n = 64) compared to a group that received standard care (n = 63). Patients were encouraged to watch the video as many times as desired and were given time for and encouraged to ask questions. After the intervention, the patients in the intervention group reported feeling more prepared and had higher satisfaction scores and lower stress even though they had a higher level of preoperative anxiety than the control group. The researchers concluded that an immersive OR environment experience with VR could minimize patients' preoperative stress.

In an RCT, Park et al[127] assigned 80 pediatric patients ages 4 to 10 years and their parents to one of two intervention groups. In the immersive mirroring intervention group (n = 40), patients watched a 4-minute VR tour via a headset while their parents watched the same 4-minute VR tour simultaneously on a separate monitor. In the second group, the patient watched the VR tour via headset, and the parents did not view the VR tour. Mirrored VR experiences significantly reduced the patients' preoperative anxiety and the parents' induction anxiety and increased parents' satisfaction. There was no difference in patient anxiety during anesthesia induction in either group, as measured by the **Induction Compliance Checklist** (ICC). The researchers concluded that having patients and their parents simultaneously watch an immersive VR tour was a feasible way to effectively reduce preoperative anxiety in pediatric patients and their parents.

In an RCT, Dehghan et al[128] assigned 40 pediatric patients ages 6 to 12 years to receive either 5 minutes of VR exposure to the OR or a control with or without a pretest (n = 10 in each group). The researchers found a significant decrease in anxiety among patients who received the VR intervention. The researchers concluded that distraction using VR decreased preoperative anxiety.

In an RCT of 69 pediatric patients ages 4 to 10 years, Ryu et al[130] found that a preoperative 4-minute VR tour of the OR (n = 35) versus standard care (n = 34) significantly lowered patients' preoperative anxiety scores and postoperative negative behaviors, as well as increasing their intraoperative ICC scores compared to the control group. The researchers concluded that VR could alleviate anxiety and increase induction compliance.

In another RCT conducted by Ryu et al[129] of 69 pediatric patients ages 4 to 10 years, the researchers found that 5 minutes of a VR gamification experience (n = 34) versus standard care (n = 35) did not have an effect on pediatric behaviors or parent satisfaction. However, preoperative anxiety and ICC scores were significantly decreased in participants in the intervention group, and the researchers concluded that the low cost and feasibility of a VR intervention can reduce pediatric anxiety.

7.5 **Stress diversion interventions may be used to reduce pain and distress in pediatric patients.**[133,134] *[Conditional Recommendation]* P

High-quality evidence supports the use of varied and multiple stress diversion interventions to reduce intraoperative pediatric pain and distress.[133,134] Types of stress diversion to reduce pain and distress include breathing exercises, cognitive behavioral therapy, distraction, information, hypnosis, memory alterations, positive suggestions,[133] and a low-stimulus environment.[134]

In an updated Cochrane review of pediatric pain and distress during needle procedures, Birnie et al[133] reviewed 20 new trials that included 5,550 pediatric patients and supported interventions such as distraction, cognitive behavioral therapy, hypnosis, increased information, breathing, positive suggestion, and memory alterations, as adjuncts to standard care.

In an RCT, Kain et al[134] assigned pediatric patients ages 2 to 7 years to receive either low sensory stimulus interventions including dim OR lights, soft background music, only one anesthesia professional, and no instrument noise from the sterile field (n = 33) or usual care (n = 37). The researchers found significantly lower anxiety scores for pediatric patients in the intervention group, lower anxiety scores for their parents, and significantly better cooperation with health care workers. The researchers concluded that manipulating the environment, including limiting loud noises, is a beneficial and low-cost intervention to reduce preoperative anxiety in pediatric patients.

7.6 **Parental presence during induction of general anesthesia for pediatric patients may be implemented, in collaboration with surgeons and anesthesia professionals, for anxiety reduction for both patients and their parents.**[135-137] *[Conditional Recommendation]* P

Moderate-quality evidence conflicts as to whether parental presence effectively reduces the parents' or pediatric patient's anxiety during anesthesia induction.[135-137] Educating perioperative team members and gaining their support for parental presence initiatives are critical for successful implementation. Without buy-in and understanding of the potential benefits, team members may be resistant or perceive the harms as outweighing the benefits. One potential roadblock may be the burden placed on the team to attend to the parent. Support for perioperative team members who facilitate parental presence may reduce the burden on the team and improve patient safety during critical phases of perioperative care. The team may also have concerns about the parent's attire because the parent's clothing may contaminate the perioperative area and increase the patient's risk for surgical site infection. Having the parent change into clean surgical attire may address this concern.

In a systematic review of RCTs of nonpharmacological interventions to assist during anesthesia induction in pediatric patients, Manyande et al[136] reviewed 28 trials (N = 2,681) and 17 interventions. They concluded that parental presence did not significantly reduce anxiety and additional studies are required to provide specific recommendations regarding other interventions, such as parental acupuncture, clown humor, playing videos of the patient's choice, low-sensory stimulation, and video game interventions.

In a quasi-experimental study, Chan and Molassiotis[137] provided 25 parents of pediatric patients ages 1 to 9 years with an education program, allowed them to be present during anesthesia induction, and allowed visitation with their children in the recovery area. The researchers found a significant decrease in parental anxiety and increase in patient satisfaction compared to parents in a control group (n = 25).

In 2014, Cagiran et al[135] surveyed 100 pediatric patients and their mothers and found that maternal anxiety was unrelated to the mother's level of education or socioeconomic status. However, the researchers found that maternal anxiety was reduced the most by increasing the mother's knowledge preoperatively (86%). Being present during transport and induction of anesthesia (37% and 46%, respectively) and speaking with other mothers (28%) were not as effective in lowering the mothers' preoperative anxiety. The researchers concluded that increasing the knowledge and experience with anesthesia processes reduced maternal anxiety.

7.7 **A procedure for patient visitation in the postanesthesia care unit may be implemented.**[54,138-140] *[Conditional Recommendation]* P

Health care organizations should weigh the benefits versus harms of including postoperative patient visitation when patient care and departmental conditions are suitable. Moderate-quality evidence supports family visits in the PACU to decrease patient anxiety and increase satisfaction for both the patient and family members.[54,138-140] Although guidelines support that family presence may benefit some patients, potential harms include an increased burden on health care workers and challenges with maintaining the privacy of other patients.

In an RCT, Burke et al[54] observed pediatric patients undergoing magnetic resonance imaging (MRI) under general anesthesia for **emergence agitation** in the recovery area with parental presence (n = 45) versus parental absence (n = 43). Parents who were present reported earlier anxiety reduction than those who were not present; however, there was no significant difference in the children's agitation with parental presence.

In a quasi-experimental study, In et al[138] found that pediatric patients ages 3 to 6 years whose parent could be present while the patient was emerging from anesthesia (n = 46) had a significant decrease in emergence delirium compared to a control group (n = 47). Though the results at each time point were not statistically significant, the researchers concluded that the parental PACU visitation program could be beneficial.

Lower-quality evidence also supports family visitation. In an observational study, Wendler et al[140] found a significant decrease in patient anxiety and increase in family satisfaction when 5- to 10-minute supervised family visits during phase I recovery were provided for patients who underwent total joint procedures with spinal anesthesia.

In an organizational experience article, Pagnard and Sarver[139] reported that after familial presence was implemented (ie, one adult was allowed 5 minutes on the phone or in the PACU for patients requiring an overnight stay), there was an increase in family satisfaction and decrease in familial stress. Pre-intervention, 70% of the 29 PACU RNs were concerned that family visitation would increase their work stress; however, 94% were positive about the intervention at the end of the trial, and the researchers concluded that the intervention was beneficial to the patient and families without increasing the strain on health care workers.

8. Visualization Interventions

8.1 Visualization interventions may be used in the perioperative environment.[26,141-146] *[Conditional Recommendation]*

Types of visualization interventions include hypnosis,[26,141] **empathetic attention**,[142] guided imagery,[143,144] and relaxation recordings.[145,146]

8.1.1 Hypnosis should only be performed by a certified hypnotherapist. *[Recommendation]* P

Hypnosis implemented by an untrained individual has the potential to increase the risk for patient injury; facilitating this intervention with education and certification is recommended.[147,148]

Chandrasegaran[147] described a case in which the unintentional use of triggering suggestions (ie, verbiage used during hypnosis that causes adverse reactions to the patient during the intervention) increased the anxiety in a previously hypnosis-relaxed patient to the point where anesthesia induction could not occur and the procedure was cancelled. Trained professionals facilitate the optimal use of hypnosis interventions and lower the risk of adverse events.

In an expert opinion article, Kuttner[148] concluded that hypnosis was effective in reducing pain and anxiety in pediatric patients undergoing anesthesia procedures, but noted that training and supervised practice is required to practice hypnosis.

8.1.2 When hypnosis is used,

- **individualize hypnotic scripts,**[26]
- **do not use hypnosis recordings, and**
- **evaluate the patient preoperatively for triggering words and do not use triggering suggestions when the patient is hypnotized.**[147]

[Recommendation]

High-quality evidence supports individualizing hypnotherapy to provide the most effective intervention without potential negative effects.[26] Low-quality evidence supports individualizing hypnotic scripts to avoid using trigger words or phrases that could negatively affect patient care.[147] Researchers found that recorded hypnosis and **therapeutic suggestion** were not as effective as live hypnosis and may increase the risk of negative outcomes because the care is not individualized to avoid words that may trigger negative emotions based on the patient's life experiences.[26]

In a meta-analysis of 26 RCTs (N = 1,890), Kekecs et al[26] found that compared to therapeutic suggestion, hypnosis decreased anxiety and pain intensity but had no effect on analgesic use or nausea. The researchers concluded that live hypnosis could decrease anxiety and postoperative pain for minor surgeries, and more research is needed to understand the efficacy of therapeutic suggestion and recorded hypnosis interventions, which have the potential to include trigger words.

In a case report, Chandrasegaran[147] described hypnosis attempted to decrease the preoperative anxiety of a patient. The patient was successfully transported to the OR; however, during the hypnosis a trigger word was said that created higher anxiety in the patient and required cancelling the procedure. Preoperative hypnosis without the patient's trigger word was attempted later, which allowed for clinical management of the patient's claustrophobia and cold sensitivity during transport to the OR and allowed for a successful anesthesia induction and surgery. Later, the patient had no recollection of the transport and anesthesia induction in the OR, which increased her satisfaction with the procedure. The researchers noted that hypnosis should be individualized, as the previous hypnosis attempt with trigger words failed, and the researchers concluded that future studies should include hypnotic scripts, especially maintenance and termination scripts.

8.1.3 Empathetic attention interventions may be used as an adjunct to hypnosis, but should not be used alone.[142] *[Conditional Recommendation]*

In an RCT, Lang et al[142] assigned patients to receive either hypnosis (n = 82), empathetic attention without hypnosis (n = 66), or standard care (n = 70). Those who received self-hypnotic relaxation had significantly lower pain and anxiety scores and consumed fewer median drug units (eg, sedative and analgesic agents, 0.5 mg midazolam plus 25 µg fentanyl). Those in the empathetic attention–only group experienced significantly more adverse events. Because the adverse events were significant early in the study, the study was discontinued, and the researchers concluded that nonspecific support that does not provide the means to manage acute pain and anxiety may be more harmful than beneficial.

8.2 Preoperative hypnosis interventions may be used.[26,141] *[Conditional Recommendation]*

Preoperative hypnosis has been associated with

- avoiding unplanned conversion to general anesthesia,[141]

- decreasing intraoperative anesthesia medication demand,[141]
- decreasing postoperative fatigue and nausea/vomiting prophylaxis demand,[141] and
- decreasing preoperative anxiety.[26]

High-quality evidence conflicts as to whether preoperative hypnosis can reduce postoperative pain.[26,141]

In a meta-analysis of 26 RCTs (N = 1,890 patients), Kekecs et al[26] found that compared to therapeutic suggestion, hypnosis decreased anxiety and pain intensity but had no effect on analgesic use or nausea. The researchers concluded that live hypnosis could decrease anxiety and postoperative pain for minor surgeries.

In an RCT that included 150 female patients undergoing minor breast cancer surgeries, Amraoui et al[141] assigned patients to either an intervention group that received 15 minutes of preoperative hypnosis (n = 75) or a control group (n = 75). The intervention group had significantly increased pain scores and LOS in the recovery area, though pain was no different at discharge or at POD 30 follow-up compared to the control group. Patients who received hypnosis had significantly lower consumption of propofol and sufentanil but higher consumption of lidocaine, significantly higher use of noninvasive laryngeal masks for anesthesia, and significantly lower nausea and vomiting prophylaxis use. The researchers concluded that hypnosis did not reduce breast incision pain and that the lower intraoperative propofol use led to higher pain scores and longer recovery times, potentially related to emergence from hypnosis. However, the researchers found that the benefits of hypnosis (ie, less anesthesia drug consumption for airway management) may improve anesthesia outcomes and concluded that the use of perioperative patient hypnosis interventions was beneficial.

8.3 **Preoperative guided imagery interventions may be used for preoperative anxiety and postoperative pain reduction.**[143,144] *[Conditional Recommendation]* P

High-quality evidence supports using guided imagery for adult[144] and pediatric patients[143] to reduce anxiety,[144] preoperative anxiety,[143] and postoperative pain[143,144] and shorten length of PACU stay.[144]

In a meta-analysis of 21 studies conducted between 1995 and 2019 (N = 1,648) that included qualitative studies, Ãlvarez-Garcia and Yaban[143] found that guided imagery was effective in relieving preoperative state anxiety in pediatric patients and preoperative trait anxiety and postoperative pain in adult patients. The researchers concluded that the intervention was easy to implement, effective, and inexpensive.

In an RCT, Gonzales et al[144] assigned 44 adult participants to either listen to a 28-minute guided imagery recording (n = 22) or receive standard care (n = 22) before head and neck procedures. The intervention significantly decreased patients' anxiety and pain at 2 hours after surgery, with an almost significant decrease in recovery time in the PACU. The researchers concluded that guided imagery could be beneficial when there is a short preoperative period, such as for emergency procedures.

8.3.1 **Relaxation recordings with guided imagery interventions may be used for anxiety reduction.**[145,146] *[Conditional Recommendation]*

Low-quality evidence supports the use of relaxation recordings with guided imagery to decrease anxiety.[145,146]

In a quasi-experimental study, Lin[146] assigned a convenience sample of patients undergoing total joint procedures to use a relaxation recording with guided imagery and breath work for 20 minutes preoperatively and on POD 3 (n = 45). The researcher found significantly lower severity of anxiety and difference in anxiety before and after the intervention and on POD 1 for the intervention group compared to a control group (n = 48). There was also a significant difference in systolic blood pressure, and the researcher concluded that relaxation therapy was effective.

Ko and Lin[145] conducted an observational study with a convenience sample of 80 patients who listened to a relaxation recording with deep breathing and guided imagery and meditation before surgery. The researchers found a significant improvement in respiratory rate, heart rate, systolic blood pressure, and anxiety after the intervention, and a significant decrease in anxiety when patients increased the time in the listening session. The researchers also found that female patients' anxiety scores changed significantly more than male patients' scores. The researchers determined they would include use of the relaxation recording as part of their standard care.

8.4 **Intraoperative hypnosis interventions**[149-151] **may be used to**

- **decrease emotional distress,**[149]
- **reduce medication consumption,**[149-151]
- **shorten operative time,**[149,150]
- **decrease pain,**[149,150]
- **improve physiological parameters,**[149,150] **and**
- **decrease recovery time (eg, length of PACU stay,**[149] **length of ICU stay).**[151]

[Conditional Recommendation]

High-quality evidence conflicts on the benefits of intraoperative hypnosis.[149-151]

In a meta-analysis of 34 RCTs (N = 2,597 patients), Tefikow et al[149] found that emotional distress, pain, medication consumption, physiological parameters, recovery, and operative time were clinically improved by hypnosis; however, the results were not statistically significant. The researchers concluded that a single session of hypnosis was as beneficial as multiple sessions for adults undergoing surgical or invasive procedures.

In an RCT, Lang et al[150] compared **self-hypnosis relaxation** (ie, hypnotic relaxation based on self-generated imagery) (n = 82) with structured attention (ie, an intervention with eight key components) (n = 80) and standard care (n = 79). The researchers found that the self-hypnotic relaxation technique significantly decreased pain, drug use, intraoperative hemodynamic instability, and procedure time compared to structured attention or standard care. They concluded that the intervention was associated with lower opioid consumption, which positively affected oxygen saturation. The researchers concluded that hypnosis was beneficial.

In a quasi-experimental study of 143 consecutive patients who underwent transfemoral transcatheter aortic valve implantation under conscious sedation with either hypnotherapy (n = 36) or standard care (n = 107), Takahashi et al[151] found significantly decreased LOS in the ICU and lower consumption of propofol, remifentanil, and norepinephrine in the intraoperative phase in the hypnotherapy group. The researchers concluded that adjunctive hypnotherapy may facilitate perioperative care.

8.4.1 **Hypnosis interventions may be used for pediatric patients before anesthesia induction in the preoperative[152] or intraoperative area.[52,152]** *[Conditional Recommendation]* P

Evidence conflicts as to whether preinduction hypnosis in pediatric patients reduces postoperative anxiety and pain.[52,152]

In an RCT conducted by Duparc-Alegria et al,[152] pediatric patients ages 10 to 18 years who received a 5- to 10-minute preinduction personalized hypnosis imaginary journey (n = 59) had significantly reduced postoperative anxiety compared to patients who received standard care (n = 60). The researchers concluded that preoperative patient interviews conducted to personalize hypnosis imagery and practitioner education regarding hypnosis interventions may decrease postoperative anxiety.

In an RCT of 29 pediatric patients ages 5 to 12 years, Huet et al[52] found those who received hypnosis (n = 14) had significantly lowered anxiety and pain, and significantly more patients rated pain as "no pain" or "mild pain" than those in the control group (n = 15). Though this study was limited to pediatric patients undergoing dental procedures who had low baseline anxiety, the researchers concluded that hypnosis was beneficial.

9. Aromatherapy Interventions

9.1 **Aromatherapy interventions may be combined with other complementary care interventions to reduce pain[43] and nausea severity[44] and increase patient satisfaction.[43]** *[Conditional Recommendation]*

High-quality evidence supports combining citrus essential oil and music to reduce pain and increase patient satisfaction[43] and combining isopropyl alcohol and slow deep breathing to reduce nausea severity.[44]

In an RCT, Nagata et al[43] assigned 224 patients scheduled for screening colonography procedures into one of four groups (n = 56 in each group). The groups received either a combination of music and aromatherapy with citrus bergamia essential oil, the interventions separately, or standard care. Researchers found no significant difference in pain or satisfaction between the groups; however, there were a significant number of requests to use aromatherapy for the next colonography (ie, 67.3% in the group that received both interventions; 72.5% in the group that received aromatherapy alone). The researchers concluded that the intervention might be more beneficial if patients could select the music and aromatherapy scent.

In an RCT conducted by Cronin et al[44] that included female patients undergoing laparoscopic procedures, the patients who experienced nausea in the postoperative recovery setting were taught controlled breathing exercises to perform with (n = 41) or without (n = 41) isopropyl alcohol aromatherapy. The researchers found that nausea severity decreased significantly for both groups. Even though there were inconsistencies in the implementation and outcome evaluation (ie, a verbal scale difficult to use), the researchers concluded that breathing and isopropyl alcohol aromatherapy interventions are inexpensive, self-regulated, and low-risk methods to prevent or reduce PONV.

9.2 **Preoperative aromatherapy interventions may be used for patient anxiety and stress reduction.[153-157]** *[Conditional Recommendation]*

High-quality evidence supports using aromatherapy preoperatively to improve outcomes, including

- reduced self-reported anxiety (lavender[153,155]) and

- improved vital signs and cortisol levels (lavender,[154] lavender essential oil,[156,157] neroli oil[156]).

In an RCT, Trambert et al[153] sought to evaluate the effects of two essential oils on postoperative vital signs and anxiety in 87 women undergoing breast biopsy procedures. The researchers divided the participants into three groups that received either lavender-sandalwood aromatherapy (n = 30), orange-peppermint aromatherapy (n = 30), or a placebo (n = 27). The intervention was performed in the preoperative setting for an average of 64 minutes. The researchers found no postoperative changes in vital signs or anxiety in the orange-peppermint group but found a significant decrease in anxiety in the lavender-sandalwood group. They concluded that aromatherapy with lavender-sandalwood was beneficial.

In an RCT, Hosseini et al[154] compared patients who received two drops of lavender in distilled water for inhalation aromatherapy given the morning of surgery (n = 45) with a standard care control group (n = 45). The researchers found a significant decrease in anxiety scores and cortisol levels, as well as a significant correlation between cortisol levels and anxiety scores before and after the intervention. There was a significant improvement of vital signs in both groups, and the researchers found that 10.8% of anxiety changes and 69.9% of cortisol changes were correlated to the intervention in a post hoc analysis.

In an RCT that included 50 patients undergoing gastroscopy procedures, Hoya et al[157] diffused lavender essential oil and played a soothing recording of natural environmental images and sounds (n = 26) in the preoperative area. The intervention was compared with standard care (n = 24). The researchers found that neither the patients' self-reported anxiety nor systolic blood pressure increased significantly before gastroscopy in the intervention group and concluded that the intervention was effective in reducing preoperative anxiety.

In an RCT of 27 patients undergoing colonoscopy procedures, Hu et al[156] compared the effects of a sunflower oil placebo (n = 13) with neroli oil (eg, citrus aurantium, floral scent with citrus overtone) (n = 14) on procedural anxiety. The researchers found significantly lowered postprocedure systolic blood pressure in the intervention group, which also had a significantly higher preoperative systolic blood pressure. The researchers concluded that neroli oil may promote stress reduction and pain relief, though there were not significant improvements in anxiety or pain scores.

In a pilot observational study conducted by Januzel et al,[155] 30 women undergoing breast or axillary procedures received a lavender aromatherapy patch on the mid-sternal area in the preoperative setting, which was removed before transport to the OR. The average time worn was 58.1 minutes. Though there was no significant change in the patients' vital signs, the researchers found a significant decrease in preoperative anxiety scores after patients wore the patch for 15 minutes.

9.3 **Postoperative aromatherapy interventions may be used for anxiety, pain, and nausea reduction.**[1,9,21,158-161] ***[Conditional Recommendation]***

High-quality evidence[158-161] and guidelines[1,21] support the postoperative use and cost-effectiveness[9] of aromatherapy interventions to reduce anxiety[160] and pain.[159] High-quality evidence conflicts on the efficacy of aromatherapy interventions to reduce patients' experience of and incidence of PONV.[9,158]

In a systematic review of 60 RCTs (N = 1,326), Asay et al[9] concluded that aromatherapy (ie, peppermint; ginger oils; a combination of lavender, peppermint, ginger, and spearmint oils) is a cost-effective method to alleviate PONV and should be considered for use, even though the researchers found nonsignificant results.

In a Cochrane review of 11 RCTs and five controlled trials (N = 1,036), Hines et al[158] found that aromatherapy (eg, isopropyl alcohol, peppermint oil, ginger, combinations) significantly reduced the patients' requests for rescue antiemetics but was not effective in reducing the patient-reported severity of PONV. The use of isopropyl alcohol aromatherapy significantly decreased nausea by 50% during the study and also significantly reduced rescue antiemetic requirements but produced no difference in patient satisfaction. The researchers concluded that the aromatherapy interventions were beneficial.

Lane et al[161] conducted an RCT of 35 women who underwent cesarean delivery and received either peppermint spirits aromatherapy (n = 22), a placebo of colored water (n = 8), or standard care including antiemetic medication only (n = 5). The researchers found a significantly lower nausea score at 2 and 5 minutes postoperatively in the intervention group. The researchers concluded that aromatherapy was useful for decreasing nausea.

In an RCT, Olapour et al[159] assigned 30 women in labor to receive 5 minutes of lavender aromatherapy when they were dilated at 10 cm and again after the baby was delivered. The researchers found significantly lower pain at 4, 8, and 12 hours with a significant decrease in heart rate and an increase in patient satisfaction compared to a control group (n = 30). The researchers also found a significantly higher use of diclofenac in the control group. The researchers concluded that aromatherapy could be used as an adjunct but not treatment.

Braden et al[160] conducted an RCT to evaluate the effects of lavandin essential oil on preoperative anxiety. The researchers assigned 150 patients to one of three groups: one intervention group received lavandin essential oil in a cotton ball on the upper chest or shoulder of their gown as well as near the pedal pulse point, the second group received jojoba oil as a placebo in the same place, and the third group received standard care (n = 50 in each group). The researchers found that the placebo group reported significantly reduced anxiety; however, the intervention group had significantly higher anxiety at baseline, which may have confounded these results. The researchers concluded that use of lavandin essential oil for aromatherapy is a low-risk and cost-effective way to improve patient outcomes and satisfaction.

10. Acupoint Interventions

10.1 Acupoint interventions may be used in the perioperative environment.[4,7,8,39,47,58,69,162-187] *[Conditional Recommendation]*

10.1.1 Implementation considerations for acupoint interventions may include[162-164,184]

- **selecting the delivery method for the specific intervention,**[4,7,8,172,182,183]
- **reviewing applicable regulations about scope of practice,**[7,8,39,172]
- **selecting delivery devices and settings,**
- **determining the length of the intervention, and**
- **recognizing precautions and warnings, such as**
  - **skin reactions**[39,168,185] **and**
  - **the need for forceful removal in the case of a wristband.**[186]

[Conditional Recommendation] P

High-quality evidence[162-164] supports that use of acupoint interventions improves both adult and pediatric[164] patient outcomes, including PONV[162,163] and pain.[164]

In a 2015 Cochrane review, Lee et al[162] investigated 59 RCTs that included 7,667 patients, 727 of whom were pediatric patients, for the effect of PC6 (eg, Nei guan inner wrist) acupoint interventions on PONV. The researchers found that perioperative acupoint stimulation interventions (eg, acupressure, acupuncture, electrical acupoint stimulation) significantly decreased the incidence of PONV and patients' requests for antiemetics compared to patients who received sham acupoint stimulation. However, the intervention did not significantly decrease PONV.

The researchers also found that combining acupoint interventions with antiemetic drugs reduced the incidence of vomiting but not of nausea. In this Cochrane review, 14 trials reported adverse events related to acupuncture that were minor, transient, and self-limiting; these included skin irritation, blisters, redness, and pain where the intervention was applied to the patient's body. The researchers concluded that PC6 acupuncture stimulation efficacy was comparable to antiemetic drug use and that more high-quality trials are required to better understand the effects of PC6 acupoint interventions on PONV.[162]

In a systematic review of the efficacy of acupuncture on PONV that included an additional four reviews not included in the 2015 Cochrane review,[162] Yang et al[164] found that acupuncture in general may reduce PONV (N = 1,552 pediatric patients). The researchers concluded that additional high-quality research is warranted based on small sample sizes and methodology limitations of the RCTs included in the review.

Albooghobeish et al[163] conducted an RCT that included 63 patients undergoing strabismus procedures. Patients underwent 60 seconds of either laser acupuncture of acupoint P6 (n = 21) or of acupoints BL10, BL11, and GB34 (n = 21), compared with a control group that received acupuncture of a sham acupoint without laser (n = 21). The acupuncture was applied 15 minutes before anesthesia induction and 15 minutes after admission to the PACU. The researchers found no significant differences between the acupoint groups except that the incidence of emesis at postoperative hours 2 and 24 was reduced in the P6 acupoint intervention group. The researchers concluded that administering acupuncture to the P6 acupoint was more effective.

10.2 When wristbands are used for acupressure interventions, implement measures to prevent adverse events (eg, pressure injury related to intraoperative positioning, risk for burns when electrosurgery is used, skin reactions,[39,168,185] **need for forceful removal**[186]**).** *[Recommendation]*

High-quality evidence conflicts on the efficacy[39,186] and safety[168,185] of using wristbands to administer acupressure.

In a meta-analysis of RCTs that included 14 trials of 1,009 women undergoing breast procedures, Sun et al[168] concluded that non-needle stimulation may be used for PONV and that 30 minutes before induction to the end of the surgery seemed to be the most effective time frame. The researchers also concluded that the stimulation wristband was most likely to cause adverse reactions (eg, redness, swelling, tenderness, paresthesia).

Cooke et al[186] conducted a pilot RCT in 2015 in which 80 patients received a beaded acupoint wristband (n = 38) or a placebo wristband with no acupressure (n = 42). Researchers found that use of the wristband was well-tolerated, feasible, and justified. Rescue antiemetic consumption, PONV, and quality of recovery were significantly improved; however, the incidence of adverse events was also high (ie, six wristbands becoming tight, with one requiring forceful removal). The researchers recommended additional research with a larger sample size to prove efficacy of the wristbands.

In an RCT, Majholm and Møller[185] assigned 112 healthy, nonsmoking, female patients to receive either P6 acupressure or sham-point acupressure by a wristband before induction of anesthesia to a 24-hour follow-up interview. One-third of the patients had adverse skin reactions related to the wristband (eg, 40 reported redness, 17 reported swelling, 16 reported tenderness, four reported paresthesias), three of which were removed prematurely (ie, before 24 hours post surgery). The researchers concluded that the specific brand of acupressure wristband used was ineffective for perioperative patient outcomes.

In an RCT conducted by Oh and Kim,[39] 54 female patients who underwent gynecological procedures and wore a relief band postoperatively reported decreased PONV when educated and trained experts implemented postoperative electrical acupoint stimulation.

10.3 **Preoperative acupoint interventions (eg, acupressure, acupuncture, electrical acupoint stimulation) may be used.**[4,47,165,166,184,187] *[Conditional Recommendation]* P

The use of preoperative acupoint interventions has been studied for a variety of outcomes and populations.[4,47,165-167,181,184,187] High-quality evidence supports the use of acupressure to reduce both preoperative patient anxiety[4,47,69,165-167] and preoperative parental anxiety.[47] High-quality evidence supports the use of bilateral acupressure before induction of anesthesia to decrease PONV and increase patient satisfaction.[184] High-quality evidence conflicts as to the benefits of placing auricular acupoint that remains in place during general anesthesia.[4,165,166] High-quality evidence supports using preoperative electrical stimulation to reduce postoperative complications (eg, abdominal pain, distention, duration of PACU stay, patient satisfaction, procedure acceptance).[167]

In an RCT, Acar et al[165] found that acupressure on relaxation points on the ear lowered bispectral index system (BIS) reactions, which suggests that the acupuncture intervention benefits are reduced by general anesthesia. Regardless of ear-point tape or needle use, relaxation points may be more beneficial than **traditional Chinese medicine** (TCM) points to reduce anxiety.[4,165]

In an RCT, Valiee et al[187] assigned 70 patients undergoing abdominal surgical procedures to receive either preoperative acupressure (ie, Yintang and Shenmen) or acupressure on two sham points (n = 35 in each group). The researchers found a significant decrease in anxiety scores and a significant decrease in both systolic and diastolic blood pressure after the intervention before surgery. The researchers concluded that acupressure on Yintang and Shenmen acupoints was clinically beneficial to reduce anxiety and blood pressure before abdominal surgery.

In an RCT, 100 patients classified as ASA physical status I and II who were undergoing major laparoscopic procedures received either sham-point (n = 50) or P6 (n = 50) acupressure stimulation bilaterally 30 to 60 minutes before general anesthesia induction.[184] White et al[184] found an overall significant decrease in vomiting from zero to 72 hours in the intervention group (12% versus 30% in the control group). The patients in the intervention group also reported significantly higher satisfaction and quality of postoperative recovery at 48 hours. There were no significant differences in recovery time or return to normal activity between the groups. The researchers concluded that acupressure to P6 combined with antiemetic drugs can reduce the incidence of vomiting among patients who have undergone major laparoscopic procedures.

In an RCT, Wu et al[166] compared acupuncture at four acupoints (ie, Baihui, Four God's Cleverness, Great Rush, Zu san li) (n = 17) to auricular acupuncture (ie, Shenmen) (n = 18) among patients undergoing ambulatory surgical procedures. The researchers found a significant reduction in anxiety scores in both groups but not between the groups. The researchers concluded that both auricular and body acupuncture treatment methods effectively decreased preoperative anxiety in patients.

In a pilot RCT, Wang et al[47] provided 61 parents (ie, one for each pediatric patient) with either 20 minutes of Yintang (n = 28) or sham-point (n = 33) acupuncture before their child's procedure. The researchers found significantly lower anxiety scores in the intervention group, but no significant difference in BIS, heart rate, and arterial blood pressure between the intervention and control groups. The researchers concluded that acupuncture may decrease the anxiety of parents of pediatric patients undergoing surgery.

In an RCT, Wang et al[4] compared responses to auricular acupuncture at different points. In this study, 91 adults with ASA physical status classification of I or II undergoing elective ambulatory surgery received auricular acupuncture at three points for 30 minutes at either TCM points (n = 31), relaxation points (n = 32), or sham/non-anxiety points (n = 27). The researchers found a significant decrease in anxiety scores with relaxation-point acupuncture and a nonsignificant but lower anxiety in the TCM group compared to the control group. The researchers concluded that a 1-minute application of this intervention using the relaxation acupoint technique produced significant improvement. The researchers noted that TCM acupuncture is typically individualized, which was not studied and potentially the reason why there were nonsignificant results for this group.

In an RCT, Chen et al[167] assigned patients with an ASA physical status classification of I or II to receive either 30 minutes of preoperative electrical stimulation on the Jiaji acupoint (n = 114) or a sham point (n = 115) before undergoing an elective colonoscopy procedure. The researchers found significantly lower abdominal pain, distention, and PACU LOS and higher patient satisfaction and acceptance of the procedure in the Jiaji acupoint intervention group. The researchers concluded that the intervention was an effective complementary resource.

10.3.1 **Acupoint interventions that are applied in the preoperative setting, remain in place during the operative or other invasive procedure, and are removed before the patient is transferred to the PACU may be used.**[168,169,181,182] *[Conditional Recommendation]* P

Acupuncture in adults[169] and pediatric patients,[181] acupoint stimulation wristbands in adults,[168] and electrical acupoint stimulation in adults[182] can improve nausea,[169,181] PONV,[168,181] and analgesic consumption.[182]

High-quality evidence supports using acupuncture to reduce nausea and PONV in pediatric patient populations,[168,169,181] though the varied techniques used in the studies limit the ability to generalize specific techniques to other patient populations. The evidence conflicts on whether electrical acupoint stimulation is beneficial;[168,181] one study showed no improvement in emergence agitation, length of PACU stay, or postoperative pain with electrical stimulation of acupoint H7 in pediatric patients.[181]

In a meta-analysis of RCTs that included 14 trials of 1,009 women undergoing breast procedures, Sun et al[168] concluded that non-needle stimulation may be used for PONV and that 30 minutes before induction to the end of the surgery seemed to be the most effective time frame. The researchers also concluded that the stimulation wristband was most likely to cause adverse reactions (eg, redness, swelling, tenderness, paresthesia).

In an RCT, Martin et al[169] assigned 161 pediatric patients ages 3 to 9 years to receive either bilateral P6 acupuncture and antiemetics (n = 86) or antiemetics only (n = 75). The researchers found significantly lower nausea in phase I and II recovery in the acupuncture group, but no difference was found at the final measurement (ie, 24 hours). The study was discontinued after interim analysis because the significant benefit of the intervention was evident. The researchers did not assess the potential effect of the duration of the intervention, which was placed and removed under anesthesia intraoperatively. Other limitations include that the quantity and frequency of the opioids used was not included.

In an RCT conducted by Nakamura et al,[181] pediatric patients ages 18 to 95 months received unilateral electrical acupoint stimulation of the H7 acupoint on the patient's right side at 1 Hz 50 mA throughout the procedure (n = 50) compared to a control group (n = 50). The researchers found no difference in pediatric anesthesia emergence delirium, emergence agitation, emergence agitation severity, PACU LOS, or postoperative pain between the intervention group and control groups; however, the study was underpowered, as the incidence of emergence anesthesia in the control group was low.

In an RCT conducted by Huang et al,[182] 80 adult patients in four groups of patients (n = 20 in each group) undergoing video-assisted thoracic surgical lobectomy procedures received different frequencies (ie, 2 Hz, 100 Hz, intermittent between 2 and 100 Hz, control at 0 Hz) of electrical acupoint stimulation on the same four acupoints (ie, Nei guan, Hegu, Lieque, Quchi) beginning 30 minutes before surgery, throughout the procedure, and during postoperative 30-minute sessions at 24 and 48 hours. The researchers found significantly lower opioid use and a decrease in intraoperative one-lung arterial oxygen partial pressure as well as lower PONV scores, a shorter time to extubation, and shorter PACU LOS when stimulation was set to intermittently change between 2 Hz and 100 Hz.

10.4 **Acupoint interventions may be used in the intraoperative phase of care.**[170,171] *[Conditional Recommendation]*

High-quality evidence supports the intraoperative use of electrical acupoint stimulation (ie, unilateral P6, bilateral P6) to reduce the incidence of PONV in women undergoing operative and other invasive procedures.[170,171] In contrast, other high-quality studies suggest that interoperative acupoint interventions are not effective and the costs may outweigh the benefits.[168,181]

In an RCT conducted by Carr et al,[170] 56 women received antiemetics either with (n = 26) or without (n = 27) P6 electrical acupoint stimulation intraoperatively during a laparoscopic cholecystectomy. The researchers found significantly lower PONV at PACU admission and at 30 and 60 minutes after PACU admission in patients who received acupuncture. The researchers concluded that the intervention was clinically meaningful but suggested that future research with larger sample sizes is needed.

In a RCT, El-Deeb and Ahmady[171] assigned 450 women undergoing elective cesarean delivery to receive either P6 bilateral electrical acupoint stimulation, placebo with a sham acupoint, or ondansetron only (n = 150 in each group) 30 minutes before spinal anesthesia. The researchers found a significantly lower incidence of nausea and vomiting during and 6 hours after surgery in both the electrical acupoint stimulation and ondansetron groups. Patients who received either acupoint stimulation or ondansetron also reported higher satisfaction than patients in the control group, although there was no significant difference in PONV incidence between postoperative hours 6 to 24. The researchers concluded that electrical acupoint stimulation was comparable to ondansetron for reducing PONV and could also improve patient satisfaction.

In a quasi-experimental pilot study, Wang et al[183] recruited 11 healthy volunteers to undergo S36 acupuncture (ie, anterior and inferior to the knee) during both awake and anesthetized MRI scans. The researchers found that patients' blood oxygen level dependent signals, a neurophysiological response to acupuncture stimulation, were lowered with concurrent use of propofol with general anesthesia. They concluded that additional research is needed to determine whether acupoint stimulation is beneficial with propofol use.

10.5 **Postoperative acupoint interventions may be used.**[7,8,39,58,172-180] *[Conditional Recommendation]* P

High-quality evidence supports that acupoint interventions result in a variety of improved outcomes[7,8,39,58,172-180]; however, high variability in study methodology and acupoint interventions was found in these studies.

In a meta-analysis of 14 RCTs (N = 1,653 patients), Chen et al[177] found significantly lower postoperative nausea, antiemetic rescue, dizziness, and pruritus with postoperative electrical acupoint stimulation used on various acupoints after patients underwent procedures with general anesthesia. The researchers concluded that postoperative electrical acupoint stimulation was effective in improving the studied outcomes and that future studies are needed to focus on the effect of spinal anesthesia and cultural biases on the efficacy of postoperative electrical acupoint stimulation.

In a systematic review of 12 trials (N = 1,025 patients), Cho et al[7] found that acupuncture after tonsillectomy significantly decreased pain scores at 4 hours and for the first 48 hours in both pediatric and adult patients (ages 1 to 80 years). Acupuncture also significantly reduced PONV without adverse effects. Analgesic requirements were significantly lower, and the researchers concluded that the intervention was beneficial.

In a systematic review and meta-analysis of 30 RCTs (N = 2,534 patients), Cheong et al[178] found that all acupoint interventions applied to the PC6 acupoint were effective; however, electrical acupoint stimulation with acupressure devices was more effective in reducing PONV than other methods. Electrical acupressure devices used for PONV prevention may be more costly upfront, though they can be reusable.

In an RCT, Yang et al[179] preoperatively assigned patients undergoing gynecological laparoscopic procedures under general anesthesia to receive either dexamethasone alone (n = 50), dexamethasone combined with electrical acupoint stimulation on the Nei guan acupoint at 1 mA/2 Hz (n = 50), or dexamethasone in combination with tropisetron (n = 53). The researchers did not investigate acupoint stimulation alone. The researchers found significantly lower PONV in patients who received acupressure in the first 24 hours compared to dexamethasone alone and nonsignificantly lower PONV in patients who received dexamethasone and tropisetron compared with dexamethasone alone. The researchers concluded that including acupuncture was more beneficial than dexamethasone alone, but had similar results to combining tropisetron with dexamethasone. Even though there was no assessment of patients' request for rescue medication during the first 24 hours, the researchers concluded that acupuncture is a beneficial intervention that can reduce the need for antiemetic medication.

In an RCT, Oh and Kim[39] postoperatively assigned 54 patients who had undergone gynecological procedures (eg, hysterectomy, ovarian cystectomy) to receive either a button wristband that put pressure on the Nei guan acupoint, a relief band with electrical acupoint stimulation to the Nei guan acupoint, or standard care (n = 18 in each group). The researchers concluded that the relief band reduced PONV in women undergoing gynecological procedures when an educated and trained researcher implemented the electrical acupoint stimulation.

In an RCT, Feng et al[173] assigned 150 patients undergoing TKA procedures who were identified as being at high-risk for PONV to receive either auricular acupressure (eg, Shenmen, Point Zero, Subcortex point) (n = 50), acupressure on a sham point (ie, tape with pellets 5 mm from the acupressure point) (n = 47), or a placebo (ie, tape without pellets) (n = 53) in the immediate postoperative period. The researchers found a significantly higher nausea incidence in the placebo group compared with either the acupressure group or the sham-point group. However, differences among all groups for episodes of emesis and recovery time were not significant. The researchers concluded that though nausea measurement is subjective, there was a significant decrease in nausea in the recovery area and at 24 hours post surgery in the acupressure intervention groups.

In an RCT, He et al[174] used **vaccaria seeds** to apply auricular acupressure including the Knee Joint, Shenmen, Subcortex, and Sympathesis points for 45 patients and to four non-acupuncture points for 45 patients undergoing TKA. The researchers found significantly lower pain scores on POD 3, 4, 5, and 7. There was also significantly lower analgesic consumption, fewer adverse events, and significantly higher scores on the Hospital for Special Surgery Knee Scoring System (eg, pain, function, ROM, muscle strength, flexion deformity, instability, subtractions), indicating higher function, at 2 weeks after surgery. The researchers concluded that the intervention was easy to apply, inexpensive, and safe to decrease pain and opioid use and facilitate early rehabilitation.

In an RCT conducted by Chang et al,[175] 62 patients undergoing TKA procedures received three sessions each of auricular acupressure (n = 31) or sham acupressure (ie, tape applied without acupressure) (n = 31) over 3 days. The researchers found a significant decrease in morphine consumption and significant increase in ROM on POD 3 for the intervention group compared with the sham group. They concluded that auricular acupressure interventions can reduce opioid analgesic consumption and improve passive ROM.

In an RCT conducted by Quinlan-Woodward et al,[8] 30 female patients with breast cancer received an average of 36 minutes of acupuncture (n = 15) compared to no acupuncture as a control (n = 15). Participants in the intervention group were significantly younger (mean difference 53.7 versus 62.5 years), and the researchers reported that differences in age may affect acceptance and efficacy of acupuncture. The researchers found a significant reduction in pain, anxiety, and nausea and an increase in coping on POD 1 and pain on POD 2 in the acupuncture group. They concluded that acupuncture could be used as a nonpharmacologic intervention, although further research is needed to determine a standard of care for the intervention.

In an RCT, Zhan and Tian[180] assigned 90 patients to receive either a transversus abdominis plane block with electrical acupoint stimulation (eg, 2 to 100 Hz at the Zusanli ST 36 and Nei guan acupoints), acupuncture (ie, at the Nei guan acupoint), or standard care after open abdominal procedures (n = 30 in each group). The researchers found pain scores decreased significantly at postoperative hours 24 and 48, with significantly lower adverse reactions and hemodynamic effects in the group that received electrical acupoint stimulation. The researchers concluded that electrical acupoint stimulation is an ideal complementary postoperative analgesic strategy.

In an RCT, 18 female patients with cancer who were undergoing open gynecological procedures received either four treatments of acupuncture percutaneous electrical nerve stimulation (PENS) (ie, electrical acupoint stimulation, SP6 and SP8) (n = 9) or traditional acupuncture (ie, no electrical stimulation) at the same points (n = 9).[176] Gavronsky et al[176] found that pain relief was equivalent between the two groups; however, the patients in the PENS group had less pain from hours 24 to 48; at 48 hours there was no difference between the groups. Limitations include that the researchers did not measure opiate consumption, which could also affect bowel motility.

In an RCT, Lili et al[58] assigned patients age 60 years or older undergoing GI procedures to receive either electrical acupoint stimulation for 2 minutes every postoperative day through POD 5 (n = 20) or standard care (n = 20). The researchers found significant improvement for the intervention group in gastrin levels on POD 3 and 5 and motilin levels on POD 3, shorter time to flatus, and a lower incidence of complications (ie, abdominal pain, distention, diarrhea), as well as higher electrocardiogram frequency and amplitude on POD 5, which correlated with gastrin levels and contraction of GI muscles. The researchers concluded that acupuncture stimulation promoted GI function recovery and may decrease complications efficiently and painlessly in older adult patients.

11. Massage and Reflexology Interventions

11.1 **Massage and reflexology interventions may be used.**[188] *[Conditional Recommendation]*

Various massage (eg, general, hand, foot, back) and reflexology techniques are described in the literature, with no one technique determined to be superior.

A feasibility RCT conducted by Rosen et al[188] supports that massage interventions may decrease anxiety when used in both the preoperative and postoperative setting. Patients with cancer who were undergoing port placement received 20 minutes of massage (n = 40) or 20 minutes of structured attention (n = 20) pre- and postoperatively. The researchers found a significant decrease in anxiety in the massage intervention group after the first intervention and after adjusting for patients' baseline mean anxiety. There was also a significant decrease in pain after the procedure and on POD 1 in the massage intervention group.

11.2 **Preoperative massage and reflexology interventions may be used.**[189,190] *[Conditional Recommendation]*

High-quality evidence supports using preoperative massage to reduce pain,[189] stress,[189] and anxiety[189,190] for patients undergoing various procedures. Researchers found that compared to standard care, massage (ie, for older adults undergoing cardiovascular procedures)[190] and hand massage (ie, for women undergoing ambulatory procedures)[189] significantly decreased patients' preoperative anxiety.

In a pilot RCT, Wentworth et al[190] assigned 130 patients undergoing cardiovascular procedures (eg, radiofrequency ablation, angiogram, angiopercutaneous coronary intervention, permanent pacemaker, implantable cardioverter defibrillator) to receive either a preoperative massage (n = 64) or standard care (n = 66) for 20 to 30 minutes. The researchers found a significant decrease in pain, tension, and anxiety between the groups after adjusting for age, with a significant increase in patient satisfaction for those who received the massage. The researchers found that participants in the intervention group were significantly older (67.6 versus 59.7 years), which may influence the conclusions. A limitation of the study was that patients were not blinded to their intervention. The researchers recommended that future studies use a control match for the length of the massage and include objective outcomes.

In a quasi-experimental study, 86 female patients undergoing ambulatory procedures (eg, orthopedic, cataract, cystoscopy, colonoscopy) received preoperative hand massages performed with the room lights dimmed before attempted IV starts.[189] Brand et al[189] found that hand massages significantly lowered the patients' anxiety before surgery. They also found a significant decrease in anxiety and easier IV starts after massage. They concluded that the intervention was easy to implement and did not affect the flow or timing of the surgery. The researchers suggested that future studies evaluate vital signs and control for the patient's biological sex, family presence, and the scheduled procedure.

11.3 **Intraoperative massage and reflexology interventions may be used for patients undergoing awake procedures.**[5,46,191] *[Conditional Recommendation]*

The benefits of using intraoperative massage and reflexology interventions may outweigh the cost of implementation (eg, additional personnel, time, supplies). Intraoperative massage and reflexology interventions can reduce anxiety,[46,191] pain,[5] and vital signs.[46]

High-quality evidence indicates that intraoperative reflexology significantly lowers patients' anxiety and pain perception during varicose vein procedures.[5,46,191] Moderate-quality evidence indicates that general hand massage lowers patients' systolic blood pressure during percutaneous vertebroplasty procedures.[46] However, the evidence conflicts on whether massage significantly decreases pain and increases patient satisfaction.[46,191] Combining handholding with providing verbal step-by-step procedural information throughout the procedure was found to significantly lower anxiety for highly anxious patients but may not benefit all awake patients.[46]

In an RCT, Hudson et al[191] assigned 100 patients (ie, 83 women, 17 men) to receive either intraoperative hand reflexology (n = 50) or standard care (n = 50) during varicose vein procedures performed under local-only anesthesia. The researchers found significantly lower anxiety and shorter pain duration in patients who received the intervention compared to standard care; however, pain and patient satisfaction were not significantly improved.

In an RCT, Dolatian et al[5] assigned 120 women with low-risk pregnancies to one of three groups: 40 minutes of foot reflexology when the patient reached 4 cm to 5 cm cervical dilation during labor, emotional support, or standard care. The patients who received reflexology had significantly lower pain after the intervention, at 6 to 7 cm, and at 8 to 10 cm cervical dilation during labor, whereas emotional support did not significantly lower pain in the stages above 5-cm dilation. Although there was no standard massage technique, the researchers concluded that massage could benefit women in labor.

In a quasi-nonequivalent controlled trial conducted by Kim et al,[46] patients received handholding and information sharing (n = 30), handholding alone (n = 34), or standard care (n = 30) during percutaneous vertebroplasty procedures. The researchers found a significant decrease in systolic blood pressure in the control group, and a significant decrease in anxiety scores in both intervention groups. The researchers concluded that the intervention promotes the relationship between the patient and nurse (ie, qualitative patient responses included feeling supported and encouraged when provided with procedural information and handholding).

11.4 **Postoperative massage and reflexology interventions may be used.**[42,192-195] *[Conditional Recommendation]*

Postoperative massage and reflexology interventions can decrease anxiety,[42,192,194,195] decrease pain,[42,192,193] improve vital signs,[42] and increase patient satisfaction.[192]

High-quality evidence supports using massage techniques in the postoperative setting to improve the patient's response to pain and anxiety and to improve vital signs in the postoperative phase of care.[42,192-195] The researchers in these studies concluded that massage interventions can improve patient experiences; however, they also concluded that additional research is required to address limitations identified in these studies and understand generalizability among perioperative patients.[42,192-195]

Postoperative reflexology may reduce patients' anxiety; however, researchers report a lack of standardization in research methodologies, limiting the ability to generalize results to all patients in the postoperative setting.[194,195] Researchers have concluded that the benefits of massage may have been confounded with benefits of other complementary care interventions that were implemented simultaneously. For example, in one study, researchers found postoperative massage to be beneficial, but the intervention was combined with relaxation techniques.[42] In another study, the researchers had no control of the effects of the external environment during the intervention, which limited their ability to generalize findings about postoperative massage interventions.[192]

In an RCT conducted by Büyükyılmaz and Aştı,[42,30] patients participated in relaxation techniques (ie, breathing, exercises, and music for 30 minutes) and received a back massage (10 minutes lateral lying with lanolin oil) on POD 1 to 3 after total joint procedures. There was a significant decrease in pain intensity, anxiety, and vital signs for patients in the intervention group compared to patients in a control group (n = 30). Limitations of the study include that two interventions were studied without a way to measure the effects of each intervention independently. The researchers concluded that the combination of interventions was effective.

In a pilot RCT, Alameri et al[192] assigned 31 patients in the cardiac ICU after undergoing elective cardiac procedures to receive either a 10-minute foot massage (n = 16) or a 10-minute placebo hand massage (ie, handholding without massage technique) (n = 15). The researchers found a significant decrease in pain intensity and anxiety in the foot massage group compared to the placebo group. Even with a small sample size and potentially confounding variables, the researchers concluded that foot massage is a safe, inexpensive, nonpharmacological intervention that can reduce pain and anxiety.

In a quasi-experimental study, Ucuzal and Kanan[193] assigned patients undergoing breast procedures to receive either a foot massage (n = 35) or analgesics alone (n = 35). The researchers found a significant decrease in pain and improvement of blood pressure and heart rate in the foot massage intervention group but found no significant decrease in respiratory rate between the groups. The researchers found an increase in all the control group participants' vital signs except heart rate. They concluded that foot massages were effective in decreasing pain and that foot massage should be offered.

In an RCT conducted by Bagheri-Nesami et al,[194] 80 patients undergoing coronary artery bypass grafting were matched for age and biological sex then randomly assigned either left-foot 20-minute reflexology for POD 1 to 4 (n = 40) or a gentle 1-minute foot rub (n = 40). The researchers found a significant decrease in anxiety in the group that received reflexology. Variable intervention lengths were not controlled, which limited the researchers' conclusion that reflexology can reduce postoperative anxiety.

In an RCT conducted by Molavi Vardanjani et al,[195] male patients received either a 30-minute foot stimulation of three different reflexology points (n = 50) or a general foot massage (n = 50) on POD 1 after undergoing coronary angiography procedures. The researchers found a significant decrease in anxiety for both groups, with a significantly higher anxiety reduction in the reflexology group. The researchers found that reflexology significantly accounted for 7.5% of the patients' anxiety reduction. Even though the researchers reported a lack of methodological consistency, they recommended implementing two or more postoperative reflexology sessions.

11.5 **Individuals who are trained in specific massage and reflexology interventions may use these techniques.**[42,46,189,190,192,193,196] *[Conditional Recommendation]*

Having educated and trained personnel perform massage or reflexology interventions facilitates consistent knowledge and skills and may be regulated by the scope of practice according to local, state, and federal rules and regulations. This may include education, training, licensure, and certification.[196] If a state does not have reflexology-specific laws, personnel may be exempt or fall within the scope of massage laws.[196] Several researchers described the education and training of the personnel who performed the massage intervention, including certification[190,193] and training of research staff by a certified massage therapist.[42,46,189,192]

12. Electrical Stimulation Interventions

12.1 Postoperative pulsed electromagnetic field (PEMF) interventions may be used.[63,197-204] ***[Conditional Recommendation]***

High-quality evidence supports the use of PEMF interventions to improve incision healing[198,201] and improve mental[197] and physical recovery.[63,197,199,200] However, the evidence conflicts on the benefits of PEMF interventions in reducing postoperative pain[198,201-204] and analgesic consumption.[198,202,203] Researchers in two studies found no improvement in pain reduction.[203,204] However, they suggested that using the patients' opposite side as their own control placebo potentially did not allow for the condition of both sides to be measured separately, if electrical stimulation transferred to the placebo side.[203,204]

In a meta-analysis, Akhter et al[63] included seven RCTs of 941 spinal fusion patients receiving postoperative electrical stimulus compared with a placebo. A postoperative PEMF intervention significantly increased the chances of a successful fusion by 2.5 times in this patient population. The researchers concluded there is moderate-quality support for using this intervention adjunctively for spinal fusion surgeries, and additional research is required to determine whether the patient's age and treatment effect outcomes, as well as measure the effect of the intervention on patient's pain and function.

In an RCT that was not included in the previously discussed meta-analysis,[63] Collarile et al[197] provided 30 patients who had undergone knee procedures with biophysical stimulation with a PEMF intervention for 4 hours every day for 60 days. Patients who received the intervention had a significant improvement in International Knee Documentation Committee Subjective Knee Form (IKDC) scores (ie, knee-specific patient reported outcome measures for subjective knee complaints affecting daily activities with varying score parameters) from 6 months to 60 months and a significant improvement at 60 months in disability and quality of life scores. The researchers also found a significant decrease in pain at 1, 2, and 60 months. They concluded that the PEMF intervention improved clinical outcomes.

In an RCT, Khooshideh et al[198] randomly assigned patients who underwent a cesarean delivery to either an intervention group that received a postoperative PEMF intervention (n = 36) or a control group (n = 36). The intervention group had significantly lower pain scores, fewer reports of severe pain at 24 hours, 1.9 times lower analgesic consumption, lower total doses of analgesics during POD 1 to 7, and significantly better incision healing at POD 7 than patients in the control group. The researchers concluded that lower amounts of exudate and edema increased patient satisfaction in addition to these benefits.

In an RCT, Osti et al[199] assigned patients to receive either a PEMF intervention (n = 32) or a placebo (n = 34) after arthroscopic rotator cuff repair procedures. The researchers found that the intervention group had significant improvement in ROM scores and UCLA scores (ie, pain, function, active forward flexion, strength of forward flexion, patient satisfaction). The researchers concluded that the intervention was safe and reduced pain, analgesic consumption, and stiffness in short-term recovery. Limitations of the study methodology included a delay in providing the intervention (ie, it was not done immediately after surgery), no postoperative objective assessment of improvement (eg, MRI), and no cost analysis of the PEMF intervention.

In an RCT, Osti et al[200] assigned patients undergoing arthroscopic knee microfracture repair procedures to receive either a PEMF intervention (n = 34) or a placebo intervention (n = 34) 6 hours a day for 60 days. The researchers found significant improvement in both groups from baseline to 2 years and significant improvement in IKDC scores (eg, knee symptoms, function, sports activities) and Lysholm improvements (ie, 8 sections of common knee complaints for a total out of 100 points and a visual scale for pain related to each knee). At 5 years, a larger percentage of patients in the PEMF intervention group were still active at their preoperative level (82% compared to 68% in the placebo group). The study was limited by subjective outcomes to evaluate the quality of the repair. The researchers concluded that PEMF application can improve the effectiveness of microfracture repair in the long term.

In an RCT of patients undergoing dental molar extraction procedures, Stocchero et al[201] found significantly fewer incidents of dehiscence and decreased pain scores for patients who received a PEMF intervention (n = 38) compared with those who received a placebo (n = 38) and a control group (n = 38). The researchers also found a significant relationship between pain scores and the length of the intervention or medication (ie, NSAIDs). Though the difficulty of extraction and tooth sectioning may have confounded the results of this study, the researchers concluded that the PEMF intervention improved soft tissue healing and has the potential to be used for pain management in the future.

In an RCT of 32 patients undergoing breast reconstruction procedures, patients who received a PEMF intervention (n = 16) reported significantly lower pain at postoperative hours 6, 12, 24, 48, and 72 compared with a group that received the device but with no electromagnetic field initiated (n = 16).[202] Rhode et al[202] also found a significant decrease in overall narcotic use, interleukin-1b (ie, an inflammatory cytokine associated with pain hypersensitivity) concentration in wound exudate, and exudate volume at 24 hours in the intervention group. The researchers concluded that a PEMF intervention can positively affect the speed and quality of wound repair.

In an RCT of 49 healthy women who underwent elective cosmetic breast augmentation procedures, Svaerdborg et al[203] found no difference in pain scores or medication consumption through POD 7 when a PEMF device was placed intraoperatively to cover each breast implant incision immediately after skin closure and dressing application.

In an RCT of 11 patients undergoing oral surgical procedures, Menini et al[204] applied a PEMF device to both sides of the patient's face immediately after skin closure and dressing application but only activated the device on one cheek so the opposite side would act as a self-control. After 48 hours, there were no significant differences in swelling or pain scores, though the researchers concluded that the active benefits of the PEMF intervention may carry across to the other cheek related to their proximity. The researchers also did not account for the level of disease in each cheek and concluded that additional trials in more invasive surgeries should be studied.

12.2 Transcutaneous electrical nerve stimulation (TENS) interventions may be used in the preoperative and postoperative periods for patients who have undergone thoracic[61,205] or total joint procedures.[96,206] *[Conditional Recommendation]*

Outcomes improved by TENS interventions include decreased analgesic consumption, pain, and systolic blood pressure[61] when used preoperatively and decreased anxiety[206] and pain[96,205,206] and increased ROM[206] when used postoperatively.[205,206]

High-quality evidence supports the use of TENS to improve postoperative outcomes in several patient populations,[96,206] including patients undergoing thoracic[61,205] and TKA procedures.[206] Researchers in two RCTs found a significant decrease in pain after TENS was applied postoperatively for 30 minutes at 85 Hz/180 us pulse width,[61] for 20 minutes at 150 pps/150 us pulse duration,[206] and pre-procedurally at 80 Hz for 45 minutes.[61] Systolic blood pressure improved with diclofenac administration and TENS application after chest tube removal.[61]

In an RCT, 40 patients undergoing thoracic procedures received TENS (n = 20) compared with a control group (n = 20).[205] Erden and Senol Celik[205] found significantly lower pain and analgesic consumption in the intervention group and concluded that TENS was an easy and reliable analgesic method.

In an RCT conducted by Rakel et al,[206] patients who underwent TKA procedures received TENS 1 to 2 times per day (n = 122) compared with a placebo (n = 123) or standard care (n = 72). The researchers found significantly lower pain during active knee extension and fast walking and significantly lower anxiety and greater pain reduction during ROM exercises at 6 weeks in the intervention group. Patients in the TENS and placebo groups had significantly lower hyperalgesia compared with the control group. The researchers concluded that there was a placebo influence, and that the benefits of the intervention were reduced after 6 weeks. However, they also concluded that the patient's anxiety and pain reduction increased the overall benefit of using this intervention. Additional research may include potential tolerance buildup to the TENS effect and measurements for long-term results, general pain, and influence of ethnic diversity.

In an RCT, Chandra et al[61] compared patients who received diclofenac combined with 45 minutes of TENS while undergoing a pleurodesis procedure (n = 30) to a control group (n = 30). The researchers found significantly lower systolic blood pressure and pain scores at postoperative hours 4, 6, and 8 in the intervention group. The researchers also found that additional diclofenac dose requests were significantly lower in the intervention group. The researchers concluded that TENS may be used as an adjunct treatment to reduce postoperative pain.

12.3 Postoperative brain stimulation interventions (eg, transcranial direct current stimulation [tDCS],[207] noninvasive interactive neurostimulation [NIN][208]) may be implemented after TKA procedures. *[Conditional Recommendation]*

The benefits of postoperative brain stimulation may include increased ROM among patients who have undergone total hip and knee procedures[208] and decreased pain[208] and opioid analgesic consumption.[207]

High-quality evidence supports that using tDCS in TKA procedures decreases opioid use without increasing adverse events.[207] High-quality evidence also supports that the use of NIN after TKA procedures may decrease pain and increase ROM.[208]

In an RCT conducted by Borckardt et al,[207] patients undergoing TKA received standard care (n = 20) or 20 minutes of tDCS (ie, stimulation applied under the scalp dermis) in the PACU, 4 hours later, and in the morning and afternoon of POD 1, for 80 total minutes (n = 20). The researchers found significantly lower opioid use in the intervention group, though there was no subjective measurement of patients' mood or pain to confirm this outcome.

In an RCT, Nigam et al[208] assigned patients to receive either NIN (ie, TENS applied directly to the scalp only) (n = 28) or standard care (n = 30) after undergoing TKA procedures. The researchers found a significant decrease in pain and increase in ROM in the intervention group. There was a greater baseline ROM deficit in the intervention group, and the final difference was nonsignificant, indicating a larger increase in ROM is clinically significant. The researchers concluded that there was clear benefit of NIN, though limitations of the study include varied intervention lengths and the potential that NSAID consumption may have influenced study outcomes by further reducing swelling and pain results.

13. Biofield Interventions

13.1 **Biofield interventions (eg, Reiki, healing touch therapy, therapeutic touch, energy field work) may be used in the perioperative environment.**[6,209-214] *[Conditional Recommendation]*

The results of studies of perioperative patients indicate that the benefits of using biofield interventions may outweigh potential harms.[6,209-213] In one review of literature, the authors concluded that qigong (ie, regulation of breath rhythm and pattern, body movement and posture, meditation mind-body exercises) may aid in managing stress and emotion, strengthen respiratory muscles, reduce inflammation, and enhance the immune system in the older adult population,[214] and this intervention might also benefit perioperative patients.

Evidence indicates that there is low risk of patient harm when practitioners are trained and certified to perform biofield interventions.[6,209-214]

13.2 **Reiki interventions may be used preoperatively and intraoperatively for anxiety and pain reduction.**[6,209,210] *[Conditional Recommendation]*

High-quality evidence supports the use of Reiki interventions to reduce preoperative anxiety[209] and to reduce preoperative pain and pain on POD 1 to 3[6,209] in patients undergoing breast and joint procedures. Moderate-quality evidence is inconclusive about whether Reiki is associated with a reduction in intraoperative medication consumption[6,210] and decreased pain in the PACU.[6]

In a systematic review of 12 studies on the effect of Reiki interventions on pain and anxiety in patients undergoing noninvasive, invasive, and operative procedures, Thrane and Cohen[209] found there was a large effect size in anxiety reduction in patients undergoing breast biopsy procedures. The researchers concluded that Reiki may be effective in reducing patient pain and anxiety.

In an RCT pilot conducted by Notte et al,[6] patients undergoing TKA received either five Reiki sessions including 20 minutes of Reiki before surgery, 30 minutes in the PACU after their admission assessment, and 20 minutes on POD 1 to 3 (n = 23) or standard care (n = 20). The researchers found a significant decrease in preoperative pain and a significant decrease in pain on POD 1 to 3 in the Reiki group. Even with variations in the intervention that included the use of relaxing music during Reiki therapy and differences in Reiki technique, the researchers concluded that Reiki was beneficial.

In a quasi-experimental cohort comparison study, Bourque et al[210] evaluated the effect of Reiki delivered during colonoscopy on opioid consumption. Twenty-five patients received 10 minutes of Reiki during the procedure compared with 5 patients who received a placebo Reiki intervention and a control group that received standard care (n = 30). Although there was no significant difference in total meperidine administration, 16% of those patients who received Reiki received less than 50 mg of meperidine, whereas every patient in the control group received more than 50 mg. The researchers concluded that additional research is needed to determine the efficacy of Reiki interventions to reduce opioid consumption.

13.3 **Perioperative healing touch interventions may be used.**[211,215] *[Conditional Recommendation]*

High-quality evidence supports preoperative healing touch interventions as a method to reduce postoperative anxiety.[211,215]

In an RCT conducted by Goldberg et al,[211] women received **magnetic clearing**, a healing touch therapy intervention, 15 minutes before undergoing a breast biopsy procedure (n = 42) compared with a control group (n = 31). The researchers found a significant decrease in anxiety on POD 1 and trait anxiety post intervention compared to POD 1, as well as a significant decrease in scores for emotional and spiritual coping. Even though there is a potential bias related to studying the effect of only one healer in this study, the researchers concluded that the intervention may reduce anxiety.

In an RCT, MacIntyre et al[215] assigned patients undergoing coronary artery bypass grafting procedures to receive either healing touch from one of two certified healing touch providers (n = 99), a visit from a volunteer (n = 94), or standard care (n = 97). For the intervention groups, a certified healing touch provider or a volunteer visited each patient the day before and the day after surgery for 20 to 60 minutes and immediately before surgery for 60 to 90 minutes. Preoperative anxiety in both inpatients and outpatients and LOHS in outpatients were significantly reduced in the healing touch group. The researchers estimated that a $500,000 reduction in cost related to LOHS in this sample population offset the cost of hiring an experienced, certified healing touch provider. The researchers concluded that healing touch was a beneficial complementary intervention that warrants further research.

13.4 Postoperative biofield interventions (eg, therapeutic touch,[212] energy field work[213]) may be used. *[Conditional Recommendation]*

High-quality evidence supports using biofield interventions to reduce postoperative pain and increase participation with postoperative occupational therapy.[213] Moderate-quality evidence supports reduction of pain and improved biobehavioral stress markers with the use of therapeutic touch for patients who have undergone vascular procedures.[212]

In an RCT, McCormack[213] assigned 90 older adult postoperative patients to receive either energy field work for 10 minutes by a trained occupational therapy student, a metronome comparison (ie, an audible, rhythmic sound that can elicit a relaxation response), or standard care as a control (n = 30 in each group). The researchers found a significant decrease in pain and better participation with postoperative occupational therapy in the group that received energy field work. Though the study did not use a certified noncontact therapeutic touch practitioner, the researcher concluded that the intervention was beneficial.

In a quasi-experimental study, Bulette Coakley and Duffy[212] assigned 21 postoperative patients to either a group that received a **Krieger's therapeutic touch intervention** (n = 12) or a control group (n = 9). The researchers found significantly lower subjective pain as well as lower levels of cortisol and higher natural killer cells in the intervention group but determined that additional research is needed.

14. Education

14.1 Health care organizations should design and implement complementary care intervention education for perioperative team members specific to perioperative complementary care interventions used in the organization.[9,25] *[Recommendation]*

14.2 Provide education, training, and supervised practice to personnel before implementing specific complementary care interventions (eg, interactive music therapy, hypnosis).[27,147,148] *[Recommendation]* P

Low-quality evidence supports utilizing providers with suitable education, training, and supervised practice.[147,148]

Chandrasegaran[147] described that unintentional use of triggering suggestions during hypnosis to facilitate anesthesia induction in the OR increased anxiety in a previously hypnosis-relaxed patient. After hypnosis, as the patient was being wheeled into the OR, a member of the care team said "you are being wheeled [into the] OR now, but you don't have to worry, as you are in safe hands of doctors and nurses here, you can remain to be relaxed and enjoy your favorite safe place. If you feel cold, it is just the coldness of the operating room, which will be a signal for you to relax more and you could imagine hugging or wrapping yourself in a warm blanket."[147(p296)] At the mention of the OR and feeling cold, the patient began frowning and shivering and refused surgery. The anxiety induced by inadvertent use of the triggering suggestion was prohibitive to induction of general anesthesia and the procedure was cancelled. Trained professionals facilitate the optimal use of hypnosis interventions and lower the risk of adverse events related to triggering suggestions.

In an expert opinion article, Kuttner[148] described perianesthesia hypnosis techniques used for pediatric patients as either a patient-directed or a "magic glove" technique, and noted that education and supervised practice is required for both. For the patient-directed hypnosis technique, patients are invited to imagine that they are in their favorite place during the preoperative period and then can be reminded to visit their favorite place in the PACU. Health care workers are encouraged to gently touch the patient's shoulder when they recognize that the patient is experiencing discomfort in the hypnotic state. The magic glove is a quick technique to decrease perianesthesia pain and anxiety in which the patient is instructed to put on an imaginary glove that protects his or her hand (eg, the one in which IV insertion will be attempted). A hypnotic trance develops as the patient is guided to experience lower pain sensation in the gloved hand, including during the procedure. In the preoperative and postoperative setting, positive language and comfort-enhancing communication promotes hypnosis. The author concluded that though education is required, hypnosis promotes a positive experience for patients undergoing anesthetic procedures.

14.3 Provide safety education and resources to health care workers who care for patients receiving complementary care interventions.[9,16,25,160] *[Recommendation]*

High-quality evidence supports providing health care workers with education,[9,25] to develop and deliver efficient and safe complementary care interventions.[25] In a systematic review of the effect of aromatherapy on PONV, researchers concluded that inconsistent education led to inconsistent implementation and realized benefits of aromatherapy interventions.[9] In another systematic review regarding preoperative counseling, researchers concluded that health care workers' education should focus on developing and delivering complementary care interventions to substantiate the claims of an intervention's efficacy.[25] Braden et al[160] recommended that nurses collaborate with certified aromatherapists to develop safe protocols for the use of essential oils in their practice. The researchers concluded that to increase safety when using essential oils, the health care organization should provide education on suppliers, patch testing, plan of care application, safety data sheets, and fire safety.

Including members of the interdisciplinary team who will be involved in the care of patients receiving complementary care in the design and implementation of education facilitates interdisciplinary implementation of the interventions. For example, changes in dietary and herbal intervention timing parameters may confuse personnel and create unnecessary delays if nursing departments, preoperative nurses, and anesthesia providers are not all aware of the changes.[16]

14.4 **Maintain records of health care personnel education and competency verification related to complementary care interventions that are used in the organization.**[23,216,217] *[Recommendation]*

Many complementary care interventions can be implemented independently by nurses; however, the nature of the interventions requires collaboration with other caregivers and taking into account their interventions as well as the patients' current or desired lifestyle practices.[23,216,217]

14.5 **Maintain records of current licensure and certification that are required for practitioners of certain complementary care interventions by local, state, and federal regulations.**[23,216,217] *[Recommendation]*

Some complementary care interventions (eg, hypnosis, biofield interventions, aromatherapy, physical therapy, acupoint, massage, reflexology) may require education and certification in addition to nursing licensure and perioperative training for health care workers who care for perioperative patients.[23,216,217]

15. Policies and Procedures

15.1 **Develop and review policies and procedures for complementary care interventions that will be used in the health care organization. Policies should include**

- **local, state, and federal regulations (eg, safety practices);**
- **required education, certification, or licensure for health care personnel implementing complementary care interventions;**
- **a method for obtaining informed consent for complementary care interventions;**
- **clinical documentation requirements (eg, informed consent, patient assessment, response to interventions);**
- **a method for obtaining orders, when required; and**
- **which tools will be used in the organization for assessment of patients receiving complementary care interventions.**

[Recommendation]

Glossary

Acupuncture: Use of sharp, thin needles that are inserted at specific acupoints of the patient's body that are believed to positively adjust the body's energy flow.

Binaural beat: Use of an auditory phenomenon in which two tones of slightly different frequencies are played in separate ears simultaneously (usually via headphones) so the human brain perceives the creation of a new, third tone, whose frequency is equivalent to the difference between the two tones being played; the tone being perceived can alter brain waves through a process called entrainment, promoting specific states of mind.

Biofield: A complex endogenously generated sphere energy activity and information that surrounds a living system engaged in the generation, maintenance, and regulation of biological homeodynamics and physiological functions of the system it surrounds. The biofield is a massless field, not necessarily electromagnetic, that surrounds and permeates living bodies and affects the body.

Biofield (therapy) interventions: Noninvasive, practitioner-mediated therapies in which the biofield of both the practitioner and patient is used to stimulate a healing response (eg, healing touch, Johrei, Pranic healing, Reiki, qigong, therapeutic touch, non-contact therapeutic touch).

Complementary: Non-mainstream practice used together with conventional medicine.

Complementary and alternative medicine: Inclusion of complementary and alternative interventions in medical care.

Continuous passive motion: An exercise in which the affected foot is fixed in a track and the leg is moved toward and away from the body.

Conventional treatment: Treatment of symptoms and diseases by health care professionals using drugs, radiation, and surgery. Synonyms: Western medicine, allopathic, mainstream, orthodox, biomedicine

Emergence agitation: Observable agitation in pediatric patients during anesthesia emergence. Synonym: emergence delirium

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Emergence delirium: Observable agitation in pediatric patients during anesthesia emergence. Synonym: emergence agitation

Empathetic attention: Eight behaviors—matching verbal preference, adapting to nonverbal communication pattern, listening attentively, providing perception of control, swift response to requests, encouragement, avoiding negative language, and using emotionally neutral descriptors.

Enhanced recovery after surgery (ERAS): Pathways for a surgical specialty and facility culture created by interdisciplinary teams for evidence-based and patient-centered care; an integrated continuum from pre-hospital to home again to optimize patients' physiologic function, surgical stress response, and recovery.

Essential oils: Natural chemicals from the essence of certain plants whose scent is used for perfumes, food flavorings, medicine, and aromatherapy; usually distilled using steam or pressure.

Functional status: An evaluation of a person's ability to perform activities of daily living and fulfill personal roles through assessment of elements of the person's being. This assessment includes elements of the person's social support, living environment, mental and emotional state, physical health, economic situation, and use of assistive services.

Healing touch therapy: Energetic therapy to rebalance the energy field for healing of the body, mind, and spirit; the practitioner's hands may be placed on or above the client's body.

Holistic: Comprehension of the parts of something as intimately interconnected and explicable only by reference to the whole; treatment of the whole person, taking into account mental and social factors rather than just the symptoms of the disease.

Induction Compliance Checklist: An observational scale of compliance during induction of anesthesia, with higher scores indicating less compliance.

Integrative treatments: Conventional and complementary approaches used together in coordinated way between different providers and institutions; a holistic, patient-focused approach to treating the whole person.

Interactive music: An intervention developed that includes the use of different instruments and songs to promote the patient's emotional expression and physical release of preoperative anxiety.

Kern light music: Music by Kevin Kern, classified in the light music genre (eg, Western classical music).

Krieger's therapeutic touch: Therapeutic touch wherein the practitioner uses their health energy field to stimulate the patient's own immunological system to repattern itself.

Magnetic clearing: A healing touch technique that focuses on relaxation by releasing congested energy to reduce anger, fear, worry, tension, and anxiety.

Older adults: Persons ages 65 years or older. Further subdivided into

- Young old: ages 65 to 74 years;
- Middle old: ages 75 to 84 years;
- Old old: ages 85 years and older.

Postoperative nausea and vomiting: A common complication associated with perioperative surgical and anesthesia care.

Prehabilitation: The process of improving the functional capability of a patient before a surgical procedure so the patient can withstand any postoperative inactivity and associated decline.

Reflexology: A manual technique on specific reflex points on specific zones within main microcosms on the body (ie, feet, hands, and ears). This brings about psychological and physiological normalization of the total body.

Self-hypnosis relaxation: Hypnotic relaxation based on self-generated imagery.

Simple calculation and reading aloud: Simple arithmetic and reading cognitive skills training, used to improve mental capacity, especially for older adults after they have undergone general anesthesia.

Therapeutic suggestion: Suggestions given without hypnotic induction.

Traditional Chinese medicine: Practices in traditional Chinese medical care; non-Western medical care (eg, acupuncture, herbal supplementation, meditation).

Vaccaria seeds: Plant seeds that are used to stimulate acupressure, typically taped to auricular acupressure points to be worn for multiple days.

References

1. Gan TJ, Diemunsch P, Habib AS, et al. Consensus guidelines for the management of postoperative nausea and vomiting. *Anesth Analg.* 2014;118(1):85-113. [IVA]
2. Donoghue TJ. Herbal medications and anesthesia case management. *AANA J.* 2018;86(3):242-248 [VA]
3. Fatma A, Serife K. Experience of pain in patients undergoing abdominal surgery and nursing approaches to pain control. *Int J Caring Sci.* 2017;10(3):1456-1464. [IIIB]
4. Wang SM, Peloquin C, Kain ZN. The use of auricular acupuncture to reduce preoperative anxiety. *Anesth Analg.* 2001;93(5):1178-1180. [IA]
5. Dolatian M, Hasanpour A, Montazeri S, Heshmat R, Alavi Majd H. The effect of reflexology on pain intensity and duration of labor on primiparas. *Iran Red Crescent Med J.* 2011;13(7):475-479. [IA]
6. Notte BB, Fazzini C, Mooney RA. Reiki's effect on patients with total knee arthroplasty: a pilot study. *Nursing.* 2016;46(2):17-23. [IB]
7. Cho HK, Park IJ, Jeong YM, Lee YJ, Hwang SH. Can perioperative acupuncture reduce the pain and vomiting experienced after tonsillectomy? A meta-analysis. *Laryngoscope.* 2016;126(3):608-615. [IA]
8. Quinlan-Woodward J, Gode A, Dusek JA, Reinstein AS, Johnson JR, Sendelbach S. Assessing the impact of acupuncture on pain, nausea, anxiety, and coping in women undergoing a mastectomy. *Oncol Nurs Forum.* 2016;43(6):725-732. [IA]
9. Asay K, Olson C, Donnelly J, Perlman E. The use of aromatherapy in postoperative nausea and vomiting: a systematic review. *J Perianesth Nurs.* 2019;34(3):502-516. [IA]
10. Complementary, alternative, or integrative health: what's in a name? National Center for Complementary and Integrative Health. https://www.nccih.nih.gov/health/complementary-alternative-or-integrative-health-whats-in-a-name. Accessed May 15, 2021. [VA]

11. Black LI, Clarke TC, Barnes PM, Stussman BJ, Nahin RL. Use of complementary health approaches among children aged 4-17 years in the United States: National Health Interview Survey, 2007-2012. *Natl Health Stat Report.* 2015;(78):1-19. [IIIA]

12. Ho TF, Rowland-Seymour A, Frankel ES, Li SQ, Mao JJ. Generational differences in complementary and alternative medicine (CAM) use in the context of chronic diseases and pain: baby boomers versus the silent generation. *J Am Board Fam Med.* 2014;27(4):465-473. [IIIA]

13. Clarke TC, Black LI, Stussman BJ, Barnes PM, Nahin RL. Trends in the use of complementary health approaches among adults: United States, 2002-2012. *Natl Health Stat Report.* 2015(79):1-16. [IIIA]

14. de Moura LA, Dias IMG, Pereira LV. Prevalence and factors associated with preoperative anxiety in children aged 5-12 years. *Rev Lat Am Enfermagem.* 2016;24:e2708. [IIIA]

15. ERAS society. https://erassociety.org/. Accessed May 15, 2021. [IVA]

16. Position statement: Care coordination and registered nurses' essential role. June 11, 2012. American Nurses Association. https://www.nursingworld.org/~4afbf2/globalassets/practiceandpolicy/health-policy/cnpe-care-coord-position-statement-final--draft-6-12-2012.pdf. Accessed May 15, 2021. [IVA]

17. Bischoff-Ferrari HA, Dawson-Hughes B, Platz A, et al. Effect of high-dosage cholecalciferol and extended physiotherapy on complications after hip fracture: a randomized controlled trial. *Arch Intern Med.* 2010;170(9):813-820. [IA]

18. Chapuy MC, Arlot ME, Duboeuf F, et al. Vitamin D3 and calcium to prevent hip fractures in elderly women. *N Engl J Med.* 1992;327(23):1637-1642. [IB]

19. Seidi J, Ebnerasooli S, Shahsawari S, Nzarian S. The influence of oral ginger before operation on nausea and vomiting after cataract surgery under general anesthesia: a double-blind placebo-controlled randomized clinical trial. *Electron Physician.* 2017;9(1):3508-3514. [IA]

20. Aytekin A, Doru Ö, Kucukoglu S. The effects of distraction on preoperative anxiety level in children. *J Perianesth Nurs.* 2016;31(1):56-62. [IIA]

21. Hooper VD. SAMBA Consensus Guidelines for the Management of Postoperative Nausea and Vomiting: an executive summary for perianesthesia nurses. *J Perianesth Nurs.* 2015;30(5):377-382. [IVB]

22. Rajabpour S, Rayyani M, Mangolian Shahrbabaki P. The relationship between Iranian patients' perception of holistic care and satisfaction with nursing care. *BMC Nurs.* 2019;18:48. [IIIA]

23. Position Statement: Position on the Role of Nurses in the Practice of Complementary & Integrative Health Approaches (CIHA). June 2016. American Holistic Nurses Association. https://www.ahna.org/Portals/66/Docs/Committees/Corrected%20Position%20Statment%20on%20the%20Role%20of%20Nurses%20in%20the%20Practice%20of%20Complementary%20%20Integrative%20Health%20Approaches%20(CIHA)%202016.pdf?ver=HNA-u5c6jHUDiJCXgifuZg%3d%3d. Accessed May 15, 2021. [IVA]

24. Best M, Butow P, Olver I. Do patients want doctors to talk about spirituality? A systematic literature review. *Patient Educ Couns.* 2015;98(11):1320-1328. [VA]

25. Guo P. Preoperative education interventions to reduce anxiety and improve recovery among cardiac surgery patients: a review of randomised controlled trials. *J Clin Nurs.* 2015;24(1-2):34-46. [IA]

26. Kekecs Z, Nagy T, Varga K. The effectiveness of suggestive techniques in reducing postoperative side effects: a meta-analysis of randomized controlled trials. *Anesth Analg.* 2014;119(6):1407-1419. [IA]

27. Kain ZN, Caldwell-Andrews A, Krivutza DM, et al. Interactive music therapy as a treatment for preoperative anxiety in children: a randomized controlled trial. *Anesth Analg.* 2004;98(5):1260-1266. [IC]

28. Institute of Medicine (US). Committee on the Use of Complementary and Alternative Medicine by the American Public. Appendix E. Model guidelines for the use of complementary and alternative therapies in medical practice. In: *Complementary and Alternative Medicine in the United States.* Washington, DC: National Academies Press; 2005. [IVB]

29. Levy I, Attias S, Ben-Arye E, et al. Perioperative risks of dietary and herbal supplements. *World J Surg.* 2017;41(4):927-934. [IIIA]

30. Wang CZ, Moss J, Yuan CS. Commonly used dietary supplements on coagulation function during surgery. *Medicines (Basel).* 2015;2(3):157-185. [IIIC]

31. Herbal products and your anesthesia. American Association of Nurse Anesthetists. https://www.aana.com/patients/herbal-products-and-your-anesthesia. Accessed May 15, 2021. [IVB]

32. Faith J, Thorburn S, Tippens KM. Examining the association between patient-centered communication and provider avoidance, CAM use, and CAM-use disclosure. *Altern Ther Health Med.* 2015;21(2):30-35. [IIIA]

33. Shere-Wolfe K, Tilburt JC, D'Adamo C, Berman B, Chesney MA. Infectious diseases physicians' attitudes and practices related to complementary and integrative medicine: results of a national survey. *Evid Based Complement Alternat Med.* 2013;2013:294381. [IIIA]

34. Kramlich D. Complementary health practitioners in the acute and critical care setting: nursing considerations. *Crit Care Nurse.* 2017;37(3):60-65. [VA]

35. Calcium and vitamin D: Moderate evidence supports use of supplemental vitamin D and calcium in patients following hip fracture surgery. American Academy of Orthopaedic Surgeons. http://www.orthoguidelines.org/guideline-detail?id=1287. Accessed May 15, 2021. [IVA]

36. Guideline for positioning the patient. In: *Guidelines for Perioperative Practice.* Denver, CO: AORN; 2021:643-718. [IVA]

37. Guideline for electrosurgical safety. In: *Guidelines for Perioperative Practice.* Denver, CO: AORN; 2021:83-108. [IVA]

38. Malley A, Kenner C, Kim T, Blakeney B. The role of the nurse and the preoperative assessment in patient transitions. *AORN J.* 2015;102(2):181.e1-181.e9. [IIIB]

39. Oh H, Kim BH. Comparing effects of two different types of Nei-Guan acupuncture stimulation devices in reducing postoperative nausea and vomiting. *J Perianesth Nurs.* 2017;32(3):177-187. [IA]

40. Sankar-Maharaj S, Chen D, Hariharan S. Postoperative shivering among cannabis users at a public hospital in Trinidad, West Indies. *J Perianesth Nurs.* 2018;33(1):37-44. [IIIB]

41. Herbal and dietary supplements and anesthesia. 2018. American Society of Anesthesiologists. https://www.asahq.org/madeforthismoment/wp-content/uploads/2018/10/ASA_Supplements-Anesthesia.pdf. Accessed May 15, 2021. [IVB]

42. Büyükyılmaz F, Aştı T. The effect of relaxation techniques and back massage on pain and anxiety in Turkish total hip or knee arthroplasty patients. *Pain Manag Nurs.* 2013;14(3):143-154. [IB]

43. Nagata K, Iida N, Kanazawa H, et al. Effect of listening to music and essential oil inhalation on patients undergoing screening CT colonography: a randomized controlled trial. *Eur J Radiol.* 2014;83(12):2172-2176. [IA]

44. Cronin SN, Odom-Forren J, Roberts H, Thomas M, Williams S, Wright MI. Effects of controlled breathing, with or without aromatherapy, in the treatment of postoperative nausea. *J Perianesth Nurs.* 2015;30(5):389-397. [IIB]

45. Hasanzadeh F, Kashouk NM, Amini S, et al. The effect of cold application and lavender oil inhalation in cardiac surgery patients undergoing chest tube removal. *EXCLI J.* 2016;15:64-74. [IA]

46. Kim BH, Kang HY, Choi EY. Effects of handholding and providing information on anxiety in patients undergoing percutaneous vertebroplasty. *J Clin Nurs.* 2015;24(23-24):3459-3468. [IIA]

47. Wang SM, Gaal D, Maranets I, Caldwell-Andrews A, Kain ZN. Acupressure and preoperative parental anxiety: a pilot study. *Anesth Analg.* 2005;101(3):666-669. [IB]

48. Çetinkaya F. Effect of listening to music on postoperative cognitive function in older adults after hip or knee surgery: a randomized controlled trial. *J Perianesth Nurs.* 2019;34(5):919-928. [IA]

49. Sultan J, Griffin SM, Di Franco F, et al. Randomized clinical trial of Omega-3 fatty acid-supplemented enteral nutrition versus standard

enteral nutrition in patients undergoing oesophagogastric cancer surgery. *Br J Surg.* 2012;99(3):346-355. [IA]

50. Grode LB, Søgaard A. Improvement of nutritional care after colon surgery: the impact of early oral nutrition in the postanesthesia care unit. *J Perianesth Nurs.* 2014;29(4):266-274. [IB]

51. Fujita T, Daiko H, Nishimura M. Early enteral nutrition reduces the rate of life-threatening complications after thoracic esophagectomy in patients with esophageal cancer. *Eur Surg Res.* 2012;48(2):79-84. [IA]

52. Huet A, Lucas-Polomeni MM, Robert JC, Sixou JL, Wodey E. Hypnosis and dental anesthesia in children: a prospective controlled study. *Int J Clin Exp Hypn.* 2011;59(4):424-440. [IC]

53. Liguori S, Stacchini M, Ciofi D, Olivini N, Bisogni S, Festini F. Effectiveness of an app for reducing preoperative anxiety in children: a randomized clinical trial. *JAMA Pediatr.* 2016;170(8):e160533. [IA]

54. Burke CN, Voepel-Lewis T, Hadden S, et al. Parental presence on emergence: effect on postanesthesia agitation and parent satisfaction. *J Perianesth Nurs.* 2009;24(4):216-221. [IA]

55. Ozkalayci O, Araz C, Cehreli SB, Tirali RE, Kayhan Z. Effects of music on sedation depth and sedative use during pediatric dental procedures. *J Clin Anesth.* 2016;34:647-653. [IA]

56. Kassai B, Rabilloud M, Dantony E, et al. Introduction of a paediatric anaesthesia comic information leaflet reduced preoperative anxiety in children. *Br J Anaesth.* 2016;117(1):95-102. [IA]

57. van Stijn MF, Korkic-Halilovic I, Bakker MS, van der Ploeg T, van Leeuwen PA, Houdijk AP. Preoperative nutrition status and postoperative outcome in elderly general surgery patients: a systematic review. *JPEN J Parenter Enteral Nutr.* 2013;37(1):37-43. [IIIA]

58. Lili H, Lei X, Yan S, Fen G. Effect of electric acupoint stimulation on gastrointestinal hormones and motility among geriatric postoperative patients with gastrointestinal tumors. *J Tradit Chin Med.* 2016;36(4):450-455. [IB]

59. Myint MW, Wu J, Wong E, et al. Clinical benefits of oral nutritional supplementation for elderly hip fracture patients: a single blind randomised controlled trial. *Age Ageing.* 2013;42(1):39-45. [IA]

60. Kulason K, Nouchi R, Hoshikawa Y, Noda M, Okada Y, Kawashima R. The beneficial effects of cognitive training with simple calculation and reading aloud (SCRA) in the elderly postoperative population: a pilot randomized controlled trial. *Front Aging Neurosci.* 2018;10:68. [IB]

61. Chandra A, Dixit MB, Banavaliker JN, Thakur V., Ranjan R. Transcutaneous electrical nerve stimulation as an adjunct to non-steroidal anti-inflammatory medications for pain management during pleurodesis. *Anaesth Pain Intensive Care.* 2013;17(2):154-157. [IB]

62. Bice AA, Wyatt TH. Holistic comfort interventions for pediatric nursing procedures: a systematic review. *J Holist Nurs.* 2017;35(3):280-295. [IIA]

63. Akhter S, Qureshi AR, El-Khechen HA, et al. Efficacy of electrical stimulation for spinal fusion: a systematic review and meta-analysis of randomized controlled trials. *Sci Rep.* 2020;10(1):4568. [IA]

64. Silva RPJ, Rampazzo ARP, Nascimento LA, Fonesca LF. Discomfort patients expect and experience in the immediate postoperative period. *Rev Baiana Enferm.* 2018;32:e26070. [IIIB]

65. American College of Surgeons Statements on Principles. American College of Surgeons. https://www.facs.org/about-acs/statements/stonprin. Updated April 12, 2016. Accessed May 15, 2021. [IVA]

66. Hole J, Hirsch M, Ball E, Meads C. Music as an aid for postoperative recovery in adults: a systematic review and meta-analysis. *Lancet.* 2015;386(10004):1659-1671. [IA]

67. Stallings-Welden LM, Doerner M, Ketchem EL, Benkert L, Alka S, Stallings JD. A comparison of aromatherapy to standard care for relief of PONV and PDNV in ambulatory surgical patients. *J Perianesth Nurs.* 2018;33(2):116-128. [IA]

68. Kiberd MB, Clarke SK, Chorney J, d'Eon B, Wright S. Aromatherapy for the treatment of PONV in children: a pilot RCT. *BMC Complement Altern Med.* 2016;16(1):450. [IB]

69. Lee YR, Shin HS. Effectiveness of ginger essential oil on postoperative nausea and vomiting in abdominal surgery patients. *J Altern Complement Med.* 2017;23(3):196-200. [IIB]

70. Zhang ZS, Wang XL, Xu CL, et al. Music reduces panic: an initial study of listening to preferred music improves male patient discomfort and anxiety during flexible cystoscopy. *J Endourol.* 2014;28(6):739-744. [IA]

71. Mifflin KA, Hackmann T, Chorney JM. Streamed video clips to reduce anxiety in children during inhaled induction of anesthesia. *Anesth Analg.* 2012;115(5):1162-1167. [IA]

72. Pedroso JL, Henriques Aquino CC, Escórcio Bezerra ML, et al. Ginkgo biloba and cerebral bleeding: a case report and critical review. *Neurologist.* 2011;17(2):89-90. [VA]

73. Dunne DF, Jack S, Jones RP, et al. Randomized clinical trial of prehabilitation before planned liver resection. *Br J Surg.* 2016;103(5):504-512. [IA]

74. Sadati L, Pazouki A, Mehdizadeh A, Shoar S, Tamannaie Z, Chaichian S. Effect of preoperative nursing visit on preoperative anxiety and postoperative complications in candidates for laparoscopic cholecystectomy: a randomized clinical trial. *Scand J Caring Sci.* 2013;27(4):994-998. [IC]

75. Bagheri H, Ebrahimi H, Abbasi A, Atashsokhan G, Salmani Z, Zamani M. Effect of preoperative visitation by operating room staff on preoperative anxiety in patients receiving elective hernia surgery. *J Perianesth Nurs.* 2019;34(2):272-280. [IIA]

76. Donker JM, de Vries J, de Lepper CC, et al. A novel finding: the effect of nurse practitioners on the relation to quality of life, anxiety, and depressive symptoms in vascular surgery. *Ann Vasc Surg.* 2014;28(3):644-650. [IIB]

77. Perry JN, Hooper VD, Masiongale J. Reduction of preoperative anxiety in pediatric surgery patients using age-appropriate teaching interventions. *J Perianesth Nurs.* 2012;27(2):69-81. [IIIA]

78. Rhodes L, Nash C, Moisan A, et al. Does preoperative orientation and education alleviate anxiety in posterior spinal fusion patients? A prospective, randomized study. *J Pediatr Orthop.* 2015;35(3):276-279. [IA]

79. Fernandes SC, Arriaga P, Esteves F. Providing preoperative information for children undergoing surgery: a randomized study testing different types of educational material to reduce children's preoperative worries. *Health Educ Res.* 2014;29(6):1058-1076. [IA]

80. Batuman A, Gulec E, Turktan M, Gunes Y, Ozcengiz D. Preoperative informational video reduces preoperative anxiety and postoperative negative behavioral changes in children. *Minerva Anestesiol.* 2016;82(5):534-542. [IA]

81. West AM, Bittner EA, Ortiz VE. The effects of preoperative, video-assisted anesthesia education in Spanish on Spanish-speaking patients' anxiety, knowledge, and satisfaction: a pilot study. *J Clin Anesth.* 2014;26(4):325-329. [IB]

82. Tou S, Tou W, Mah D, Karatassas A, Hewett P. Effect of preoperative two-dimensional animation information on perioperative anxiety and knowledge retention in patients undergoing bowel surgery: a randomized pilot study. *Colorectal Dis.* 2013;15(5):e256-e265. [IB]

83. Zhang CY, Jiang Y, Yin QY, Chen FJ, Ma LL, Wang LX. Impact of nurse-initiated preoperative education on postoperative anxiety symptoms and complications after coronary artery bypass grafting. *J Cardiovasc Nurs.* 2012;27(1):84-88. [IB]

84. Bahrami N, Soleimani MA, Sharifnia H, Shaigan H, Sheikhi MR, Mohammad-Rezaei Z. Effects of anxiety reduction training on physiological indices and serum cortisol levels before elective surgery. *Iran J Nurs Midwifery Res.* 2013;18(5):416-420. [IA]

85. Amini K, Alihossaini Z, Ghahremani Z. Randomized clinical trial comparison of the effect of verbal education and education booklet on preoperative anxiety. *J Perianesth Nurs.* 2019;34(2):289-296. [IA]

86. Lin SY, Huang HA, Lin SC, Huang YT, Wang KY, Shi HY. The effect of an anaesthetic patient information video on perioperative anxiety: a randomised study. *Eur J Anaesthesiol.* 2016;33(2):134-139. [IA]

87. Lindstrom D, Azodi OS, Wladis A,et al. Effects of a perioperative smoking cessation intervention on postoperative complications: a randomized trial. *Ann Surg.* 2008;248(5):739-745. [IA]

88. Emilsson S, Svensk AC, Tavelin B, Lindh J. Support group participation during the post-operative radiotherapy period increases levels of coping resources among women with breast cancer. *Eur J Cancer Care (Engl).* 2012;21(5):591-598. [IIA]

89. Li C, Carli F, Lee L, et al. Impact of a trimodal prehabilitation program on functional recovery after colorectal cancer surgery: a pilot study. *Surg Endosc.* 2013;27(4):1072-1082. [IIA]

90. Carli F, Charlebois P, Stein B, et al. Randomized clinical trial of prehabilitation in colorectal surgery. *Br J Surg.* 2010;97(8):1187-1197. [IA]

91. Gillis C, Li C, Lee L, et al. Prehabilitation versus rehabilitation: a randomized control trial in patients undergoing colorectal resection for cancer. *Anesthesiology.* 2014;121(5):937-947. [IA]

92. Murata K, Yoshimoto M, Takebayashi T, Ida K, Nakano K, Yamashita T. Effect of cryotherapy after spine surgery. *Asian Spine J.* 2014;8(6):753-758. [IIB]

93. Adie S, Kwan A, Naylor JM, Harris IA, Mittal R. Cryotherapy following total knee replacement. *Cochrane Database Syst Rev.* 2012(9):CD007911. [IIB]

94. Demoulin C, Brouwers M, Darot S, Gillet P, Crielaard JM, Vanderthommen M. Comparison of gaseous cryotherapy with more traditional forms of cryotherapy following total knee arthroplasty. *Ann Phys Rehabil Med.* 2012;55(4):229-240. [IA]

95. Quinlan P, Davis J, Fields K, et al. Effects of localized cold therapy on pain in postoperative spinal fusion patients: a randomized control trial. *Orthop Nurs.* 2017;36(5):344-349. [IA]

96. Curry AL, Goehring MT, Bell J, Jette DU. Effect of physical therapy interventions in the acute care setting on function, activity, and participation after total knee arthroplasty: a systematic review. *J Acute Care Phys Ther.* 2018;9(3):1-14. [IIB]

97. Marx W, McKavanagh D, McCarthy AL, et al. The effect of ginger (Zingiber officinale) on platelet aggregation: a systematic literature review. *PLoS One.* 2015;10(10):e0141119. [IIIB]

98. Begtrup KM, Krag AE, Hvas AM. No impact of fish oil supplements on bleeding risk: a systematic review. *Dan Med J.* 2017;64(5): A5366. [IIA]

99. Chen B, Zhou Y, Yang P, Wan H, Wu X. Safety and efficacy of fish oil–enriched parenteral nutrition regimen on postoperative patients undergoing major abdominal surgery: a meta-analysis of randomized controlled trials. *JPEN J Parenter Enteral Nutr.* 2010;34(4):387-394. [IA]

100. Kinross JM, Markar S, Karthikesalingam A, et al. A meta-analysis of probiotic and synbiotic use in elective surgery: does nutrition modulation of the gut microbiome improve clinical outcome? *JPEN J Parenter Enteral Nutr.* 2013;37(2):243-253. [IA]

101. Arumugam S, Lau CSM, Chamberlain RS. Probiotics and synbiotics decrease postoperative sepsis in elective gastrointestinal surgical patients: a meta-analysis. *J Gastrointest Surg.* 2016;20(6):1123-1131. [IA]

102. Short V, Herbert G, Perry R, et al. Chewing gum for postoperative recovery of gastrointestinal function. *Cochrane Database Syst Rev.* 2015(2):CD006506. [IA]

103. Yoshikawa K, Shimada M, Nishioka M, et al. The effects of the Kampo medicine (Japanese herbal medicine) "Daikenchuto" on the surgical inflammatory response following laparoscopic colorectal resection. *Surg Today.* 2012;42(7):646-651. [IIB]

104. Yasunaga H, Miyata H, Horiguchi H, Kuwabara K, Hashimoto H, Matsuda S. Effect of the Japanese herbal Kampo medicine Dai-kenchu-to on postoperative adhesive small bowel obstruction requiring long-tube decompression: a propensity score analysis. *Evid Based Complement Alternat Med.* 2011;2011:264289. [IIIB]

105. Al Sebaee HA, Elhadary SM. Effectiveness of a care bundle on postoperative thirst relief and oral condition among patients undergoing abdominal surgeries. *IOSR J Nurs Health Sci.* 2017;6(5):82-90. [IIA]

106. Garcia AK, Fonseca LF, Aroni P, Galvão CM. Strategies for thirst relief: Integrative literature review. *Rev Bras Enferm.* 2016;69(6):1215-1222. [IIIA]

107. Garcia AKA, Furuya RK, Conchon MF, Rossetto EG, Dantas RAS, Fonseca LF. Menthol chewing gum on preoperative thirst management: randomized clinical trial. *Rev Lat Am Enfermagem.* 2019;27:e3180. [IA]

108. Tóth B, Lantos T, Hegyi P, et al. Ginger (Zingiber officinale): an alternative for the prevention of postoperative nausea and vomiting. A meta-analysis. *Phytomedicine.* 2018;50:8-18. [IA]

109. Soltani E, Jangjoo A, Afzal Aghaei M, Dalili A. Effects of preoperative administration of ginger (Zingiber officinale Roscoe) on postoperative nausea and vomiting after laparoscopic cholecystectomy. *J Tradit Complement Med.* 2017;8(3):387-390. [IA]

110. 21 CFR 501.22. Animal foods; labeling of spices, flavorings, colorings, and chemical preservatives. US Food and Drug Administration. https://www.ecfr.gov/cgi-bin/text-idx?SID=5f3a5f2bf61663047e77fc6506971be5&mc=true&node=pt21.6.501&rgn=div5#se21.6.501_122. Accessed May 17, 2021.

111. Hsu CC, Chen SR, Lee PH, Lin PC. The effect of music listening on pain, heart rate variability, and range of motion in older adults after total knee replacement. *Clin Nurs Res.* 2019;28(5):529-547. [IIA]

112. Forooghy M, Mottahedian Tabrizi E, Hajizadeh E, Pishgoo B. Effect of music therapy on patients' anxiety and hemodynamic parameters during coronary angioplasty: a randomized controlled trial. *Nurs Midwifery Stud.* 2015;4(2):e25800. [IA]

113. Martindale F, Mikocka-Walus AA, Walus BP, Keage H, Andrews JM. The effects of a designer music intervention on patients' anxiety, pain, and experience of colonoscopy: a short report on a pilot study. *Gastroenterol Nurs.* 2014;37(5):338-342. [IIA]

114. Palmer JB, Lane D, Mayo D, Schluchter M, Leeming R. Effects of music therapy on anesthesia requirements and anxiety in women undergoing ambulatory breast surgery for cancer diagnosis and treatment: a randomized controlled trial. *J Clin Oncol.* 2015;33(28):3162-3168. [IA]

115. Ko CH, Chen YY, Wu KT, et al. Effect of music on level of anxiety in patients undergoing colonoscopy without sedation. *J Chin Med Assoc.* 2017;80(3):154-160. [IA]

116. Weeks BP, Nilsson U. Music interventions in patients during coronary angiographic procedures: a randomized controlled study of the effect on patients' anxiety and well-being. *Eur J Cardiovasc Nurs.* 2011;10(2):88-93. [IC]

117. Padmanabhan R, Hildreth AJ, Laws D. A prospective, randomised, controlled study examining binaural beat audio and preoperative anxiety in patients undergoing general anaesthesia for day case surgery. *Anaesthesia.* 2005;60(9):874-877. [IA]

118. Klassen JA, Liang Y, Tjosvold L, Klassen TP, Hartling L. Music for pain and anxiety in children undergoing medical procedures: a systematic review of randomized controlled trials. *Ambul Pediatr.* 2008;8(2):117-128. [IA]

119. Bradt J, Dileo C, Shim M. Music interventions for preoperative anxiety. *Cochrane Database Syst Rev.* 2013;(6):006908. [IIA]

120. Bashiri M, Akçalı D, Coşkun D, Cindoruk M, Dikmen A, Çifdalöz BU. Evaluation of pain and patient satisfaction by music therapy in patients with endoscopy/colonoscopy. *Turk J Gastroenterol.* 2018;29(5): 574-579. [IA]

121. Jiménez-Jiménez M, García-Escalona A, Martín-López A, De Vera-Vera R, De Haro J. Intraoperative stress and anxiety reduction with music therapy: a controlled randomized clinical trial of efficacy and safety. *J Vasc Nurs.* 2013;31(3):101-106. [IA]

122. Wu J, Chaplin W, Amico J, et al. Music for surgical abortion care study: a randomized controlled pilot study. *Contraception.* 2012;85(5):496-502. [IC]

123. Easter B, DeBoer L, Settlemyer G, Starnes C, Marlowe V, Tart RC. The impact of music on the PACU patient's perception of discomfort. *J Perianesth Nurs.* 2010;25(2):79-87. [IA]

124. Hosseinpour M, Memarzadeh M. Use of a preoperative playroom to prepare children for surgery. *Eur J Pediatr Surg.* 2010;20(6):408-411. [IA]

125. Kerimoglu B, Neuman A, Paul J, Stefanov DG, Twersky R. Anesthesia induction using video glasses as a distraction tool for the management of preoperative anxiety in children. *Anesth Analg.* 2013;117(6):1373-1379. [IA]

126. Bekelis K, Calnan D, Simmons N, MacKenzie T, Kakoulides G. Effect of an immersive preoperative virtual reality experience on patient reported outcomes: a randomized controlled trial. *Ann Surg.* 2017;265(6):1068-1073. [IA]

127. Park JW, Nahm FS, Kim JH, Jeon YT, Ryu JH, Han SH. The effect of mirroring display of virtual reality tour of the operating theatre on preoperative anxiety: a randomized controlled trial. *IEEE J Biomed Health Inform.* 2019;23(6):2655-2660. [IA]

128. Dehghan F, Jalali R, Bashiri H. The effect of virtual reality technology on preoperative anxiety in children: a Solomon four-group randomized clinical trial. *Perioper Med (Lond).* 2019;8:5. [IA]

129. Ryu JH, Park JW, Nahm FS, et al. The effect of gamification through a virtual reality on preoperative anxiety in pediatric patients undergoing general anesthesia: a prospective, randomized, and controlled trial. *J Clin Med.* 2018;7(9):284. [IA]

130. Ryu JH, Park SJ, Park JW, et al. Randomized clinical trial of immersive virtual reality tour of the operating theatre in children before anaesthesia. *Br J Surg.* 2017;104(12):1628-1633. [IA]

131. Agostini F, Monti F, Neri E, Dellabartola S, de Pascalis L, Bozicevic L. Parental anxiety and stress before pediatric anesthesia: a pilot study on the effectiveness of preoperative clown intervention. *J Health Psychol.* 2014;19(5):587-601. [IA]

132. Berger J, Wilson D, Potts L, Polivka B. Wacky Wednesday: use of distraction through humor to reduce preoperative anxiety in children and their parents. *J Perianesth Nurs.* 2014;29(4):285-291. [IIB]

133. Birnie KA, Noel M, Chambers CT, Uman LS, Parker JA. Psychological interventions for needle-related procedural pain and distress in children and adolescents. *Cochrane Database Syst Rev.* 2018(10):CD005179. [IA]

134. Kain ZN, Wang SM, Mayes LC, Krivutza DM, Teague BA. Sensory stimuli and anxiety in children undergoing surgery: a randomized, controlled trial. *Anesth Analg.* 2001;92(4):897-903. [IA]

135. Cagiran E, Sergin D, Deniz MN, Tanattı B, Emiroglu N, Alper I. Effects of sociodemographic factors and maternal anxiety on preoperative anxiety in children. *J Int Med Res.* 2014;42(2):572-580. [IIIA]

136. Manyande A, Cyna AM, Yip P, Chooi C, Middleton P. Non-pharmacological interventions for assisting the induction of anaesthesia in children. *Cochrane Database Syst Rev.* 2015(7):CD006447. [IC]

137. Chan CS, Molassiotis A. The effects of an educational programme on the anxiety and satisfaction level of parents having parent present induction and visitation in a postanaesthesia care unit. *Pediatr Anaesth.* 2002;12(2):131-139. [IIA]

138. In WY, Kim YM, Kim HS, et al. The effect of a parental visitation program on emergence delirium among postoperative children in the PACU. *J Perianesth Nurs.* 2019;34(1):108-116. [IIB]

139. Pagnard E, Sarver W. Family visitation in the PACU: an evidence-based practice project. *J Perianesth Nurs.* 2019;34(3):600-605. [VA]

140. Wendler MC, Smith K, Ellenburg W, Gill R, Anderson L, Spiegel-Thayer K. "To see with my own eyes": experiences of family visits during phase 1 recovery. *J Perianesth Nurs.* 2017;32(1):45-57. [IIIA]

141. Amraoui J, Pouliquen C, Fraisse J, et al. Effects of a hypnosis session before general anesthesia on postoperative outcomes in patients who underwent minor breast cancer surgery: The HYPNOSEIN randomized clinical trial. *JAMA Netw Open.* 2018;1(4):e181164. [IB]

142. Lang EV, Berbaum KS, Pauker SG, et al. Beneficial effects of hypnosis and adverse effects of empathic attention during percutaneous tumor treatment: when being nice does not suffice. *J Vasc Interv Radiol.* 2008;19(6):897-905. [IA]

143. Ãlvarez-García C, Yaban ZŞ. The effects of preoperative guided imagery interventions on preoperative anxiety and postoperative pain: a meta-analysis. *Complement Ther Clin Pract.* 2020;38:101077. [IA]

144. Gonzales EA, Ledesma RJA, McAllister DJ, Perry SM, Dyer CA, Maye JP. Effects of guided imagery on postoperative outcomes in patients undergoing same-day surgical procedures: a randomized, single-blind study. *AANA J.* 2010;78(3):181-188. [IB]

145. Ko YL, Lin PC. The effect of using a relaxation tape on pulse, respiration, blood pressure and anxiety levels of surgical patients. *J Clin Nurs.* 2012;21(5-6):689-697. [IIIB]

146. Lin PC. An evaluation of the effectiveness of relaxation therapy for patients receiving joint replacement surgery. *J Clin Nurs.* 2012;21(5-6):601-608. [IIA]

147. Chandrasegaran A. Case study: clinical management of claustrophobia and cold sensitivity towards operating room environment with preoperative hypnosis. *Sleep Hypn.* 2018;20(4):294-298. [VB]

148. Kuttner L. Pediatric hypnosis: pre-, peri-, and post-anesthesia. *Pediatr Anesth.* 2012;22(6):573-577. [VB]

149. Tefikow S, Barth J, Maichrowitz S, Beelmann A, Strauss B, Rosendahl J. Efficacy of hypnosis in adults undergoing surgery or medical procedures: a meta-analysis of randomized controlled trials. *Clin Psychol Rev.* 2013;33(5):623-636. [IA]

150. Lang EV, Benotsch EG, Fick LJ, et al. Adjunctive non-pharmacological analgesia for invasive medical procedures: a randomised trial. *Lancet.* 2000;355(9214):1486-1490. [IA]

151. Takahashi M, Mouillet G, Khaled A, et al. Perioperative outcomes of adjunctive hypnotherapy compared with conscious sedation alone for patients undergoing transfemoral transcatheter aortic valve implantation. *Int Heart J.* 2020;61(1):60-66. [IIA]

152. Duparc-Alegria N, Tiberghien K, Abdoul H, Dahmani S, Alberti C, Thiollier AF. Assessment of a short hypnosis in a paediatric operating room in reducing postoperative pain and anxiety: a randomised study. *J Clin Nurs.* 2018;27(1-2):86-91. [IB]

153. Trambert R, Kowalski MO, Wu B, Mehta N, Friedman P. A randomized controlled trial provides evidence to support aromatherapy to minimize anxiety in women undergoing breast biopsy. *Worldviews Evid Based Nurs.* 2017;14(5):394-402. [IA]

154. Hosseini S, Heydari A, Vakili M, Moghadam S, Tazyky S. Effect of lavender essence inhalation on the level of anxiety and blood cortisol in candidates for open-heart surgery. *Iran J Nurs Midwifery Res.* 2016;21(4):397-401. [IA]

155. Jaruzel CB, Gregoski M, Mueller M, Faircloth A, Kelechi T. Aromatherapy for preoperative anxiety: a pilot study. *J Perianesth Nurs.* 2019;34(2):259-264. [IIA]

156. Hu PH, Peng YC, Lin YT, Chang CS, Ou MC. Aromatherapy for reducing colonoscopy related procedural anxiety and physiological parameters: a randomized controlled study. *Hepatogastroenterology.* 2010;57(102-103):1082-1086. [IC]

157. Hoya Y, Matsumura I, Fujita T, Yanaga K. The use of nonpharmacological interventions to reduce anxiety in patients undergoing gastroscopy in a setting with an optimal soothing environment. *Gastroenterol Nurs.* 2008;31(6):395-399. [IC]

158. Hines S, Steels E, Chang A, Gibbons K. Aromatherapy for treatment of postoperative nausea and vomiting. *Cochrane Database Syst Rev.* 2018;3(3):CD007598. [IIA]

159. Olapour A, Behaeen K, Akhondzadeh R, Soltani F, Al Sadat Razavi F, Bekhradi R. The effect of inhalation of aromatherapy blend containing lavender essential oil on cesarean postoperative pain. *Anesth Pain Med.* 2013;3(1):203-207. [IA]

160. Braden R, Reichow S, Halm MA. The use of the essential oil lavandin to reduce preoperative anxiety in surgical patients. *J Perianesth Nurs.* 2009;24(6):348-355. [IA]

161. Lane B, Cannella K, Bowen C, et al. Examination of the effectiveness of peppermint aromatherapy on nausea in women post C-section. *J Holist Nurs.* 2012;30(2):90-104. [IA]

162. Lee A, Chan SK, Fan LT. Stimulation of the wrist acupuncture point PC6 for preventing postoperative nausea and vomiting. *Cochrane Database Syst Rev.* 2015;(11):CD003281. [IA]

163. Albooghobeish M, Mohtadi A, Saidkhani V, et al. Comparative effects of the stimulation of BL10, BL11, and GB34 acupuncture points with P6 point using a low-level laser on the prevention of vomiting after strabismus surgery: a randomized, double-blind, controlled clinical trial. *Iran Red Crescent Med J.* 2019;21(2):e64713. [IA]

164. Yang C, Hao Z, Zhang LL, Guo Q. Efficacy and safety of acupuncture in children: an overview of systematic reviews. *Pediatr Res.* 2015;78(2):112-119. [IA]

165. Acar HV, Cuvas O, Ceyhan A, Dikmen B. Acupuncture on Yintang point decreases preoperative anxiety. *J Altern Complement Med.* 2013;19(5):420-424. [IA]

166. Wu S, Liang J, Zhu X, Liu X, Miao D. Comparing the treatment effectiveness of body acupuncture and auricular acupuncture in preoperative anxiety treatment. *J Res Med Sci.* 2011;16(1):39-42. [IC]

167. Chen Y, Wu W, Yao Y, Yang Y, Zhao Q, Qiu,L. Transcutaneous electric acupoint stimulation at Jiaji points reduce abdominal pain after colonoscopy: a randomized controlled trial. *Int J Clin Exp Med.* 2015;8(4):5972-5977. [IA]

168. Sun R, Dai W, Liu Y, et al. Non-needle acupoint stimulation for prevention of nausea and vomiting after breast surgery: a meta-analysis. *Medicine (Baltimore).* 2019;98(10):e14713. [IA]

169. Martin CS, Deverman SE, Norvell DC, Cusick JC, Kendrick A, Koh J. Randomized trial of acupuncture with antiemetics for reducing postoperative nausea in children. *Acta Anaesthesiol Scand.* 2018;63(3):292-297. [IA]

170. Carr KL, Johnson FE, Kenaan CA, Welton JM. Effects of P6 stimulation on postoperative nausea and vomiting in laparoscopic cholecystectomy patients. *J Perianesth Nurs.* 2015;30(2):143-150. [IB]

171. El-Deeb AM, Ahmady MS. Effect of acupuncture on nausea and/or vomiting during and after cesarean section in comparison with ondansetron. *J Anesth.* 2011;25(5):698-703. [IA]

172. Kwon JH, Shin Y, Juon HS. Effects of Nei-guan (P6) acupressure wristband: on nausea, vomiting, and retching in women after thyroidectomy. *Cancer Nurs.* 2016;39(1):61-66. [IIA]

173. Feng C, Popovic J, Kline R, et al. Auricular acupressure in the prevention of postoperative nausea and emesis: a randomized controlled trial. *Bull Hosp Jt Dis (2013).* 2017;75(2):114-118. [IA]

174. He BJ, Tong PJ, Li J, Jing HT, Yao XM. Auricular acupressure for analgesia in perioperative period of total knee arthroplasty. *Pain Med.* 2013;14(10):1608-1613. [IA]

175. Chang LH, Hsu CH, Jong GP, Ho S, Tsay SL, Lin KC. Auricular acupressure for managing postoperative pain and knee motion in patients with total knee replacement: a randomized sham control study. *Evid Based Complement Alternat Med.* 2012;2012:528452. [IA]

176. Gavronsky S, Koeniger-Donohue R, Steller J, Hawkins JW. Postoperative pain: acupuncture versus percutaneous electrical nerve stimulation. *Pain Manag Nurs.* 2012;13(3):150-156. [IB]

177. Chen J, Tu Q, Miao S, Zhou Z, Hu S. Transcutaneous electrical acupoint stimulation for preventing postoperative nausea and vomiting after general anesthesia: a meta-analysis of randomized controlled trials. *Int J Surg.* 2020;73:57-64. [IA]

178. Cheong KB, Zhang JP, Huang Y, Zhang ZJ. The effectiveness of acupuncture in prevention and treatment of postoperative nausea and vomiting—a systematic review and meta-analysis. *PLoS One.* 2013;8(12):e82474. [IA]

179. Yang XY, Xiao J, Chen YH, et al. Dexamethasone alone vs in combination with transcutaneous electrical acupoint stimulation or tropisetron for prevention of postoperative nausea and vomiting in gynaecological patients undergoing laparoscopic surgery. *Br J Anaesth.* 2015;115(6):883-889. [IA]

180. Zhan W, Tian W. Addition of transcutaneous electric acupoint stimulation to transverse abdominis plane block for postoperative analgesia in abdominal surgery: a randomized controlled trial. *Eur J Integr Med.* 2020;35:101087. [IA]

181. Nakamura N, Mihara T, Hijikata T, Goto T, Ka K. Unilateral electrical stimulation of the heart 7 acupuncture point to prevent emergence agitation in children: a prospective, double-blinded, randomized clinical trial. *PLoS One.* 2018;13(10):e0204533. [IA]

182. Huang S, Peng W, Tian X, et al. Effects of transcutaneous electrical acupoint stimulation at different frequencies on perioperative anesthetic dosage, recovery, complications, and prognosis in video-assisted thoracic surgical lobectomy: a randomized, double-blinded, placebo-controlled trial. *J Anesth.* 2017;31(1):58-65. [IA]

183. Wang SM, Constable RT, Tokoglu FS, Weiss DA, Freyle D, Kain ZN. Acupuncture-induced blood oxygenation level-dependent signals in awake and anesthetized volunteers: a pilot study. *Anesth Analg.* 2007;105(2):499-506. [IIB]

184. White PF, Zhao M, Tang J, et al. Use of a disposable acupressure device as part of a multimodal antiemetic strategy for reducing postoperative nausea and vomiting. *Anesth Analg.* 2012;115(1):31-37. [IB]

185. Majholm B, Møller AM. Acupressure at acupoint P6 for prevention of postoperative nausea and vomiting: a randomised clinical trial. *Eur J Anaesthesiol.* 2011;28(6):412-419. [IA]

186. Cooke M, Rapchuk I, Doi SA, et al. Wrist acupressure for postoperative nausea and vomiting (WrAP): a pilot study. *Complement Ther Med.* 2015;23(3):372-380. [IA]

187. Valiee S, Bassampour SS, Nasrabadi AN, Pouresmaeil Z, Mehran A. Effect of acupressure on preoperative anxiety: a clinical trial. *J Perianesth Nurs.* 2012;27(4):259-266. [IB]

188. Rosen J, Lawrence R, Bouchard M, Doros G, Gardiner P, Saper R. Massage for perioperative pain and anxiety in placement of vascular access devices. *Adv Mind Body Med.* 2013;27(1):12-23. [IB]

189. Brand LR, Munroe DJ, Gavin J. The effect of hand massage on preoperative anxiety in ambulatory surgery patients. *AORN J.* 2013;97(6):708-717. [IIB]

190. Wentworth LJ, Briese LJ, Timimi FK, et al. Massage therapy reduces tension, anxiety, and pain in patients awaiting invasive cardiovascular procedures. *Prog Cardiovasc Nurs.* 2009;24(4):155-161. [IB]

191. Hudson BF, Davidson J, Whiteley MS. The impact of hand reflexology on pain, anxiety and satisfaction during minimally invasive surgery under local anaesthetic: a randomised controlled trial. *Int J Nurs Stud.* 2015;52(12):1789-1797. [IA]

192. Alameri R, Dean G, Castner J, Volpe E, Elghoneimy Y, Jungquist C. Efficacy of precise foot massage therapy on pain and anxiety following cardiac surgery: pilot study. *Pain Manag Nurs.* 2020;21(4):314-322. [IB]

193. Ucuzal M, Kanan N. Foot massage: effectiveness on postoperative pain in breast surgery patients. *Pain Manag Nurs.* 2014;15(2):458-465. [IIB]

194. Bagheri-Nesami M, Shorofi SA, Zargar N, Sohrabi M, Gholipour-Baradari A, Khalilian A. The effects of foot reflexology massage on anxiety in patients following coronary artery bypass graft surgery: a randomized controlled trial. *Complement Ther Clin Pract.* 2014;20(1):42-47. [IA]

195. Molavi Vardanjani M, Masoudi Alavi N, Razavi NS, Aghajani M, Azizi-Fini E, Vaghefi SM. A randomized-controlled trial examining the effects of reflexology on anxiety of patients undergoing coronary angiography. *Nurs Midwifery Stud.* 2013;2(3):3-9. [IA]

196. Laws Across the USA. Reflexology Association of America. https://reflexology-usa.org/wp-content/uploads/reflexology_laws_by_state2.pdf. Revised October 2013. Accessed May 17, 2021.

197. Collarile M, Sambri A, Lullini G, Cadossi M, Zorzi C. Biophysical stimulation improves clinical results of matrix-assisted autologous chondrocyte implantation in the treatment of chondral lesions of the knee. *Knee Surg Sports Traumatol Arthrosc.* 2018;26(4):1223-1229. [IA]

198. Khooshideh M, Latifi Rostami SS, Sheikh M, MD P, Ghorbani Yekta B, Shahriari A. Pulsed electromagnetic fields for postsurgical pain management in women undergoing cesarean section: a randomized, double-blind, placebo-controlled trial. *Clin J Pain.* 2017;33(2):142-147. [IA]

199. Osti L, Del Buono A, Maffulli N. Pulsed electromagnetic fields after rotator cuff repair: a randomized, controlled study. *Orthopedics.* 2015;38(3):e223-e228. [IA]
200. Osti L, Del Buono A, Maffulli N. Application of pulsed electromagnetic fields after microfractures to the knee: a mid-term study. *Int Orthop.* 2015;39(7):1289-1294. [IB]
201. Stocchero M, Gobbato L, De Biagi M, Bressan E, Sivolella S. Pulsed electromagnetic fields for postoperative pain: a randomized controlled clinical trial in patients undergoing mandibular third molar extraction. *Oral Surg Oral Med Oral Pathol Oral Radiol.* 2015;119(3):293-300. [IA]
202. Rohde CH, Taylor EM, Alonso A, Ascherman JA, Hardy KL, Pilla AA. Pulsed electromagnetic fields reduce postoperative interleukin-1[beta], pain, and inflammation: a double-blind, placebo-controlled study in TRAM flap breast reconstruction patients. *Plast Reconstr Surg.* 2015;135(5):808e-817e. [IB]
203. Svaerdborg M, Momsen OH, Damsgaard TE. Pulsed electromagnetic fields for postoperative pain treatment after breast augmentation: a double-blind, placebo-controlled study. *Aesthet Surg J.* 2016;36(6):NP199-NP201. [IB]
204. Menini M, Bevilacqua M, Setti P, Tealdo T, Pesce P, Pera P. Effects of pulsed electromagnetic fields on swelling and pain after implant surgery: a double-blind, randomized study. *Int J Oral Maxillofac Surg.* 2016;45(3):346-353. [IB]
205. Erden S, Senol Celik S. The effect of transcutaneous electrical nerve stimulation on post-thoracotomy pain. *Contemp Nurse.* 2015;51(2-3):163-170. [IA]
206. Rakel BA, Zimmerman BM, Geasland K, et al. Transcutaneous electrical nerve stimulation for the control of pain during rehabilitation after total knee arthroplasty: a randomized, blinded, placebo-controlled trial. *Pain.* 2014;155(12):2599-2611. [IA]
207. Borckardt JJ, Reeves ST, Robinson SM, et al. Transcranial direct current stimulation (tDCS) reduces postsurgical opioid consumption in total knee arthroplasty (TKA). *Clin J Pain.* 2013;29(11):925-928. [IA]
208. Nigam AK, Taylor DM, Valeyeva Z. Non-invasive interactive neurostimulation (InterXTM) reduces acute pain in patients following total knee replacement surgery: a randomised, controlled trial. *J Orthop Surg Res.* 2011;6:45. [IA]
209. Thrane S, Cohen SM. Effect of Reiki therapy on pain and anxiety in adults: an in-depth literature review of randomized trials with effect size calculations. *Pain Manag Nurs.* 2014;15(4):897-908. [IA]
210. Bourque AL, Sullivan ME, Winter MR. Reiki as a pain management adjunct in screening colonoscopy. *Gastroenterol Nurs.* 2012;35(5):308-312. [IIC]
211. Goldberg DR, Wardell DW, Kilgarriff N, Williams B, Eichler D, Thomlinson P. An initial study using healing touch for women undergoing a breast biopsy. *J Holist Nurs.* 2016;34(2):123-134. [IA]
212. Bulette Coakley A, Duffy ME. The effect of therapeutic touch on postoperative patients. *J Holist Nurs.* 2010;28(3):193-200. [IIA]
213. McCormack GL. Using non-contact therapeutic touch to manage post-surgical pain in the elderly. *Occup Ther Int.* 2009;16(1):44-56. [IA]
214. Feng F, Tuchman S, Denninger JW, Fricchione GL, Yeung A. Qigong for the prevention, treatment, and rehabilitation of COVID-19 infection in older adults. *Am J Geriatr Psychiatry.* 2020;28(8):812-819. [VA]
215. MacIntyre B, Hamilton J, Fricke T, Ma W, Mehle S, Michel M. The efficacy of healing touch in coronary artery bypass surgery recovery: a randomized clinical trial. *Altern Ther Health Med.* 2008;14(4):24-32. [IA]
216. *AORN Position Statement on Patient Safety.* 2017. AORN, Inc. https://aorn.org/guidelines/clinical-resources/position-statements. Accessed May 17, 2021. [IVA]
217. *AORN Position Statement on a Healthy Perioperative Practice Environment.* 2015. AORN, Inc. https://aorn.org/guidelines/clinical-resources/position-statements. Accessed May 17, 2021. [IVA]

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